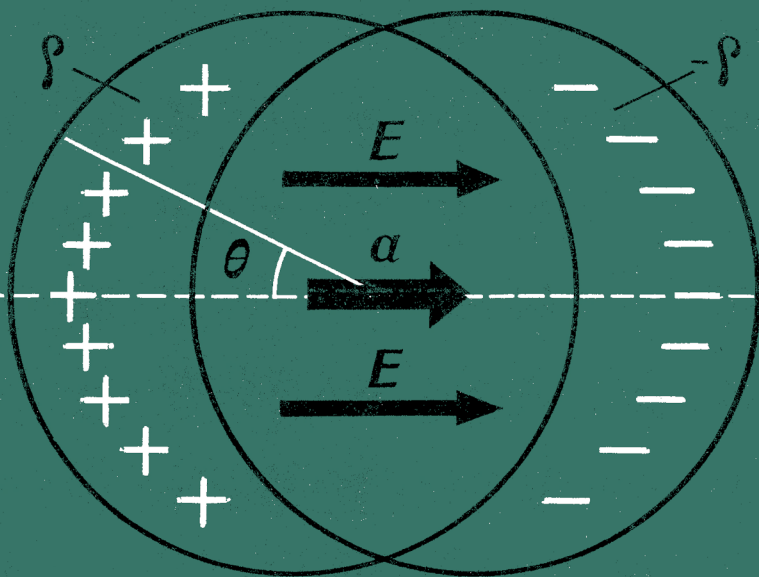


S.KOZEL, E.RASHBA, S.SLAVATINSKII

COLLECTED PROBLEMS IN PHYSICS



MIR PUBLISHERS MOSCOW

**COLLECTED
PROBLEMS
IN
PHYSICS**

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CONTENTS

Preface	6
Authors' Note	9
Problems	11
I. Mechanics	11
II. Thermodynamics and Molecular Physics	19
III. Electricity and Magnetism	27
IV. Optics	41
V. Atomic Physics	50
VI. Radiation	56
VII. Solid State	57
VIII. Nuclear Physics	63
Answers and Solutions	69
Appendices	173
I. Constants and Tables	173
II. Some Non-System Units	175
III. Units of Some Physical Quantities in the SI and CGS Systems and Their Relationships	176
IV. Elementary Particle Table	178
V. Prefixes and Powers for Decimal Multiple and Fractional Units	180
Periodic Table of Elements	182

PREFACE

Just as fairy-tales assist children to learn about the adult world, so problems serve the same purpose by introducing a student to physics. Problems truly do have much in common with folklore, giving the student an idea of the world of physics, how it is described, and how it can be understood. Elementary problems deal with an imaginary world of point masses, weightless threads, ideal gases, and other perfect bodies, in the same way as fairy-tale worlds are populated with fierce serpents and handsome princes who travel on magic flying carpets in search for firebirds. In such a world the Powers of Good and Evil are distinct and moral problems are notable for their clarity and single answers. Problems also allow us to consider situations that are almost unreal or even fantastic, and hence like fairy-tales stimulate our imagination. Step by step more advanced problems bring us nearer to our complicated picture of the reality of scientific research, where in many cases a great deal of effort is needed even to formulate the questions and where in the end deeper investigation often results in an expansion of our understanding, enabling us to reinterpret our initial problem. The same is true with the problems in this collection. In many cases a more profound analysis will require either new calculations or more serious reflections.

Problems and fairy-tales are also alike in respect of their authorship. It is rare to know exactly who originated a particular problem while established sections such as mechanics contain a collection of traditional plots which are rearranged by each generation in its own fashion. Most of these problems have been extracted from the archives of the Moscow Physico-Technical Institute (MPTI). The compilers of this collection, Professors S. M. Kozel, E. I. Rashba, and S. A. Slavutinskii, have accomplished a great deal by systematically editing the whole of this voluminous archives for the first time: until now problems were only published individually. The compilers

have succeeded in preserving the vivid and ingenious approach which seems to be characteristic of modern active research physicists.

This problems book exemplifies the "Phystech System", in which active scientists take direct part in the education and training of the next generation of researchers and engineers. This direct link is especially important when teaching the fundamental branches of the natural sciences of which physics and mathematics are the principal ones. The gap between a student's speciality and general education is thus narrowed and eliminates the delay between demands and supply which arises when general courses are taught exclusively by full-time educators. The problems book thus continues the traditions of teaching physics laid down by Academicians P. I. Kapitza and L. D. Landau when they founded the MPTI. We are all aware of the originality of P. I. Kapitza's own problems, as they have been published in a number of books, and the position given to problems in the theoretical physics curriculum devised by L. D. Landau and E. M. Lifshitz.

To a certain extent the problems themselves define the standards required of students on the general physics course at the Moscow Physico-Technical Institute. This is because problems are a much better measure of the demands placed on a student than curricula. It is thus no coincidence that many problems have been taken from the Final Examination, which is the summit of an MPTI course of general physics. The Final Examination is set by a commission of physicists from all the basic chairs and departments at the MPTI. A student must present both written solutions to the problems set and an essay on a subject he or she has selected. The problems suggested for this examination are designed so as to include several branches of physics at once. When solving a problem a student must have all the skills of general physics, viz. the ability to idealize phenomena, to apply conservation laws, symmetry and invariance conditions, and similarity principles, to isolate dimensionless parameters, and to interpret results under limiting conditions. Another demand consistently required of students is that the solution be carried through to a numerical answer. It is important that the consequent numbers have not only a real sense, but are useful and help the student remember necessary data.

New physical problems mirror the time and circumstances of their formulation much better than do the curricula of courses in general physics. In fact, we can and must reflect every significant advance in physics in the problems we set if we are

to teach students at the pace of the advance in scientific thought. The two most active branches of fundamental research, i.e. the physics of outer space and astrophysics, on one hand, and elementary particle physics, on the other, are an inexhaustible source of topics and inspiration for problems in general physics. The many years of mankind's experience of learning how to understand natural phenomena demonstrate that the most important fundamental properties of matter become clear when it is in some limiting form. The extremum conditions of nature engender a vast scope for imagination and can be used to test estimates and conclusions by direct and often instructive applications of the principal laws of physics. It is to be hoped, therefore, that some problems will form the basis of more thorough analyses in seminars, during which the solutions given by the compilers can be developed.

Finally, I would ask both instructors and, especially, students to search for and invent their own problems. With our modern, somewhat pragmatic, predilection in education, or rather in teaching methods, solving ready-made problems is given much thought, but it is important that in the necessary inculcation of the habits needed to solve problems, in which are sown the seeds of scientific work, we should not inhibit imagination or suppress creative intuition.

Of all tasks the most important is to correctly state the problem. This proposition is valid both when a scientist is being educated and during his later career. Yet to teach someone how to do this is the hardest of all tasks.

S. P. Kapitza

AUTHORS' NOTE

This collection includes problems that have been set over many years to physics students at the Moscow Physico-Technical Institute. The initiator of this project was Professor S. P. Kapitza, the chairman of the Physics Chair.

The problems we chose from the department's voluminous archives are those which seemed the most interesting from the physical point of view. There is a rather large proportion of advanced problems, and so many of the problems are provided with solutions or detailed instructions. At the same time we felt in a number of cases that there is a need for some to remain without full solutions and they only have final answers. The SI system of units is preferred though the absolute Gaussian system is also used.

Note that most of the problems are original and have been suggested by the staff of the department. In this sense the collection is the joint work of the whole department although all responsibility for potential oversights is entirely ours. We distributed the work involved, thus Sections III and IV were prepared by S. M. Kozel, Sections II, VI, and VII by E. I. Rashba, while Sections I, V, and VIII were prepared by S. A. Slavatinskii. Many of the problems in the collection were suggested by Professors M. D. Galanin, A. D. Gladun, L. L. Gol'din, B. G. Erozolinskii, S. P. Kapitza, L. A. Mikaelyan, and D. V. Sivukhin, and by Assistant Professors D. B. Dialtrotov, A. P. Kiryanov, L. B. Luganskii, A. B. Franzesson, and I. F. Shchyogolev. We are especially grateful to Assistant Professor D. A. Zaikin for his help.

We have freely used several problems books which specialised in the different branches of physics and which were published in small quantities at the MPTI between 1976 and 1977. A large amount of the work of preparing these problems books was performed by L. P. Bakanina, N. S. Be-

ryuleva, A. V. Stepanov, M. A. Tulaikova, and I. A. Fomin. The preparation of the manuscript for publication was mainly performed by T. A. Seliverstova and L. N. Shustova. We express our gratitude to all those mentioned. We are also grateful to our editor B. S. Belikov, whose remarks greatly improved the manuscript.

When compiling the collection we preferred the more practical problems originating from the outcome of experiments. Such problems involve real physical objects and not idealized schemes. Many problems require approximations and should assist students in developing a clarity of physical thought and a sense-of-scale of physical quantities and phenomena. In order to solve the majority of the problems both a formal knowledge of the laws and a broad physical outlook are required. Some problems make no pretence at being original, they are rather of a general physical interest.

We hope that this collection will be useful for physics students, postgraduates, and physics instructors at higher educational institutions.

Since the publication of the Russian edition of this collection we, our colleagues, and our students have detected a number of errors. All the errors so far identified have been corrected and we should like to express our warm gratitude to all those who drew them to our attention.

PROBLEMS

I. Mechanics

- 1.1. Is it possible to measure the gravitational constant γ with a relative error of $\delta = 10\%$ by rolling a lead ball to a test weight suspended from the pan of an analytical spring balance whose maximum relative sensitivity is $\eta = 10^{-7}$ (0.1 mg/kg)?
- 1.2. A binary star has a period $T = 3$ years, while the distance L between its components is two astronomical units. Express the mass of the star in terms of the mass of the Sun.
- 1.3. Find the tension in a 63 m cord used to connect a 70 kg astronaut to a satellite whose mass M is much larger than that of the astronaut. The satellite closely orbits the Earth. Which arrangement of the astronaut, satellite and the Earth causes the tension to be at a maximum?
- 1.4. Determine the mass of the planet Mars in terms of the parameters of the elliptic orbit of the Soviet space probe "Mars-2" around the planet. The maximum distance between the probe and the planet surface at apogee is $a = 25\,000$ km, the minimum distance at perigee is $b = 1380$ km, the revolution period T_M being 18 h 00 min. Mars's diameter D_M is 6800 km, the necessary parameters of the Earth are taken to be known. In your calculations, take the Earth's mass to be unit mass.
- 1.5. By considering the Earth to be a uniform ball of radius R_E with density ρ , find the gravitational pressure p as a function of the distance from the Earth's centre. Calculate the pressure at the centre assuming that $R_E = 6370$ km, $\rho = 5.5$ g/cm³.
- 1.6. Estimate the difference due to the rotation of the Earth between the distance from the centre of the Earth to the sea level R_p at the pole and that at the equator R_e .
- 1.7. Determine the direction and magnitude of the escape velocity v at which a spacecraft will leave the solar system if launched from the Earth with a velocity v_0 whose di-

rection is perpendicular to the line connecting the Earth and the Sun, and in the direction of the Earth's rotation around the Sun.

1.8. Find the minimum velocity with which an uncontrolled spacecraft launched from the Earth along an Earth-Moon trajectory strikes the surface of the Moon. The motion of the Earth and the Moon should be neglected.

1.9. Evaluate the mean time t between the collisions of two Sun-like stars in a cluster of stars if their average relative velocity u is 60 km/s and the number of stars per cubic light year is $N \approx 10$. Define more accurately your concept of collision. The mass of the Sun and its radius are $M_S = 2 \times 10^{30}$ kg and $R_S = 7 \times 10^8$ m, respectively.

1.10. Two satellites A and B are 45 km apart and move in a common circular orbit of the Earth. In order to dock, the satellites must approach each other and yet continue to move along their common orbit. Which simple sequence of short bursts of the lagging satellite's engine can accomplish this manoeuvre, if the engine is oriented at a tangent to the orbit and each burst can change the satellite's velocity by a Δv not exceeding 8 km/h?

1.11. At a certain altitude the density of the atmosphere is $\rho_a \approx 1.6 \times 10^{-11}$ kg/m³. Evaluate the aerodynamic drag experienced by a satellite with a cross-sectional area $S = 0.5$ m² and mass $m = 10$ kg orbiting at this height. How will the velocity and height of the satellite change per revolution?

1.12. In a science fiction story the Earth rotational velocity was changed by launching a projectile along a tangent to the equator. What should the difference ($c - v$) be between the velocities of light and the projectile in order to stop the Earth rotating around its axis? The Earth radius is $R_E = 6370$ km, and its mass is $M_E = 6 \times 10^{24}$ kg. The Earth's moment of inertia relative to its rotational axis, with its density nonuniformity properly incorporated, is quite accurately given by the formula $I \approx M_E R_E^2/3$. Compare kinetic energies of the projectile and the Earth's rotation. The projectile's rest mass can be taken to be $m = 10^6$ kg.

1.13. Determine the power needed by a photon rocket to move outside the solar system with a nonrelativistic speed and a constant acceleration of $a = 10$ m/s². The rocket's mass is $M = 10^3$ kg. Compare the power of such a rocket with that of the Bratsk hydroelectric power station (4.5×10^6 kW).

1.14. A toy rocket is filled with water, which occupies a small part of its internal cavity. The remainder of the cavity is pumped full of air until the pressure is p . Estimate the rocket's final altitude h , assuming the mass of water m is much smaller than that of the rocket M , that the time taken by the water to discharge is much smaller than the duration of the flight, and that the cross section of the rocket's nozzle is much smaller than that of cavity.

1.15. A satellite revolves around a planet (or other body with a spherically symmetric mass distribution) in the immediate vicinity of its surface. Show that the period of revolution T depends on the planet's average density ρ . Calculate the period of revolution of a satellite around a neutron star assuming that latter's density is the same as the density of matter inside atomic nuclei ($\rho \approx 10^{14} \text{ g/cm}^3$).

1.16. A satellite of mass m moves along a circular orbit around a star of mass M , m being much less than M . At a certain instant the star explodes like a supernova and ejects a mass qM . Assuming that this mass goes into orbit instantaneously, describe the possible kinds of motion the satellite may have subsequently.

1.17. A spaceship nears the Moon along a parabolic trajectory that almost touches the Moon's surface. In order to transfer into a circular orbit a retro engine fires at the instant of the closest approach. The engine ejects gas at a speed of 4 km/s relative to the spaceship in its direction of motion. What fraction of total mass should the fuel burnt make up? Estimate the temperature of combustion if the specific heat capacity of the ejected gas is $c_p = 2.2 \times 10^3 \text{ J/K} \cdot \text{kg}$.

1.18. A rocket is launched from the top of the highest mountain on the Moon. The angle between the direction of gas jet and the horizon is maintained at $\varphi = 0.1 \text{ rad}$. The speed of the jet relative to the rocket is $u = 4 \text{ km/s}$. How must the rocket's mass M be changed in time for the rocket to move horizontally? How long will it take to reach orbital velocity? By how many times will the rocket's mass be reduced during this interval? What will the astronaut's overload be? The Moon's radius is $R_M = 1700 \text{ km}$, and the free-fall acceleration near the Moon's surface $g_M = 1.7 \text{ m/s}^2$.

1.19. A satellite flies along a circular orbit close to the Earth's surface. The satellite's mass is 50 kg. A small meteorite which has a mass of 0.1 g and is moving towards the centre of the Earth at 80 km/s hits the satellite and adheres to it.

Assuming that the impact is central, estimate the difference between the apogee a and the perigee b of the new orbit.

1.20. A spaceship takes off from the North Pole at a tangent to the Earth's surface. Find the rotation angle of the Earth's axis due to the missile's launch. The Earth's mass is 6×10^{24} kg and its radius R_E 6400 km, the missile's mass is 1000 t and its launch speed 15 km/s. The engine can be assumed only to operate at launch.

1.21. A "launching-pad" with a spaceship is moving along a circular orbit of the Moon whose radius R is triple that of the Moon R_M . The ship leaves the launching-pad with a relative velocity equal to the launching-pad's initial orbital velocity v_0 , and the launching-pad then falls to the Moon. Determine the angle φ at which the launching-pad crashes onto the surface if its mass is twice that of the spaceship m .

1.22. Determine the minimum fuel reserve necessary for a soft lunar landing of a rocket. A retro engine is fired for a time τ at a height h above the Moon's surface. Assume that the rocket's velocity at infinity is much less than the velocity v_0 at height h and that v_0 is gained by the rocket due to the gravity of the Moon. The jet velocity relative to the rocket is u and the mass of the fuelled rocket before the engine is started is m_0 . Assume that $h \ll R_M$, where R_M is the Moon's radius.

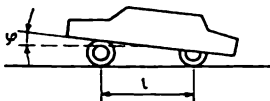


Fig. 1

✓1.23. A pendulum clock is located on the Earth's surface over a subway tunnel. The tunnel's diameter is $D = 2R = 10$ m and its axis is at a depth $h = 15$ m. Taking the average soil density to be $\rho = 2$ g/cm³, estimate the relative change in clock's period due to the presence of the tunnel.

1.24. By what angle φ will a car (Fig. 1) be tilted as it brakes? The centre of mass is equidistant from the front and rear wheels and is 0.4 m above the ground. The sliding friction coefficient f is 0.8 and the distance between the wheel axles l is 5 m. The elasticity of all the suspension springs is the same and such that if the car was resting on a horizontal platform the deflection would be $\Delta = 10$ cm.

1.25. A gymnast falls 12 m into an elastic net. By how many times will the maximum force, which is exerted upon the gymnast by the net, exceed his initial weight if the net sags $\Delta_0 = 1$ m under the initial weight of the gymnast?

1.26. An α -particle flying at a velocity v_0 is elastically scattered from a resting nucleus through 90° relative to its initial direction of motion. For which ratio between the mass of the α -particle m and that of the nucleus M is this possible? Determine the velocities of the α -particle v and nucleus u after the collision. Determine also the angle φ between the direction of the escaped nucleus and the initial direction of motion of the α -particle.

1.27. By how many times is the nucleus recoil energy in the α -decay of ^{226}Ra larger than in its γ -decay if the total energy released in α -decay is $E_1 = 4.9$ MeV and that released in the γ -decay is $E_\gamma = 0.2$ MeV?

1.28. Is it possible for a cesium-133 atom to ionize by colliding with an oxygen-16 atom having a kinetic energy T_0 of 4 eV? Cesium-133's ionization energy is 3.9 eV.

1.29. At which kinetic energies T of α -particles can they be inelastically scattered from ^{14}N nuclei if the energy of the first excited state of this nucleus is $U = 2.31$ MeV?

1.30. In what manner will the speed of a pocket watch be changed if laid on a horizontal, absolutely smooth table? Assume that the axis of the torsional pendulum passes through the watch's centre while the watch's moment of inertia I_0 is 500 times larger than that of the pendulum I .

1.31. In an earlier classical model, an electron was considered to be a rotating ball of mass m uniformly filling a sphere of radius $r_0 = e^2/mc^2$, where e is the electron charge. The electron's intrinsic angular momentum (spin) was $\hbar/2$. Show that this model contradicts the special theory of relativity, since the linear velocity of the electron's rotation v exceeds the velocity of light c .

1.32. Estimate the value of the angular momentum L of a bicycle wheel (adult size) moving at the 30 km/h speed.

1.33. A man sits on a swing moving with a small angular amplitude φ_0 . When the swing passes through the equilibrium position, the man stands quickly and at the instant of the swing's maximum deviation he sits again. How will the angular amplitude of the swing be changed during a period? The man's centre of mass moves up and down through a height h . The swing's ropes are l m long. Assume in the calculations that $l \gg h$ and that mass of the swing is negligibly small compared to the mass of the man.

1.34. A tennis-ball hits a heavy racket at 60° to the normal and rebounds elastically. The ball's mass is negligibly small compared to that of the racket. How must the racket move

to make the ball rebound at right angles to its initial trajectory?

1.35. A bar is firmly fixed to a vertical axle at an angle φ to the latter. Two identical masses m are located at distances l from the fixing point. The axle is rotated at an angular velocity ω . The distances from the fixing point to axle's ball-bearings are a and b . Find the reaction forces on the ball-bearings.

1.36. A horizontal disc of radius R and mass m is suspended in a gas by an elastic thread with torsional modulus f at a distance h ($h \ll R$) from a horizontal fixed surface. Find the internal friction coefficient η in the gas as a result of measuring the logarithmic damping factor d of torsional oscillations. Assume that the gas's motion between the disc and fixed surface is laminar and neglect edge effects.

1.37. A hoop of radius R is thrown with a linear velocity v_0 and simultaneously is given an angular velocity ω_0 . Determine the minimum angular velocity at which the hoop will roll backwards after moving forward and slipping. Find the final velocity v if $\omega_0 > \omega_{0 \text{ min}}$. Rolling friction can be neglected.

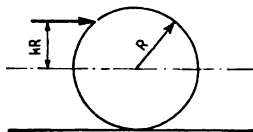


Fig. 2

1.38. A ball of mass m and radius R (Fig. 2) lying on a horizontal table experiences a short horizontal impulse which

imparts a momentum p . The height of the impact point above the centre is kR ($k \leq 1$). Find the energy of translational and rotational motions of the ball. At what k will the ball roll without sliding?

1.39. A stool is fitted so that it rests with two legs on the floor, then it is freed and falls on all four legs. Estimate how far the stool will move across the floor.

1.40. A wheel has fallen off a car and rolled along the ground. The wheel was observed to have travelled in a circle of radius R . Determine the angle of inclination between the wheel axle and the horizon. The car moved at a velocity v . Assume all the mass of the wheel is concentrated at its periphery. It is known that R is much larger than the radius of the wheel.

1.41. Determine the maximum gyroscopic pressure of a rapid turbine that can be installed in a ship. The ship pitches

with an angular amplitude of $\varphi = 9^\circ$ and a period $T = 15$ s around the axis perpendicular to the rotor axis. The turbine rotor, whose moment of inertia is $I = 1260 \text{ kg}\cdot\text{m}^2$, rotates at $\nu = 3000$ rpm. The distance between the rotor bearings is $l = 2$ m.

1.42. A motor-bike moves along the inner surface of a cylinder. To be able to move in a horizontal plane, the bike has a flywheel. This has a moment of inertia I and spins around an axis that becomes to be vertical when the bike is standing on the ground. The mass of the bike and a driver is m , while the centre of mass is located at a distance l from

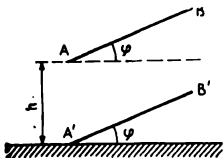


Fig. 3

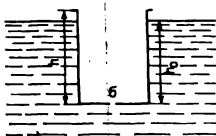


Fig. 4

the wall (the radius of the cylindrical wall is much larger than the dimensions of the bike). What must the coefficient of rotation transmission from the wheels to the flywheel k be for the bike to move without the wheels slipping? The coefficient of friction between the wheels and the wall is f and the wheel radius is r .

1.43. A rod of length $2l$ is inclined at an angle φ to the horizontal. It falls without rotating from a certain height (Fig.3) to a horizontal table and hits its surface initially on the left-hand end and then on the right-hand end. The impact on the rod is elastic and friction can be neglected. When struck on the right-hand end the rod will again be at an angle φ to the horizontal. Find the height h . For what angles φ can this situation occur?

1.44. A method of determining absolutely the energy radiated by quantum pulse generators is to measure the mechanical momentum due to the reflection or absorption of the radiation light by a solid surface. An appropriate instrument is a torsion ballistic pendulum. Calculate the oscillation period of a pendulum that has a sensitivity of $\eta = 10 \text{ mm}/(\text{J}\cdot\text{m})$. The pendulum's moment of inertia can be taken as $I = 2 \times 10^{-6} \text{ kg}\cdot\text{m}^2$, and the distance between the rota-

tional axis and reflecting mirror is $r = 50$ mm. Calculate the diameter D of the quartz suspension thread of length $l = 0.1$ m that will ensure this sensitivity. The shear modulus of quartz is $G = 10^{11}$ N/m².

1.45. Two identical steel bars of length $l = 0.1$ m and cross section 10×10 mm² (density $\rho = 7800$ kg/m³, Young's modulus $E = 2 \times 10^{11}$ N/m²) collide end to end. Estimate the duration of the collision by considering elastic waves. At what velocities will inelastic phenomena arise if the steel's limit of elasticity is $p = 2 \times 10^8$ N/m²?

1.46. Estimate the duration of a collision of a football kicked gently against a wall.

1.47. A rapidly rotating uniform ball (angular velocity ω_0) is placed onto a horizontal board so that its rotational axis is at an angle φ with the vertical. Determine the ball velocity v and angular velocity ω of its rotation established as soon as the ball stops slipping on the board. Rolling friction is to be neglected. The ball's radius is R .

1.48. A billiard ball rolls without slipping across a horizontal table and hits a vertical wall. During the impact both the ball and the wall deform elastically. The coefficient of friction of the ball against the wall is f . Neglecting both gravity during the impact and the rolling friction, determine at which angle to the horizontal the ball will rebound from the wall. Investigate the dependence of the result on the magnitude of the friction coefficient f .

1.49. A cylindrical tank with a base area S is filled with water to a height h . A small hole of area σ appears in the bottom of the tank. Determine how long it takes the water to run out from the tank.

1.50. A cylindrical can of height h is immersed in water to a depth h_0 (Fig. 4). A small hole of area σ appears in the bottom of the can. Determine how quickly the can sinks.

1.51. Determine the maximum pressure that can be produced by water during freezing. The density of ice is $\rho = 0.92$ g/cm³, Young's modulus is $E = 2.8 \times 10^{10}$ N/m², and Poisson's coefficient is $\mu = 0.3$.

1.52. A 1 : 100 scale model of a ship is being tested in a basin. The design speed of the ship is 36 km/h. At what speed must the model be towed to have gravitational waves like those of the full-scale ship?

1.53. Estimate by the dimensional method the phase velocity of waves over the surface of a liquid by neglecting the effect of surface tension and finite depth.

II. Thermodynamics and Molecular Physics

2.1. Calculate the heat capacity $C(V)$ of one mole of an ideal gas undergoing the process shown in Fig. 5. Assume the value of the parameter $\gamma = C_p/C_v$ is known. What is the maximum temperature that will be reached by the gas during this process? Draw the heat capacity plot $C(V)$. Indicate the polytropic processes whose plots in the p - V -plane are tangential to the straight line from Fig. 5 at the points corresponding to $C(V) = 0$ and $C(V) = \infty$.

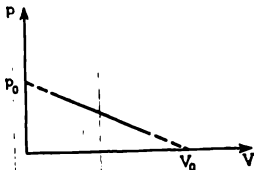


Fig. 5

2.2. An ideal gas is compressed by a piston in a cylinder so that the heat dissipated into the environment is equal to a change in the internal energy of the gas. Determine the work

expended by compressing one mole of the gas to half its volume. What is the heat capacity in this process? The initial temperature of the gas is T_0 .

2.3. Prove that for a substance having an arbitrary equation of state the temperature coefficient of volume expansion β , the temperature coefficient of pressure γ , and the isothermal coefficient of omnidirectional compression α are related thus $\beta = \gamma\alpha$.

2.4. Find the velocity of the adiabatic exhaust of an ideal gas jet from a vessel through a small hole into a vacuum, given the velocity of sound in the gas is s .

2.5. Calculate how many times greater than the final mass M should the launching mass of a single-stage rocket M_0 be in order for the rocket to reach orbital velocity v_1 . Assume the combustion products of rocket engine exhaust into a vacuum via an ideal adiabatic process with adiabatic index $\gamma = 1.2$. The average molar mass of the combustion products is $\mu = 30$ g/mole, and the temperature of combustion is 3000 K. Neglect the effects of gravity and air friction during the acceleration of the rocket.

2.6. Calculate how much greater the efficiency of an internal-combustion engine will be if its compression ratio is increased from 5 to 10. Replace the real cycle of the engine by an ideal closed cycle composed of 2 isochores and 2 adiabats, and assume the working substance is a polyatomic ideal gas.

2.7. The atmosphere of the Earth may be considered to be a giant heat engine whose heat source and heat sinks are the equator and poles respectively, while solar radiation serves as the energy source. Assuming that the total flux of solar energy arriving at the Earth is $Q = 1.7 \times 10^{17}$ W and the efficiency of the "engine" is an order of magnitude smaller than the maximum possible, estimate the average power available for forming winds per square kilometre of the Earth's surface. Discuss the physical reasons that cause $\eta < \eta_{\max}$.

2.8. Prove that for a substance with an arbitrary equation of state given by a single-valued function $T = T(p, V)$, two polytropes cannot be encountered in more than one state.

2.9. During the adiabatic compression of a liquid the relative volume change is $\Delta V/V = 0.1\%$ and the temperature is raised by $\Delta T = 1$ K. Find C_p/C_v from these data, if the temperature coefficient of the liquid's volume expansion is $\beta = 10^{-4}$ K $^{-1}$. By how much does the pressure in a liquid whose temperature coefficient of pressure is $\gamma = 10^6$ N/(m $^2 \times$ K) change?

2.10. A silver wire 1 mm in diameter is adiabatically loaded at room temperature by a force $F = 10$ N. Assuming that the specific heat capacity is $c = 234$ J/(K·kg), the density $\rho = 10$ g/cm 3 , and the temperature coefficient of linear expansion $\alpha = 1.9 \times 10^{-5}$ K $^{-1}$, determine the change in the temperature of the wire.

2.11. A heat-conducting piston of mass m is held inside a vertical thermally insulated cylinder of radius r and subdivides the cylinder into two equal parts. Each part contains ν moles of the same ideal gas at a pressure p and a temperature T . When the piston is released it falls due to gravity. Determine the change in the system's entropy ΔS at the instant of establishing equilibrium. Neglect the heat capacity of the cylinder and piston and take $\pi r^2 p \gg mg$.

2.12. A volume $V_1 = 3$ l contains $\nu_1 = 0.5$ mol of oxygen (O_2) and a volume $V_2 = 2$ l contains $\nu_2 = 0.5$ mol of nitrogen (N_2) at the temperature $T = 300$ K. Find the maximum work that could be performed by mixing these gases in the total volume $V_1 + V_2$ both isothermally and adiabatically. Assume both gases are ideal.

2.13. A closed tube of volume V_0 contains a mixture of two ideal gases, ν mol each. The initial pressure is p . Two pistons are located at the extremities of the tube (Fig. 6) and each

is transparent to one of the gases. When the pistons are moved to the middle the gases are completely separated. Calculate directly the work A performed by the external forces during an isothermal quasi-static displacement of the pistons and compare the ratio A/T (where T is the absolute temperature) with the change in the total entropy of the gases.

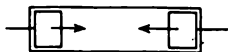


Fig. 6

2.14. An absent-minded physicist leaves in his lab a thin-walled rubber bag filled with helium 20 l in volume. The physicist returns to find that all the helium has diffused outside. Find the change in the helium's entropy. Air contains 10^7 molecules of gas per helium atom. What minimum work must be expended to retrieve from the atmosphere and into the bag the same amount of helium using an isothermal process?

2.15. In a shock wave a gas is heated so rapidly that as the temperature rises from T_0 to a high temperature T_1 only the external translational and rotational degrees of freedom are involved. Gradually the internal, viz. vibrational, degrees of freedom are then excited by a process called "vibrational relaxation". Find the temperature T_2 of a diatomic gas after vibrational relaxation and the change in gas's molar entropy during this relaxation given that $kT_1 \gg \hbar\omega$ and $kT_0 \ll \hbar\omega$, where ω is the frequency of intramolecular vibrations. Assume that the relaxation occurs at a constant pressure and under the conditions of thermal insulation.

2.16. The saturated vapour pressure of water at 290 K is 0.02 atm. Some vapour takes up a volume of 10 l. Find the change in the free energy ΔF and entropy ΔS after isothermally compressing it to 5 l. Assume the vapour is an ideal gas. The specific heat of vapourization at this temperature is $\lambda = 2460$ J/g.

2.17. Depict on the p - V -plane and using thermodynamic identities analyze the Carnot cycle for a heat engine working with a substance for which $(\partial p / \partial T)_V < 0$ throughout its working range. Indicate the "hot" and "cold" isotherms, mark in the signs of the heat flows on both isotherms, etc.

2.18. A mole of a Van der Waals gas, whose constant b is known, is contained in a vertical cylinder under a massive piston of area S . Find the frequency of the small-amplitude oscillations of the piston around its equilibrium position assuming that the compression-rarefaction process is

isothermal with $T = 27^\circ\text{C}$. Set the equilibrium gas volume under the experimental conditions to be equal to the critical volume. Neglect external pressure.

2.19. Find the adiabatic velocity of sound at the critical point of a Van der Waals gas. Assume the constants a and b of the gas and its molar mass μ are known. The heat capacity C_V is given and independent of temperature.

2.20. A fissure $\delta = 10^{-4}$ cm wide and $l = 5$ cm long has been found at the bottom of a container of liquid helium. The thickness of the bottom is $d = 0.5$ mm. Find the maximum velocity of the helium $v_{\max}$ in the fissure and the total consumption of liquid per unit time m if the helium column height is $h = 20$ cm. The helium's density and coefficient of dynamic viscosity are $\rho = 0.15$ g/cm³ and $\eta = 3.2 \times 10^{-6}$ g/(cm·s), respectively. Assume the flow is laminar.

2.21. A wire of radius $R_1 = 1$ mm is pulled at a constant velocity $v_0 = 10$ cm/s along the axis of a tube of radius $R_2 = 1$ cm that is filled with a liquid having a viscosity $\eta = 0.01$ P. Determine the friction force F per unit length of the wire and find the velocity distribution of the liquid across tube's radius, assuming the flow is laminar.

2.22. A long vertical tube of length l and radius R is filled with a liquid whose coefficient of kinematic viscosity is ν . In what time τ will the whole liquid flow out of the tube due to gravity? Surface tension is to be neglected and assume that the establishment of the liquid's velocity is instantaneous and the flow is laminar.

2.23. Water flows out of a wide, open vessel through a horizontal cylindrical capillary. The capillary has a radius $R = 1$ mm and a length $l = 10$ cm and is located near the bottom of the vessel. How much power N is lost in heat release if the water's height in the vessel is $h = 5$ cm? The flow is laminar and the coefficient of dynamic viscosity is $\eta \approx 10^{-2}$ P.

2.24. A rocket with a cylindrical thermally insulated cabin that is h metres long is accelerating along the cylinder's axis. The air mass inside the cabin is m . How will the air temperature and entropy be changed inside the cabin if the engine is switched off? Air is to be considered an ideal gas with molar mass μ . Assume that $\mu ah/RT_0 \ll 1$, where T_0 is the air temperature in the cabin. Consider two cases: (1) the engine is switched off slowly so that the processes occur quasistatically, and (2) the engine is switched off instantaneously.

2.25. One mole of an ideal gas is contained in a thermally insulated cylindrical vessel of radius r . How will the entropy of the gas be changed if the vessel is rotated until it has an angular velocity of $\omega \ll \frac{1}{r} \sqrt{\frac{RT}{\mu}}$, where μ is the gas's molar mass?

2.26. Estimate the radius R of an aluminium particle suspended in a liquid of density 1 g/cm^3 and viscosity 1 P if the viscous fall velocity of the particle is equal to the velocity of its thermal motion at room temperature. State whether such particles would be precipitated from an aluminium paint.

2.27. A motionless thin-walled vessel of volume V is filled with a monatomic ideal gas and maintained at a constant temperature T . The vessel's wall has a hole of area S through which gas molecules escape into a vacuum. The hole's dimensions are small compared with those of the vessel and with the gas's mean free path length. Derive the relationship between time and the heat flow Q that must be supplied to the vessel to maintain it at a constant temperature.

2.28. A monatomic ideal gas fills a motionless thin-walled vessel of volume V provided with thermally insulated walls. The vessel's wall has a hole of area S through which molecules flow out into a vacuum. The hole's dimensions are small compared with those of the vessel and the gas's mean free path length. Derive the relationship between time and the gas's temperature in the vessel. Assume the initial temperature is T_0 and all the gas's parameters are known. Neglect the heat capacity of the vessel's walls.

2.29. Estimate the time needed for water to evaporate from a tube $l = 10 \text{ cm}$ long that is soldered shut at one end. The temperature is $T = 300 \text{ K}$ and initially the tube was half-full of water. The relative air humidity is 50% and the saturated vapour pressure $p_{\text{sat}} \approx 27 \text{ mm Hg}$. The mean free path Λ in the air-vapour system is about 10^{-6} cm . Assume the vapour near the water's surface is saturated and neglect capillary phenomena.

2.30. Knowing that the mean free path length of a singly charged ^{40}Ar ion in a gas is $\Lambda = 10^{-6} \text{ cm}$, find the average ion drift velocity v_{dr} in the gas due to a uniform electric field of 300 V/cm . The gas is at room temperature.

2.31. A vessel is evacuated through a long tube by an ideal vacuum pump (viz. it traps every molecule entering it).

Because of leaks in the vessel's walls the internal pressure does not fall to zero. After prolonged pumping the pressure stabilizes such that the mean free path length $\Lambda \gg d$, where d is the tube's diameter. In what way would the final pressure be changed if the tube's diameter is halved?

2.32. Air is pumped from a large vessel at a pressure of 10^{-4} mm Hg and a temperature of 300 K and pumped through a pipeline of length $l = 2$ m and radius $R = 10$ cm. The pump has a capacity of $V_1 = 1000$ l/s. What would the pump capacity be if the vessel is evacuated at the same speed and the pump is attached directly to the vessel?

2.33. From the spread of radioactive gases after nuclear explosions it is known that turbulence causes the Earth's atmosphere to intermix entirely within about a year. How many times faster does atmospheric turbulence mix the atmosphere than molecular diffusion?

2.34. Gaseous helium leaks into a vacuum through a hole of area $S = 0.1$ mm² from a thin-walled vessel in which the pressure and temperature are maintained at $p = 10^{-6}$ atm and $T = 273$ K, respectively. The total number of atoms N that leak through the hole is measured for a time interval of $t = 10^{-3}$ s. Find the relative root-mean-square fluctuation of this number. Also, find the probability that no atoms leak out during an experimental run.

2.35. Calculate the relative root-mean-square fluctuation of the potential energy of intramolecular vibrations for: (a) a diatomic molecule, (b) an N -atomic molecule, (c) a mole of an ideal gas composed of diatomic molecules. The vibrations are assumed to be harmonic.

2.36. Two identical communicating vessels are filled with a gas under normal conditions. What should the volume V of each vessel be to ensure that the probability of a state in which the pressure in the vessels is isothermally changed by $10^{-7}\%$ is e^{100} times smaller than the probability of the initial state?

2.37. To measure temperature fluctuations in a helium-filled vessel of volume V , a small control volume $v \ll V$ is isolated. At what value of v will the root-mean-square temperature fluctuation amount to $10^{-4}\%$? The helium is under normal conditions.

2.38. A vessel of volume 1 l at room temperature contains the same number N of the atoms of two different gases. Estimate the value of N for which the probability that the gases will separate at least once during the lifetime

of the observable Universe ($\approx 10^{10}$ years) is comparable to unity.

2.39. A vessel contains a Van der Waals gas such that the average molar volume is equal to the critical volume, and the temperature T exceeds the critical temperature T_{cr} . Find the root-mean-square isothermal volume fluctuation $(\Delta V)^2$ of a small element of the gas having the equilibrium volume V .

2.40. The thermoregulator of a car engine is a cylindrical vessel with corrugated walls (bellows) and filled with ethyl

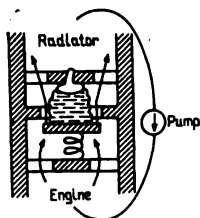


Fig. 7

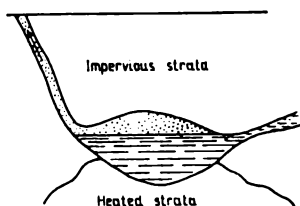


Fig. 8

alcohol and its vapour (Fig. 7). At a low temperature the pressure in the bellows is low so it compresses and shuts a valve, thus attenuating the circulation rate of the water in the engine's cooling system. As the water temperature rises, the bellows expand and open the valve so that the water starts circulating more rapidly thus cooling the engine, etc. What must the pressing force F of a valve spring be to open the valve at $T = 363$ K? The bellows' diameter is $D = 2$ cm, the boiling point of alcohol at 1 atm is $T_0 = 351$ K, alcohol's specific heat of vaporization is $\lambda = 850$ J/g, and its chemical formula C_2H_5OH .

2.41. Geysers can be considered to originate from large underground reservoirs filled with ground water and boiled by subterranean heat (Fig. 8). The reservoirs are connected to the Earth's surface via a narrow channel that is filled with water during a "quiet" period. Assuming that an "active" period starts when the water in the underground reservoir boils and that during the eruption the channel is filled

solely with vapour which does reach the outside, estimate what fraction of water is lost from a geyser reservoir during one discharge. The channel's depth is $h = 90$ m, and the latent heat of vaporization of water is $\Lambda = 41$ kJ/mol.

2.42. At normal pressure helium-3 (^3He) remains a liquid down to absolute zero. The minimum pressure under which it is solidified is $p_{\min} = 28.9$ atm. The entropy of liquid helium-3 per mole over the temperature range of interest is described thus $S_l = R(T/\Theta)$, where R is the gas constant and $\Theta = 0.22$ K (cf. Problem 7.36). The entropy of solid helium-3 over the temperature range in question is independent of temperature and is $S_s = R \ln 2$ per mole (cf. Problem 7.9). The difference between the volumes of liquid and solid helium is $\Delta V = V_l - V_s \approx 1.25$ cm³/mol. Find the temperature on the melting curve corresponding to $p_{\min}$, the pressure at which helium-3 solidifies at $T = 0$ and the temperature dependence of heat of melting.

2.43. According to one model, the Earth's centre (called the core) consists of iron. The outer layers of the core are molten while the centre itself, $R \approx 1200$ km in radius, is solid. The core cools at a rate of $u = 10^{-7}$ K/year, temperature differences within the core being neglected. How far will the radius of the solid part expand in 10^9 years? Assume that the specific heat of iron molting under the conditions at the core's surface is $q \approx 125$ J/g, the temperature at this surface being $T \approx 3700$ K, and the change in iron's density during solidification being $\Delta\rho = 0.3$ g/cm³. Because $\Delta\rho \ll \rho$ it is possible to disregard the change in the pressure distribution $p(r)$ as the core solidifies.

2.44. A closed vessel contains water in an equilibrium with its vapour at 100°C . The vapour/water mass ratio β is 0.1, the specific heat capacity of water c_0 is 4.2 J/(K·g). Find the specific heat capacity c of the system, assuming that the vapour is an ideal gas. The specific heat of vaporization is $\lambda = 2260$ J/g.

2.45. Find the specific heat capacity at constant volume c_v of a rarefied fog (viz. a saturated vapour with water droplets whose total mass is much less than that of the vapour) at 100°C . The molar heat of vaporization of water at 100°C and constant pressure is $\Lambda = 41$ kJ/mol.

2.46. Due to a fall in external pressure a soap-bubble doubles its radius. Assuming the process is isothermal, find the change in soap-bubble's entropy. The initial pressure is p_0 .

the surface tension coefficient of the soap film is σ , and the specific heat of forming a soap film surface is q . Assume that $2\sigma/r \ll p_0$.

III. Electricity and Magnetism

3.1. Electrons are omitted as a result of thermal emission from a negatively charged plate into a parallel-plate capacitor whose field strength $E = 10^3$ V/m. By taking into account the action of the electric image field, estimate how far from the emitter plate the electron velocity will be the slowest.

3.2. An ideal gas whose molecules have a polarizability of $\alpha = 4 \times 10^{-30}$ m³ is contained in a large vessel at 300 K.

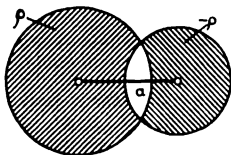


Fig. 9

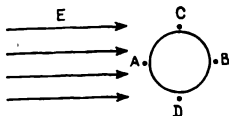


Fig. 10

A charged parallel-plate capacitor with a field strength $E = 3 \times 10^4$ V/m is placed inside the vessel. Find the relative difference between the concentrations of the molecules inside and outside the capacitor.

3.3. What is the electric field in a cavity formed by the intersection of two balls (Fig. 9). The balls carry charges with densities ρ and $-\rho$ uniformly distributed throughout their volumes. The distance between the centres of the balls is a .

3.4. How should charges $\sigma(\theta)$ be distributed on the surface of a sphere of radius R for the field inside the sphere to be uniform and equal to E ? What will be the field outside the sphere?

3.5. How much does the permittivity ϵ of an "ideal gas" composed of a large number of conducting balls of radius R differ from unity? The concentration n of the balls is small enough for $nR^3 \ll 1$.

3.6. An uncharged conducting spherical speck of dust has entered an external uniform electric field. How

will the field strength at points A , B , C , and D (Fig. 10) be changed?

3.7. Under normal conditions, gaseous argon has a permittivity $\epsilon - 1 \simeq 6 \times 10^{-4}$. Making use of this fact, calculate the displacement of the "centro of mass" of the electron shell of an argon atom from the nucleus in a static electric field with strength $E = 300$ V/cm. Argon's atomic number $Z = 18$. (In the absence of an external field the electron distribution around the nucleus is considered spherically symmetric).

3.8. In the Thomson model of the atom the positive charge e is assumed to be distributed inside a sphere of radius 10^{-8} cm. How should the positive charge density depend on the distance from the sphere's centre if an electron (a negatively charged point particle carrying a charge $-e$) inside the sphere oscillates harmonically? Find the frequency of the electron oscillations assuming that the charges do not interact.

3.9. Is there any possibility in principle for determining a rocket's acceleration by purely electrical methods (ways)? Do numerical estimates for an acceleration of 10 g and a conductor of length $l = 10$ m. Neglect external electric and magnetic fields.

3.10. A point charge q is placed inside an uncharged spherical conducting shell, a distance a from its centre. The radii of the internal and external surfaces are r and R , respectively. Find the surface density σ of the induced electric charge on the shell's external surface and the shell potential φ . Determine the surface densities σ_1 and σ_2 of the charges induced on the shell's internal surface at points lying on the same diameter as the charge q .

3.11. A dielectric plate of thickness h having a "frozen" polarization $\mathbf{P} = \text{const}$ is placed inside a parallel-plate capacitor, the plates of the capacitor being connected together. The polarization vector is perpendicular both to the side faces of the dielectric plate and the capacitor plates. Determine the electric field strength and induction inside the plate. The distance between the capacitor plates is d .

3.12. Given a thin dielectric cylinder of length $2l$ and radius r with a "frozen" polarization $\mathbf{P} = \text{const}$ (Fig. 11), find the field at the point A . By how many times is this field stronger than that at the point B ?

3.13. Due to inhomogeneous deformation, a piezodielectric

plate $2d$ thick is polarized so that the polarization in the middle of the plate is P_0 and elsewhere is $P = P_0(1 - x^2/d^2)$, where x is the distance from the plate's midplane. The polarization vector points along the x -axis (Fig. 12). Determine the electric field strength both inside and outside the plate as well as the potential difference between its lateral surfaces. Edge effects should be neglected.



Fig. 11

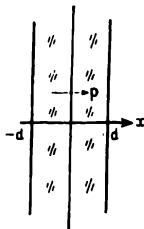


Fig. 12

3.14. A pulsed discharge from a capacitor through a rarefied gas (hydrogen) causes it to heat to a temperature T . Estimate the value of T assuming that the entire discharge energy was spent on heating the gas. Indicate the possible causes why the temperature may be lower. Make numerical estimates for $U = 3 \times 10^4$ V and $C = 1.8 \times 10^{-5}$ F assuming that the hydrogen is at room temperature $T_0 = 293$ K and occupies a volume $V = 10^{-2}$ m³ at the pressure $p = 1.29$ N/m² before the discharge. Estimate what the temperature of the capacitor plates is if the entire discharge energy was dissipated. The specific heat capacity of copper is $c = 4.2 \times 10^3$ J/(K·kg), and the mass of the plates m is 0.1 kg.

3.15. Find the attractive force between a point electric dipole, which has a dipole moment of $p = 4 \times 10^{-10}$ C·m, and an infinite metallic plate $L = 1$ cm from the dipole. The dipole axis is perpendicular to the plate. Determine the work needed to move the dipole 2 cm from the plate surface.

3.16. Derive an expression for the energy of a small dipole in an external electric field E . Consider the case of a rigid dipole with a dipole moment p and that of an elastic dipole with a polarizability α ($p = \alpha \epsilon_0 E$).

3.17. Estimate the potential difference between the head and tail of the steel core of an armour-piercing projectile

when the projectile penetrates an obstacle. Assume that the core is 25 cm long and that its lost speed on piercing armour plate $d = 5$ cm thick. The projectile's speed at the instant it hit the armour plate was 1000 m/s.

3.18. A dielectric with a permittivity ϵ fills a half-space. A point charge q in a vacuum is at a distance L from the plane boundary of the dielectric. Find the charge distribution σ over the dielectric's surface and the force F acting on the charge.

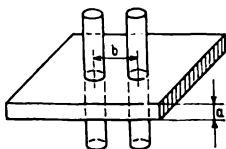


Fig. 13

3.19. A parallel-plate capacitor C is connected to the battery of e.m.f. $\mathcal{E}$ in parallel with a resistor. The capacitor plates are rapidly brought together so that the separation distance is halved. Assuming that during the movement of the plates

the charge on the capacitor remained practically unchanged, find the Joule heat dissipated by the resistor at the instant the recharge ceases. Estimate the order of magnitude of the resistance for which the hypothesis could practically be fulfilled if the time needed to bring the plates together is $\Delta t \approx 10^{-2}$ s and $C \approx 10^{-10}$ F.

3.20. Estimate the force acting on an atom $200 \text{ \AA}$ from the tip surface of a metal needle having an edge radius $R = 100 \text{ \AA}$. The potential at the needle is $\varphi = 10$ kV. The polarizability of the atom α is of the same order as its volume.

3.21. Two cylindrical conductors of radius r_0 are soldered to a large metallic sheet of thickness a , a distance b from each other (see Fig. 13). Find the resistance between the conductors, if $a \ll r_0 \ll b$. The conductivity of the conductors λ_1 is much larger than that of the sheet λ .

3.22. When the voltage across a krypton reaches 500 V, the anode temperature is 800°C . What will the anode temperature be when the anode voltage is 1000 V when: (1) the anode current is saturated at 500 V and (2) there is no saturation even at 1000 V?

Note. When there is no saturation the anode current $\mathcal{I}$ is proportional to $U^{3/2}$.

3.23. A device consists of two concentric conducting spheres, the internal being made of a radioactive material that emits fast electrons. The velocity of the electrons in the gap between the spheres and hence their ionizing action are con-

stant. After traversing the air gap, the electrons are absorbed by the external sphere. When the battery is turned off an equilibrium sets in between the charge flux carried by the fast electrons and the conduction current in the ionized air. Find the electric field in the gap between the spheres if the element e.m.f. is $\mathcal{E}$ and the radii of the spheres are R_1 and R_2 , respectively.

3.24. Determine the conductivity of the isolation in a spherical capacitor filled with a lossy dielectric.

3.25. The space between two concentric spheres is filled with a dielectric whose conductivity depends only on the distance from the point to the spheres. What must the conductivity relation be for the volume density of the Joule losses when a current is flowing to be the same at every point in the dielectric?

3.26. In a Van de Graaf generator, shown in Fig. 14, the charge is transported by a dielectric belt to charge a high-voltage spherical electrode. The surface-bound charges are transferred to the belt from the source located near the lower pulley. Estimate the maximum potential and maximum current that can be produced by such generator if the radius of the high-voltage electrode is $R = 1.5$ m, the belt moves at $v = 20$ m/s, and the belt's width is $l = 100$ cm. The belt and high-voltage electrode are placed in a gas in which the break-down occurs at an electric field strength of $E_b = 30$ kV/cm.

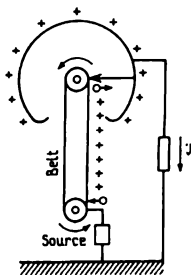


Fig. 14

3.27. A spherical capacitor with sphere radii R_1 and R_2 is filled with a weakly conducting medium. Its capacitance is C , while the potential difference across the capacitor after it was disconnected from a battery halves after a time t . Determine the permittivity ϵ of the medium and its resistivity ρ .

3.28. Modern data indicate that the experimentally permissible allowable difference between the absolute values of the charges on an electron q_e and a proton q_p is such that $|(q_p - q_e)/q_p| < 10^{-21}$.

Could this difference account for the existence of the Earth's observed geomagnetic moment? The geomagnetic field is $B_E \approx 3 \times 10^{-5}$ T and its density $\rho \approx 5 \times 10^6$ kg/m³. For the atoms constituting the Earth, the ratio of relative atomic mass A to atomic number Z is about 2.

3.29. A long thin-walled duralumin tube is charged and rapidly rotated about a longitudinal axis. What form will the resultant magnetic field have? The upper limit of the speed at which the tube can rotate is governed by duralumin's mechanical strength ($f_{\max} = 5.9 \times 10^8$ N/m²).

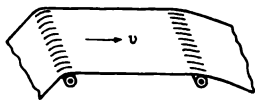


Fig. 15

What is maximum possible ratio between the volume densities of the energies of the magnetic field inside the tube and the electric field on the tube's external surface? Duralumin's density ρ is 2.7×10^3 kg/m³.

3.30. To manufacture a polyethylene film, a wide band is pulled over rollers at $v = 15$ m/s (Fig. 15). During the process the film's surface acquires a uniformly distributed charge σ mainly because of friction. Estimate the maximum values of σ and the magnetic field's flux density B near the film's surface taking into account that an electric field strength of $E = 20$ kV/cm can cause a discharge in air.

3.31. How many fixed singly charged positive ions must exist within a parallel homogeneous electron beam of a circular cross section so that the radius of the beam remains constant if the beam moves at a velocity v ? The concentration of the electrons in the beam is n_0 and collisions between electrons and ions should be neglected.

3.32. A current $I = 10^5$ A flows in the surface layer of a long plasma cylinder of radius $R = 0.05$ m. The pressure inside the plasma is $p = 10^5$ N/m². Determine the force acting per unit area of the lateral surface of the plasma cylinder. Is the plasma compressing or expanding here? Find the current required to balance the radial forces.

3.33. A plasma filament is confined by a magnetic field parallel to the filament's axis because the field does not penetrate the plasma. Estimate the magnetic field's flux density needed to confine a plasma with a particle concentration of $n = 10^{16}$ cm⁻³ and temperature $T = 10^8$ K.

3.34. A steady axially symmetric beam of electrons is accelerated by a potential difference U and formed along a long

evacuated cylindrical tube of radius R . Find the electron concentration distribution in terms of the radius r in a cross section of the beam if the magnetic field B measured as a function of r in the cross section can be described by the expression $B = B_0(r/R)^q$ for $r < R$ (B_0 and $q > 0$ being constants). Determine the electric field $E(r)$ assuming that the beam parameters do not change along the beam's axis.

3.35. A square frame with current $\mathcal{I}_2 = 1$ A is near a long straight conductor carrying a current $\mathcal{I}_1 = 10$ A (Fig. 16). The frame and the conductor lie in the same plane, a frame

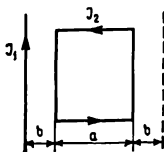


Fig. 16

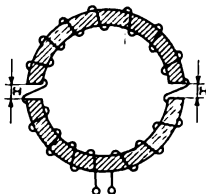


Fig. 17

side is $a = 6.8$ cm, and the distance between them is $b = 4$ cm. What work should be done to move the straight conductor into the position indicated by the dashed line?

3.36. N turns of a wire are wound onto an iron toroidal core with a magnetic permeability μ . The toroid's radius is R , the radius of the core cross section being such that $r \ll R$. The toroid is cut into halves separated by an air gap x (Fig. 17). Determine the force of attraction between the two halves when a current $\mathcal{I}$ is flowing through the coil. Consider the case of $x = 0$.

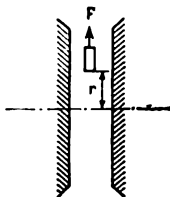


Fig. 18

3.37. To determine the magnetic susceptibility χ of a diamagnetic material, a pair of scales is used to measure the force pushing a small sample out of the gap between the poles of an electromagnet (Fig. 18). If the magnetic field in the gap is described by the relation $B = B_0 e^{-\alpha r^2}$, where

r is the distance in the radial direction from the axis of symmetry (in cm), B_0 is the field strength along the axis of 10^3 G, and α is a constant equal to 10^{-2} cm $^{-2}$. How far from the axis must the diamagnetic sample be for the expulsive force to be at a maximum and what is the maximum force for a sample in the form of a small thin disc of volume $V = 0.1$ cm 3 ? The magnetic susceptibility of the material is $\chi = -1.4 \times 10^5$ (bismuth). The disc is placed perpendicular to the magnetic field.

3.38. Superstrong magnetic fields can be produced by the explosive compression of a conductive tube in which there is an initial magnetic field B_0 . Determine the final magnetic field B and the tube's radius R , if during the compression the pressure of the magnetic field is balanced by a pressure

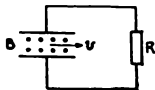


Fig. 19

of 10^6 atm caused by the explosion. Consider a case when $B_0 = 5$ T and initially the tube's radius $R_0 = 0.05$ m. Mechanical and electrical resistances should be neglected.

3.39. A simple magnetic hydrodynamic generator consists of a parallel-plate capacitor with a plate area S and a separation d which is put into a flowing conducting liquid with a specific conductivity of λ , the liquid moving parallel to the plates with constant velocity v . The capacitor is placed in a magnetic field with a flux density B and directed perpendicular to the velocity of the liquid (Fig. 19). How much power will be dissipated in an external circuit containing a resistance R ?

3.40. A compass is placed above a wire carrying a direct current at a distance $R = 0.1$ m from the wire's axis. Find the current needed to lift the compass needle over its peg. The residual induction in the needle steel is equal to the saturation induction $B_0 = 2$ T. The steel's density is $\rho = 7.8 \times 10^3$ kg/m 3 .

3.41. Determine the period of the small oscillations in a thin magnetic bar of length $l = 0.1$ m freely suspended from the middle in a geomagnetic field of $B_E = 2 \times 10^{-5}$ T. The steel density is $\rho = 7.8 \times 10^3$ kg/m 3 , and the residual induction $B_0 = 1$ T.

3.42. After a small steel ball is magnetized to saturation in an external field it is switched off. Estimate the residual

magnetization of the ball if B and H are related by the equation $B = B_0 (1 + H/H_c)$ and for the steel $H_c = 4 \times 10^5$ A/m and $B_0 = 1$ T. The demagnetization factor of the ball is $\beta = 1/3$.

3.43. The magnetization I as a function of the field strength H for a magnetic material is shown in Fig. 20. This material is used to make the core of a thin toroidal coil with N

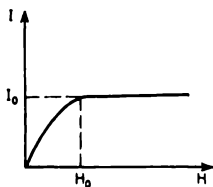


Fig. 20

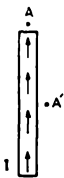


Fig. 21

turns. The coil length (perimeter) is L . There is a narrow transverse gap l in the core. Determine the current J_0 at which the core will be saturated. How will the magnetic flux density B be changed in the core's gap for $J > J_0$? I_0 and H_0 are assumed to be given.

3.44. In order to impart an angular velocity to an Earth satellite the geomagnetic field can be used. Find the angular velocity gained by the satellite if a storage battery with a capacity of $Q = 5$ A·h is discharged suddenly through a coil of $N = 20$ turns wound around the satellite's surface along the circumference of a large circle. The satellite has a mass of $m = 10^3$ kg and is a thin-walled sphere. The geomagnetic field is parallel to the winding plane and its flux density is $B = 0.5$ G.

3.45. What current should be passed through a long thin single-layer solenoid with winding density of n turns per unit length to have a flux density B throughout the solenoid so that it is equal to that of a permanent magnet of the same dimensions? The magnetization I is constant and directed along the axis.

3.46. A thin long permanent magnet of length $2l$ and radius r has a constant magnetization I (Fig. 21). Find the flux density B at point A . How much is it more than the flux density at A' ?

3.47. Determine the frequency of the transverse oscillations of the protons trapped in a relativistic electron beam with cross-sectional area $\pi R^2 = 3.14 \times 10^{-6} \text{ m}^2$ and current $J = 10^3 \text{ A}$.

3.48. Two slits S_1 and S_2 , each $d = 0.1 \text{ cm}$ wide, have been placed in an evacuated vessel and separate out a "sheet"

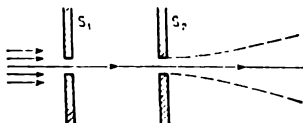


Fig. 22

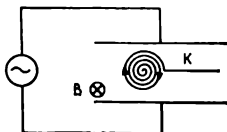


Fig. 23

electron beam with energy $W = 400 \text{ eV}$ (Fig. 22). How far x from slit S_2 will the electron beam's width have doubled due to the Coulomb repulsion between the electrons if the electron current per unit length of the slit (downstream of the slit S_2) is $J = 10^{-4} \text{ A/cm}$?

When calculating, assume the slits are infinitely long.

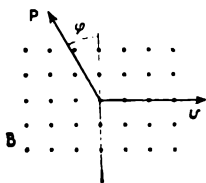


Fig. 24

3.49. In an omegatron, an ion of the residual gas untwists along a spiral in a crossed a.c. electric (amplitude $E = 1 \text{ V/cm}$) and d.c. magnetic ($B = 3 \times 10^3 \text{ G}$) fields (Fig. 23). Find the frequency at which the N_2^+ ions will reach the collector K . At this frequency, the radius of the spiral

increases until the ion reaches the collector with a radius $R = 1 \text{ cm}$. If we slightly change the frequency, the ion will untwist for some time and then begin to twist toward the source. Estimate how the current frequency must be changed to cut the current off the collector.

3.50. An electric dipole moves in a uniform magnetic field B at a velocity v perpendicular to B . The dipole moment p forms a small $\angle \varphi$ with the direction $[v \cdot B]$ (Fig. 24). Find the angular frequency ω_0 of the small amplitude oscillations of the dipole assuming its moment of inertia I_0 , velocity v , and dipole moment p , and the flux density B are known.

3.51. In an electron accelerator, a betatron, the accelerating voltage is the e.m.f. induced by changes in the magnetic flux that pierces the electrons' orbit. The electrons move along an orbit of approximately constant radius. Assuming the radius of the orbit is constant, determine the relation between the average magnetic field B_{av} piercing the orbit and the magnetic field B in the orbit (at a given instant of time). The magnetic field runs parallel to the axis of symmetry of the betatron's vacuum chamber.

3.52. In a cylindrical proportional counter a beam of particles produces volumetric ionization. Estimate the time during which the ions collect in a counter filled with argon at normal pressure. The cathode radius is $R = 1$ cm, the anode radius $r = 2 \times 10^{-3}$ cm, the potential difference between the anode and the cathode $U = 2500$ V, and the mobility of positive argon ions is $\mu^+ = 1.4 \text{ cm}^2/(\text{V} \cdot \text{s})$.

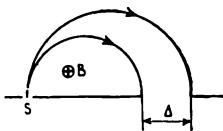


Fig. 25

3.53. A proton beam with an energy $W = 4$ MeV is extracted from an accelerator and travels 4 m through a vacuum before hitting a target. The beam expands due to Coulomb interaction between the protons. Estimate the maximum current density j of the beam if the beam radius is allowed to expand by 10% of its initial value. The proton distribution in the beam is axially symmetric and their initial transverse velocities may be neglected.

3.54. In a device for separating the isotopes ^{235}U and ^{238}U (Fig. 25) a beam of singly charged accelerated uranium ions with an energy of $W = 5$ keV moves from a source through a slit S into a homogeneous magnetic field perpendicular to the plane of the figure. Ions with different masses move along different circumferences and after transversing a semicircle they hit collectors. The design of the collectors should be such that the ^{235}U and ^{238}U beams at the exit are further than $\Delta = 5$ mm apart. What magnetic field B will satisfy this condition? Also, find the time t needed to completely separate $M = 1$ kg of natural uranium if the ion current produced by the source is $I = 5$ mA.

3.55. In a direct action (electrostatic) accelerator a proton moves inside a vacuum tube in a practically homogeneous electric field. Extraneous magnetic fields bend the trajectory

of the proton so that it may hit the wall before arriving at the end. Estimate the permissible level of an external homogeneous magnetic field for such an accelerator if the tube length is $l = 2$ m and the protons are accelerated to an energy $W = 4$ MeV. The permissible deviation of the protons from the axis at the end of the tube is $b = 1$ cm. The initial proton velocity should be neglected.

3.56. A proposed technique to produce the high temperatures needed to realize thermonuclear reactions involves what is called "magnetic thermal insulation", the fast particles

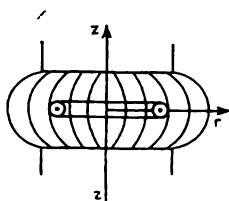


Fig. 26

being prevented from escaping from the high temperature zone by a magnetic field. Estimate the current in a gas discharge column of radius $R = 3$ cm such that electrons with an average speed of random motion corresponding to $T = 10^6$ K cannot move further than 3×10^{-3} cm from the column's surface.

3.57. A particle of mass m and charge e moves along an equilibrium circular orbit of radius r_0 in the horizontal plane of a magnetic gap (Fig. 26) in which the magnetic field falls off along the radius such that $B_z(r) = A/r^n$ ($0 < n < 1$). The orbital centre coincides with the zz symmetry axis. Determine the frequency of the vertical oscillations of the particle for small deviations from the horizontal plane.

3.58. Determine the frequency of radial oscillations of the particle in Problem 3.57 for small deviations from the equilibrium orbit.

3.59. A superconducting ball of radius R is placed in a homogeneous magnetic field with a flux density B_0 . Find both the magnetic field outside the ball and the current surface density i .

Hint. The magnetic field strength and flux density vanish inside a superconductor.

3.60. By how much does the magnetic permeability μ of an "ideal gas" composed of a large number of superconducting balls of radius R differ from unity. The ball concentration n is small enough for $R^3 n \ll 1$.

3.61. The currents flowing over the surface of a superconducting body always result in a zero magnetic field inside

the superconductor. However, if the magnetic field around the body is too high the superconductivity breaks down and the metal returns to the normal state. The critical field B_{cr} for lead at 2 K is 750 G. Estimate how large a ball can be suspended on a "magnetic cushion" at 2 K. The density of lead is $\rho = 11.3 \text{ g/cm}^3$.

3.62. A superconducting ball flies towards a solenoid along its axis. The flux density of the solenoid's magnetic field is $B_0 = 0.1 \text{ T}$. What must be the initial velocity v_0 of the ball for it to penetrate the solenoid? The ball's radius is $R = 0.02 \text{ m}$ and its mass is $m = 10^{-3} \text{ kg}$.

3.63. A superconducting solenoid is deformed so that its magnetic field is adiabatically compressed. What equation (analogous to the adiabatic equation in the kinetic theory of gases) can describe the behaviour of the magnetic pressure in terms of the change in the solenoid's cross-sectional area?

3.64. A current $\mathcal{I}_0$ is induced in a closed superconducting electromagnet's winding. The steel magnetic circuit (length

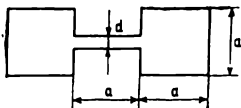


Fig. 27

L) has a magnetic permeability μ and a small gap of width l such that the magnetic field leakage in the gap can be neglected. How will the current through the winding be changed if the gap is halved by deforming the core?

3.65. A thin straight conductor with a direct current flowing through it is positioned above a superconducting plane. Assuming that the linear density of the conductor ρ is $2 \times 10^{-3} \text{ kg/m}$, find how high above the plane the conductor carrying a current of $\mathcal{I} = 20 \text{ A}$ will hover?

3.66. Estimate the resonance frequency of a toroidal klystron cavity whose cross section is illustrated in Fig. 27. The centre of the cavity is a capacitor through which a current is flowing, while the toroidal cavity is filled with the magnetic field produced by the current. The cavity's dimensions are $a = 10 \text{ cm}$ and $d = 1 \text{ cm}$.

3.67. An inductive sensor is a radio-frequency device to detect small variations in inductance. A commonly used sensor consists of an electrical oscillatory circuit with a varying inductance (Fig. 28). Estimate the minimum measurable relative variation in inductance $\Delta L/L$, if the circuit is tuned to resonance. The power supply voltage $U = 100 \text{ V}$,

the minimum measurable variation in voltage across the resistor is $\Delta V = 10 \mu V$, and the circuit quality factor (Q -factor) is $Q = 100$.

3.68. A capacitive sensor is a very sensitive device used to detect small mechanical displacements. The common capacitive sensor consists of an electrical oscillating circuit with

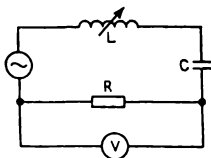


Fig. 28

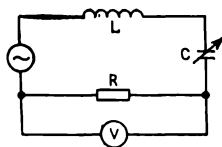


Fig. 29

an air capacitor (Fig. 29) one of whose plates is at rest. Estimate the minimum measurable displacement of the capacitor plate Δd if the circuit is tuned to resonance. The power supply voltage is $U = 100 V$, the minimum measurable

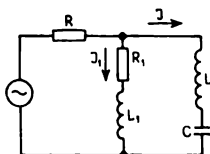


Fig. 30

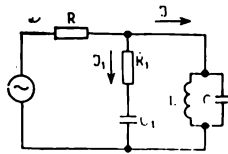


Fig. 31

variation in voltage across the resistor is $\Delta V = 10 \mu V$, the circuit's Q -factor is $Q = 10^3$, and the interplate width is $d = 1 \text{ mm}$.

3.69. There is an a.c. e.m.f. described by the relationship $\mathcal{E}(t) = \mathcal{E}_0 \cos^2 \Omega t$ in the circuit shown in Fig. 30. Determine the currents I and I_1 if the circuit's parameters are known to satisfy the relation $\Omega^2 = 1/4 LC$.

3.70. The a.c. e.m.f. in the circuit shown in Fig. 31 is described by relationship $\mathcal{E}(t) = \mathcal{E}_0 \cos^2 \Omega t$. Determine the currents I and I_1 if the circuit's parameters are known to satisfy the relation $\Omega^2 = 1/4 LC$.

3.71. Given the RC parameters of the circuit shown in Fig. 32, what will be the frequency Ω of the output voltage V_{out} when it is in phase with the input voltage V_{in} ? What will be the ratio between the amplitudes of V_{out} and V_{in} ?
 3.72. At a certain instant of time switch K in the circuit in Fig. 33 is on and a capacitor C with an initial charge q_0 starts to discharge through inductance L . When the discharge current reaches its maximum value, K is again switched off.

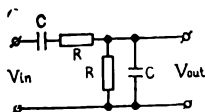


Fig. 32

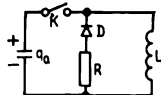


Fig. 33

How much charge will run through the resistor R ? The resistance of a diode D in the forward direction is much lower than that of R while in the backward direction it is infinitely high.

3.73. Near the coil of an oscillatory circuit with L_1CR parameters there is located a second coil with inductance L_2 . The mutual inductance between the coils is M . What will be the resonance frequency of the circuit, if the terminals of the second coil are shorted? The inductive resistance of the second coil at the circuit's oscillation frequency is assumed to be significantly larger than its active resistance.

IV. Optics

4.1. A spectrograph has a collimator objective of diameter D with a focal length F_1 and a camera objective of the same diameter with a focal length F_2 . A light source B is projected sharply onto the exit slit of the spectrograph using a condenser in the first experiment magnifying the source (the distance between the condenser and slit is b_1) and in the second experiment reducing the source (the distance between the condenser and slit is b_2). What must the condenser's diameter be for the illumination intensity on the phototube to be the same in both cases? What is the illumination intensity if the reflection and absorption losses are neglected?

4.2. A radio-frequency emission from a cosmic source with an angular size ψ is being received by a centre-fed dipole aerial. The dipole is situated on a smooth seashore at a height

h above sea-level. Treating water surface as if it were a plane mirror, determine how the signal's intensity will be changed in terms of the elevation α of the source above the horizontal. For which values of the source's angular size will

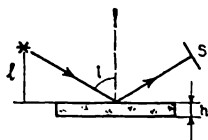


Fig. 34

the signal intensity be independent of α ? To simplify your working, carry out the calculations for small values of α and ψ . The wavelength λ is given.

4.3. A source of light S is placed a distance $l = 1$ m from a thin mica plate of thickness $h = 0.1$ mm and with refractive

index $n = 1.4$ (Fig. 34). At the same distance from the plate, a small screen S is placed at right angles to the reflected rays to observe the interference fringes. The angle of incidence is $i = 60^\circ$. (1) Determine the order of the fringe

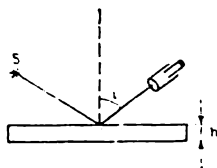


Fig. 35

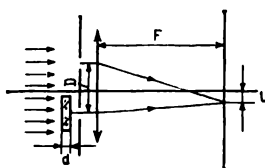


Fig. 36

located at the screen's centre. (2) Find the width of the interference fringes. (3) Estimate the admissible size of the source. (4) Estimate the admissible nonmonochromaticity $\Delta\lambda$. The experiment involves light with a wavelength $\lambda = 5600 \text{ \AA}$.

4.4. Using a telescope set to infinity, interference fringes are observed in a thin glass sheet of thickness $h = 0.2$ mm and with refractive index $n = 1.41$, the angle of observation i being from 0 to 90° (Fig. 35). (1) Find the maximum and minimum orders of the fringes observed. (2) Estimate the admissible source nonmonochromaticity for which all the interference fringes will be fairly distinct. (3) What is the admissible size of the light source in this interference experiment? Green light ($\lambda = 5600 \text{ \AA}$) is used.

4.5. In a Rayleigh interferometer (Fig. 36) a plane wave is

diffracted by two slits. A diffraction pattern is observed in the focal plane of a lens whose focal length F is 100 cm. One of the slits is closed by a plate made from a dispersive substance of thickness $d = 0.01$ mm and for which $n(\lambda) = A - B\lambda$, where A and B are constants. Here the white (achromatic) fringe is shifted 4 mm. Determine the constant A , if the slits are 1 cm apart.

4.6. How far are the interference fringes produced by a Lummer-Gehrcke plate shifted when the temperature is changed by 1°C ? The answer must be given in relative units

(with the distance between adjacent fringes taken to be unity). The plate thickness is $d = 2$ cm, the refractive index $n = 1.5$, and the temperature coefficient of linear expansion

of glass $\alpha = 8.5 \times 10^{-6} \text{ K}^{-1}$, the wavelength $\lambda = 5000 \text{ \AA}$. The temperature dependence of n must be neglected.

4.7. A Fabry-Perot interferometer (Fig. 37) consists of two plane mirrors with reflection coefficient (in intensity) $\rho = 99\%$ separated by 10 cm. A plane monochromatic wave is incident on this interferometer used as an optical cavity. Estimate the width of resonance curve (in MHz) and determine the frequency interval between two adjacent resonances.

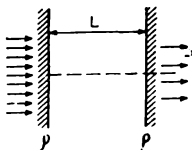
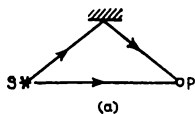
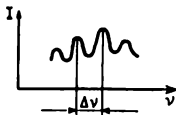


Fig. 37



(a)



(b)

Fig. 38

4.8. Two beams of white light produced by a single point source are brought together onto the entrance slit of an optical spectral device. The path-length difference is $\Delta = 300$ m. Estimate the resolving power of a spectral device capable of detecting the interference of these beams.

4.9. Two beams of white light from one source arrive at a point of observation P (Fig. 38a) with a path-length difference Δ . Using a high resolution spectroscope the energy distribution in the oscillation spectrum at P is investigated by superimposing the beams. It has been proved that there are

alternating maxima and minima of intensity $I(\nu)$, whereas the distance in frequency between adjacent maxima $\Delta\nu$ is 10 MHz (Fig. 38b). Determine the path-length difference Δ .

4.10. An electron moves in a vacuum with a velocity v near the surface of a diffraction grating with period d . The electron's velocity is in parallel to the grating's surface and perpendicular to the grating's rulings. What

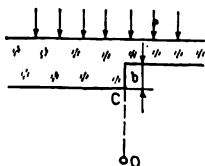


Fig. 39

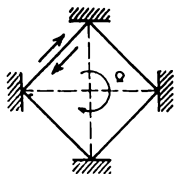


Fig. 40

wavelength may be emitted at 0 to the normal to the grating due to an interaction between the electron and the grating (the Smith-Purcell effect)?

4.11. A parabolic mirror 1 m in diameter is used as an aerial for 3-cm waves. Find the minimum distance at which a receiver must be placed to obtain radiation pattern.

4.12. A plane wave passes through a glass plate with refractive index $n = 3/2$ by falling normally onto its surface. There is a step change in the plate's thickness, the change being by $b = 2/3 \lambda$, along the straight line passing through the point C and perpendicular to plane in Fig. 39. Find the intensity of light at the point O lying in the plane that passes through C if the case of a plane-parallel plate (viz. at $b = 0$) it is I_0 .

4.13. In an optical cavity composed of 4 plane mirrors (Fig. 40), light can propagate in either direction along the perimeter of a square of length l . If the cavity is rotated with an angular velocity Ω around an axis perpendicular to the plane of the figure, then the resonance frequencies for the waves are different in each direction. Explain the phenomenon and determine the difference between the frequencies.

4.14. Estimate the duration of a light pulse from a face of an octahedral rotating mirror 200 m away from a point source of light. The light pulse is detected by a photomultiplier

placed near the source and behind a narrow slit (Fig. 41). The mirror's face is 1 cm wide. Let the wavelength of the light be $\lambda = 5000 \text{ \AA}$. The mirror rotates with a frequency $\nu = 16 \text{ Hz}$.

4.15. A lens with a focal length $F = 50 \text{ cm}$ and diameter $D = 5 \text{ cm}$ is illuminated by a parallel monochromatic beam of light with wavelength $\lambda = 6300 \text{ \AA}$. How many times greater is the light intensity at the lens's focus than the intensity of the wave incident on the lens? Estimate the size of the spot in the focal plane.

4.16. A rocket moves away from the Earth and ceases to be discernible against the background of the sky using a telescope with a $D_1 = 80 \text{ mm}$ objective when it is $2 \times 10^4 \text{ km}$ away from the Earth. How far from the Earth could the rocket be detected using a telescope with a $D_2 = 200 \text{ mm}$ objective and given the eye's contrast sensitivity is the same?

4.17. Radiation from a continuous laser at a wavelength $\lambda = 0.63 \text{ \mu m}$ and power $P = 10 \text{ mW}$ is directed onto a satellite using a telescope having an objective of diameter $D = 30 \text{ cm}$. The light reflected by the satellite is collected by another telescope and focussed onto a photodetector whose threshold sensitivity is $P_{\text{thr}} = 10^{-14} \text{ W}$. Estimate the maximum distance at which the light reflected by the satellite will still be detectable. The satellite's surface uniformly scatters the incident light with a reflecting coefficient $\rho = 90\%$. The satellite is $d = 20 \text{ cm}$ in diameter.

4.18. How far away can the naked eye see the light of a laser which generates a continuous power of 10 W at a frequency $\nu = 4 \times 10^{14} \text{ Hz}$ if the beam is formed using a parabolic mirror of diameter $D = 50 \text{ cm}$? An eye can see a source if the pupil (diameter $d = 5 \text{ mm}$) receives 60 quanta of light per second with a frequency within the green part of a spectrum.

4.19. When an aerial photograph is made of the ground an objective with focal length of 10 cm and diameter 5 cm is used. The photograph is taken using a film with a resolution of 100 lines per mm. Determine how much detail on the

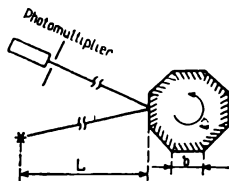


Fig. 41

ground can be resolved on a photograph if they were taken from a height of 10 km.

4.20. A telescope objective has a focal length $F_1 = 3$ m and diameter $D = 15$ cm. Determine what the focal length F_2 of the eye-piece must be in order to use completely the objective's resolving power; the eye pupil's diameter is 3 mm. Assuming that there is no aberration in the telescope-eye system, estimate how far away a book with letters $l \approx 2$ mm in size can be read using the telescope.

4.21. Some astronauts have landed on the Moon. In order to communicate the fact to the Earth, they lay a black circular canvas out over the Moon's surface. What radius must the circle have to be visible from the Earth through a telescope with a 5-m objective? The receiver's contrast sensitivity is 0.01.

4.22. A glittering metallic interplanetary spaceship 10 m in diameter has landed on the Moon during a full Moon. Estimate the diameter of a telescope's mirror needed to observe the spaceship from the Earth if the contrast for reliable eye observations is taken to be $k = 0.15$. The reflectivity of the moon's surface is $\rho_1 = 0.1$ and that of the metal surface is $\rho_2 = 1$. The distance between the Earth and the Moon is 4×10^5 km, and the observation is made in light of wavelength $\lambda = 0.6 \mu\text{m}$.

4.23. The angular aperture of an electron microscope is 10^{-4} and that of an optical one is about 1. What voltage is needed to accelerate the electrons for the resolving powers of the devices to be identical?

4.24. The spectral lines emitted by a heated gas appear broadened because atoms of the gas move with different velocities relative to an observer (Doppler effect). Assuming that the velocity distribution of the atoms is Maxwellian, estimate the dimensions of a diffraction grating to be used for studying the form of the spectral lines emitted by ^{20}Ne at 1000 K. The diffraction constant of the grating d is $1 \mu\text{m}$.

4.25. Estimate the order of the velocity with which a spaceship must move away from the Sun in order for an astronaut using a spectrometer with a diffraction grating to be able to detect the spaceship's motion relative to the Sun by observing the visible part of the Sun's hydrogen spectrum in the second diffraction order. How many rulings must the diffraction grating have? The Sun's surface temperature is 6000 K.

4.26. A binary star consists of two stars similar in mass ro-

tating with a period of 10 days and 2×10^7 km apart. Determine how many rulings a diffraction grating must have in order that the relative rotation of the stars can be detected in the second diffraction order by observing the visible part of the hydrogen spectrum in their radiation. Is it possible in principle to detect the relative rotation of the stars using this technique, if the period of rotation is 10 years? The temperature of stars' surfaces is 6000 K.

4.27. Ruby laser radiation is scattered by sound oscillations in water. When the light is scattered a Doppler frequency shift occurs. Estimate how many rulings a diffraction grating must have for the shift in the light scattered at right angles to be detected in the first diffraction order. The velocity of sound in water is 1400 m/s, and the refractive index n is 1.3. Assume that sound waves in water have every possible orientation.

4.28. How many rulings must a diffraction grating have in order to detect (during a total solar eclipse) a gravitational shift in the spectral lines of the spectrum of solar radiation? The observation is made in the second diffraction order. The data needed for a numerical calculation can be derived from the angular diameter of the Sun (observed from the Earth), viz. $\alpha \approx 0.01$ rad and the orbital velocity of the Earth, $v = 30$ km/s.

4.29. Determine the resolving power of the following type of infrared spectrometer. The radiation of an infrared source in the range $\alpha_{IR} \approx 3 \mu\text{m}$ is mixed in a nonlinear crystal with the radiation of a stabilized argon laser. A sum radiation with a frequency lying within optical range is produced. This radiation is analyzed using a Fabry-Pérot interferometer whose mirrors are 1 cm apart and have a reflection coefficient of 0.9.

4.30. A beam of plane-polarized light of wavelength $6280 \text{ \AA}$ and power 3 W is incident on a quartz four-wave plate. Under what conditions will the plate experience a torque and what will be its magnitude and direction?

4.31. A plane monochromatic wave is incident on a periodic structure composed of thin parallel dielectric plates (Fig. 42). The plate thickness is d_0 , the distance between the plates is d , while the permittivity of the plates is ϵ_1 and that of the surrounding medium ϵ . The wavelength is much longer than d_0 and d . Demonstrate that this structure is analogous to uniaxial crystal and determine the refractive indices of the ordinary and extraordinary rays.

4.32. A plane monochromatic wave of frequency ω is incident on a system of alternating polaroids and quartz plates cut parallel to their optical axis (Fig. 43). The principal directions of all the polaroids are parallel and at 45° to the optical axis of the plates. The wave is polarized along the polaroids' principal direction. Determine the wave's amplitude at the outlet of the system if at the inlet it is A_0 .

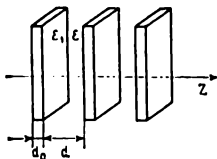


Fig. 42

The system consists of N plates and $N + 1$ polaroids. The thickness of the plates is d , $2d$, $\dots$, $2^{N-1}d$. The refractive indices of quartz are n_o and n_e . The light reflected at the interface between the plate and the polaroid must be neglected. Is such a system a spectral device or not?

4.33. Find the refractive index of gas near the surface of Venus and the derivative of the refractive index with respect to height at the surface. Venus's atmosphere consists of CO_2 molecules whose polarizability is $2.7 \times 10^{-23} \text{ cm}^3$ and its surface pressure p_0 is 100 atm, and its temperature is 500°C . Find

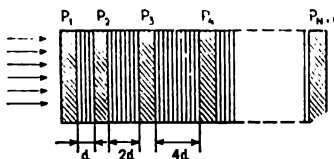


Fig. 43

the curvature of a light beam emitted horizontally. What can we say about the planet's atmospheric optics from this result?

Note. The radius of curvature R of a horizontal beam is governed by the relationship $\frac{1}{R} = \frac{1}{n} \frac{dn}{dh}$.

4.34. How much more dense must the Earth's atmosphere be for the circular refraction of light around the Earth to exist, as it does on Venus? The refractive index of air at atmospheric pressure is $n_0 = 1.0003$ (cf. Problem 4.33).

4.35. A lens with a relative aperture of $1 : 3.5$ focuses solar light onto the surface of a black ball placed in a vacuum. To what temperature could the ball be heated if its diameter equals that of the solar image, if the Sun subtends from the Earth an angle $\alpha = 0.01$ rad, and if the solar constant β is 0.14 W/cm^2 ?

4.36. How much power P must a laser beam of diameter $D = 1 \text{ mm}$ have in order to initiate an electric breakdown in

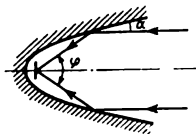


Fig. 44

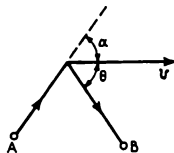


Fig. 45

a gas? Under experimental conditions the mean free path of electrons in the gas is 10^{-4} cm and its ionization potential is 10 V .

4.37. A mirror in the form of an extremely elongated paraboloid of revolution focuses soft X-rays due to total internal refraction at sliding angles of incidence α on the paraboloid far from the vortex (Fig. 44). Estimate the angle of convergence φ at the paraboloid's focus for 2 keV X-rays if the mirror is made from berillium (berillium's density is 1.82 g/cm^3).

4.38. Determine the number of free electrons per atom of silver if a film of the metal is transparent for ultraviolet light starting from 5 eV . Silver's mass number is 108 and its density 10.5 g/cm^3 .

4.39. The refractive index of the ionosphere for radiowaves of frequency 10 MHz is 0.90 . Find the electron concentration in ionosphere, and the phase and group velocities for the radiowaves.

4.40. A laser beam is focused by an ideal optical system with $F/D = 1$. Estimate the laser power P needed to donate an energy of about mc^2 to electrons in the electric field at the system's focus.

4.41. A laser beam at point A (Fig. 45) with frequency ω_0 is incident on a satellite flying with velocity v . The reflected beam is detected at point B. What will the frequency of the

detected light be? Estimate the resolving power a detecting spectral device would need in order for the relativistic correction to the frequency shift to be detected.

4.42. In order to test the theory of relativity an exact measurement of the parameters of the orbit of an Earth satellite using radiowaves is planned. However, measurement errors emerge because the radiowaves are refracted in the ionosphere, the average electron concentration there being 10^8 cm^{-3} . Estimate the minimum frequency at which the measurements should be performed.

4.43. The acceleration of a space rocket is the greater the higher the velocity of the gas ejected from its nozzle. From this point of view the best type of engine is a photon rocket which emits a photon flux from its nozzle. Calculate what velocity will be acquired by such a rocket if starting from zero velocity half its mass is converted to photons. The engine efficiency can be taken as unity.

4.44. After 16 orbits around the Earth a satellite is brought back down to the space-launch complex. What will the difference between the satellite and space-launch complex clocks be and how accurately can it be detected if both the stability and reproducibility of the clock readings amount to 10^{-13} (hydrogen maser)? The effects of trajectory curvature, gravity, and the acceleration of take-off and landing on the satellite's clock must be neglected.

V. Atomic Physics

5.1. Determine the energy of the γ -quanta which experienced a Compton backscattering ($\varphi = 180^\circ$) if the ejected electron is ultrarelativistic ($E \gg mc^2$).

5.2. What X-ray wavelength is doubled after a Compton scattering through 90° ?

5.3. A crystalline plate is used in an X-ray spectrograph that uses Bragg interference reflection. What is the minimum thickness of the plate D at which it will be possible to detect the Compton shift when a photon is scattered at an angle $\varphi = 90^\circ$ to the initial direction? The X-ray wavelength is $\lambda = 0.7 \text{ \AA}$ and the scattered X-rays are incident on the crystal at the slip angle $\theta = 30^\circ$.

5.4. The Compton scattering of a photon from an electron at rest results in the electron being given a recoil momentum p . Determine at which angles φ to the incident photon direction the electron can be ejected.

5.5. Estimate the angle of deflection due to gravity of a photon moving near the Sun's surface ($M_S = 2 \times 10^{33}$ g, $R_S = 7 \times 10^{10}$ cm).

5.6. A parallel beam of monoenergetic neutrons moving with a velocity v is incident on a plane crystal surface at a slip angle φ_0 . It then undergoes m -th order Bragg reflection

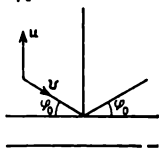


Fig. 46

(Fig. 46). The neutron source is set in motion with a constant velocity u along the normal to the reflecting plane. At what angle φ to the plane must the neutron beam be directed to observe the initial m -th order Bragg reflection?

5.7. One technique for monochromatising slow neutrons involves a solid cylinder of radius $R = 10$ cm and length $L = 1.0$ m with a spiral groove of width $b = 1$ cm and pitch angle $\varphi = 30^\circ$ (Fig. 47). The cylinder rotates at $v = 3000$ rpm. Determine the wavelength of the neutrons transmitted by

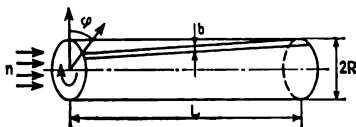


Fig. 47

the monochromator and estimate their monochromaticity. The neutron beam is directed along the cylinder axis.

5.8. Estimate the minimum diameter D of the spot produced on a detector by a beam of silver atoms emitted from an oven with a temperature $T = 1470$ K. The distance between the oven's exit slit and the detector is $L = 1$ m.

5.9. A one-dimensional well of width b and infinitely deep contains N electrons. Determine the minimum value of the total energy and pressure of the electrons upon the well's walls. Electron interaction should be neglected.

5.10. Estimate the kinetic energy of a nucleon in a nucleus assuming that the nuclear radius is $r = 10^{-12}$ cm.

5.11. Assuming nuclear forces are due to an exchange of nuclear field quanta, viz. mesons, between nucleons estimate

the range of the nuclear forces if the rest mass of the mesons is known to be $mc^2 \approx 100$ MeV and their velocity is close to that of light.

5.12. An electron moves with a velocity v perpendicularly to the bounding planes of a plane-parallel layer of a substance of thickness d and with refractive index n . The electron velocity v is greater than c/n so that Vavilov-Cherenkov radiation is observed. Determine the additional angular opening $\Delta\varphi$ of the radiation, which results from the finite thickness of the layer (Fig. 48).

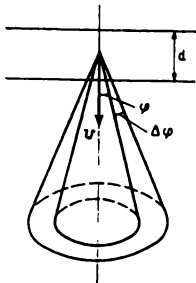


Fig. 48

5.13. Find the minimum energy of an electron at which it will easily pass over a rectangular well of depth $U = -5$ eV and width $b = 10^{-8}$ cm.

5.14. The wavelength of the H_α -line in a Balmer series is $\lambda = 6563$ Å. Determine the ionization potentials of positronium and muonium in the ground state. The muon mass is $m_\mu = 214 m_e$ (m_e is the electron mass).

5.15. Calculate the energy of the radiation emitted during

the transition of a negative muon in a hydrogen atom from the N to M shell. The muon mass is $m_\mu c^2 = 106$ MeV. In the calculation take into account that such a mass is about a tenth the mass of a proton. How large is the Bohr orbit radius in this case?

5.16. A positively charged muon that forms a hydrogen-like atom, viz. muonium, together with an electron decays into a positron and a neutrino which rapidly scatter in different directions. What is the average value of the kinetic energy of the remaining electron if the muon decay occurred when the muonium was in the $2S$ -state? The electron wave function is

$$\Psi = \frac{1}{\sqrt{8\pi r_1^3}} \left(1 - \frac{r}{2r_1}\right) e^{-r/2r_1},$$

where r_1 is the radius of the first Bohr orbit.

5.17. Atomic energy levels are usually found by assuming a point nuclear charge. In fact nuclei have finite dimensions and nuclear radii are described thus $r_N = 1.3 \times$

$10^{-13} \sqrt{A}$ cm, where A is the relative atomic mass. Estimate the sign and order of magnitude of the relative correction $\Delta E/E$ to the energy of a muon in the K -shell of a neon meso-atom ($Z = 10$, $A = 20$), which correction is due to the muon being a fraction of time inside the nucleus, viz. in a field whose potential differs from Ze^2/r . The normalized expression for the wave function of an electron's ground state in a hydrogen atom is

$$\Psi = \frac{1}{\sqrt{\pi r_1^3}} e^{-r/r_1},$$

where r_1 is the radius of the first Bohr orbit.

5.18. In Compton scattering of quanta from electrons, the phenomenon is complicated by the electrons not being at rest in the atoms. Estimate the relevant spread in ejection angle of the recoil electrons knocked out of the hydrogen atoms if the X-ray quanta of a wavelength $\lambda = 1 \text{ \AA}$ are elastically scattered strictly backwards.

5.19. A parallel beam of atoms which are all in the same excited state moves along a vacuum tube axis at $v = 10^6$ cm/s. The tube walls have the windows for detecting the radiation from the atoms as a function of the distance covered by the

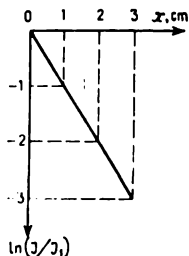


Fig. 49

beam in the tube. The results of these measurements are shown in Fig. 49. The abscissa is the distance covered by the beam along the tube and measured from the first window, while the ordinate is the natural logarithm of the ratio between the intensity of light $I(x)$ and the intensity measured by the detector in the first window. Determine the natural width of the spectral line emitted by the beam atoms.

5.20. Estimate the minimum size of a speck of iron needed for the Mössbauer effect to be observed with a transition energy $E = 14 \text{ keV}$ and a lifetime $\tau = 10^{-3} \text{ s}$ if the speck recoil results in a Doppler shift equal to the intrinsic line width.

Note. The Mössbauer effect occurs when at a rather low temperature the entire crystal (in this case the speck of dust) recoils rather than just the individual irradiated nucleus.

5.21. At what height above a source // must an absorber be placed to detect the red shift predicted by the general theory of relativity? Consider the Mössbauer effect on the ^{67}Zn isotope. The lifetime of the excited level with an energy $E = 93 \text{ keV}$ is $\tau = 10^{-8} \text{ s}$. Assume that in order to get the needed accuracy the red shift must be tenfold more than the width Γ of the resonance absorption line.

5.22. The energies of the transitions between three successive rotational energy levels in a diatomic molecule are measured experimentally (Fig. 50). Find the orbital quantum numbers l of the levels and the moment of inertia I of the molecule.

5.23. In Shull's experiments (1968), a neutron beam was observed to split into two beams when refracted at the boundary of a homogeneous magnetic field. Find the small angle θ

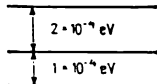


Fig. 50

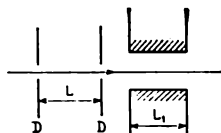


Fig. 51

between the directions of the refracted beams. The magnetic field had an induction $B = 2.5 \text{ T}$. The neutrons of wavelength $\lambda = 5 \text{ \AA}$ were incident at 30° on the sharp boundary of the magnetic field.

5.24. A beam of sodium atoms is ejected from an oven whose temperature is $T = 700 \text{ K}$. The beam is split in a transverse inhomogeneous magnetic field with a gradient $dB/dx = 5 \text{ T/cm}$ along the path of length $l = 10 \text{ cm}$. A detector is placed 65 cm apart from the magnet. Find the separation distance between the spots on the detector screen.

5.25. A beam of lithium atoms in the ground state with a maximum energy $E_k = 0.1 \text{ eV}$ passes through a Stern-Gerlach magnet $L_1 = 6 \text{ cm}$ long and having a field gradient $dB/dx = 5 \text{ T/cm}$. Two identical diaphragms D , $L = 1 \text{ m}$ apart, stand in front of the magnet (Fig. 51). How large can the diaphragms be for the components of the split beam to be completely separated?

5.26. Estimate the distance L between mirrors and a Fabry-Perot interferometer needed to observe Zeeman splitting in

a magnetic field $B = 1$ T. The interferometer mirrors are silvered in such a way that $N \approx 20$ reflections are observed between them.

5.27. Estimate how large a magnetic field can be detected for a Sun-like star (period of revolution $\tau = 10^4$ s, radius $R = 10^{10}$ cm, surface temperature $T = 6 \times 10^3$ K) on the basis of measuring the Zeeman effect in the optical part of spectrum ($\omega = 10^{15}$ s $^{-1}$).

5.28. Caesium is an alkali metal. During a P - S transition, caesium emits a broad doublet composed of two lines: $\lambda_1 = 4555$ Å and $\lambda_2 = 4593$ Å. Draw a diagram showing how the terms of the doublet's will be split in a magnetic field. Is the line splitting in a magnetic field $B = 3$ T described by the formulae for the normal or the anomalous Zeeman effect?

5.29. In lithium's spectrum the first two lines of the principal series pertain to the transitions $2^3P_{1/2} \rightarrow 2^3S_{1/2}$ and $2^3P_{3/2} \rightarrow 2^3S_{1/2}$. The wavelengths of these lines are 6707.80 and 6707.95 Å, respectively. Estimate the magnetic field that induces electron orbital motion in a lithium atom in the $2P$ -state.

5.30. Estimate (in electron-volts) the splitting of the positronium $2P$ -state induced by the interaction of the spin magnetic moments of the positron and electron.

5.31. Estimate the order of magnitude of the splitting of the leading line of the Balmer series in hydrogen's spectrum due to the interaction between the magnetic moments of the electron and the nucleus (hyperfine splitting of spectral lines). Could this splitting be detected by a spectral device if the average lifetime Δt of excited hydrogen atoms is of the order of 10^{-8} s? The magnetic moment of a proton is 2.8 nuclear magnetons.

5.32. Estimate the order of magnitude of the wavelength of the radiation of interstellar atomic hydrogen in radio-frequency range. Interstellar hydrogen is in the ground state and its radiation is due to electron spin reorientation.

5.33. Free atoms can possess a magnetic moment but have no dipole electric moment. Under certain conditions, atoms forming a crystalline lattice can also have such a moment. In this case paraelectric resonance is possible, which is analogous to the paramagnetic one. Find the atomic dipole moment if it is known that the resonance absorption of electromagnetic waves of wavelength $\lambda = 5$ mm is observed in

a stationary electric field strength $E = 2.5 \times 10^3$ kV/m. Estimate the size of the atomic dipole.

5.34. Liquid helium possesses a negative work function for excess electrons. Therefore an electron introduced into the helium repels the atoms of the liquid and forms a spherical cavity, which is, for the electron, a potential well with practically an infinitely high wall. Calculate the cavity radius R when the electron occupies the lowest quantum level in the cavity. The surface tension of liquid helium is $\sigma = 0.35$ dyn/cm. Assume the external pressure to be zero. Neglect the polarizability of helium (due to the smallness of its dielectric susceptibility).

5.35. A region of high pressure forms around an electrically charged particle, e.g. an Ho^+ ion, in liquid helium, and in the immediate vicinity of the charge the helium solidifies. The increase in pressure is caused by the attraction of the helium atoms to the electric charge because of the polarizability of their electron shell. Find the radius R of the ball formed by solid helium if the helium density $\rho = 0.145$ g/cm³, the molar polarizability $\alpha N_A = 0.125$ cm³/mol, and the solidification pressure $p_s = 25$ atm. Neglect helium's compressibility; external pressure is absent and assume that the ball size significantly exceeds an atomic radius.

VI. Radiation

6.1. A circular disc at a temperature T_0 is the source of thermal radiation. The disc is placed at a height H above a plane blackened on both sides. The disc radius $R \ll H$ and the disc is oriented parallel to the plate. Find the stationary temperature distribution across the plane. Assume that the system is in vacuum, thermal radiation background is absent, and thermal conductivity along the plane can be neglected.

6.2. Find the ratio of the energy flux densities W between corpuscular and wave radiations of the Sun at the Earth. Assume that the corpuscular radiation is a neutral plasma composed of electrons and protons with a concentration of particles of each sort $n = 5$ cm⁻³ and a flux velocity $v = 300$ km/s, and that the Sun radiates as an absolutely black body with a temperature $T = 6000$ K. The angular dimension of the Sun is to be taken as $\alpha \approx 10^{-3}$ rad.

6.3. At what concentration n of gas molecules is the gas-kinetic pressure at $T = 100$ K equal to the pressure exerted

by equilibrium thermal radiation at the same temperature?

6.4. Gaseous neon in a closed vessel of constant volume is in equilibrium with the thermal radiation at $T = 500$ K. At what pressure will the neon's heat capacity be comparable with the heat capacity of thermal radiation?

6.5. Starting from the photon picture derive the equation of state for electromagnetic radiation, $p = 1/3 u$. Using this equation and applying thermodynamic laws to electromagnetic radiation, find the dependence of energy and entropy per unit volume (u and s) on temperature T . Relate the integration constant that emerges in this calculation to the Stefan-Boltzmann constant σ .

6.6. Find the heat capacity C_p and the adiabatic equation for the equilibrium radiation enclosed in a vessel of variable volume.

6.7. An excited atom with excitation energy $E = 1$ eV is in a field of equilibrium radiation with temperature $T = 300$ K. Find the probability ratio between induced and spontaneous atomic radiations. Find an analogous ratio for the electron spin in a magnetic field $B = 10^3$ G.

6.8. Estimate the probability w_{sp} for a molecule to undergo a spontaneous transition from an excited level E_m into a level E_n when the molecule is inside a cavity tuned to a frequency $\omega = (E_m - E_n)/\hbar$. The corresponding probability of spontaneous radiation in free space is w_{sp}^0 . The volume of the cavity is V and its quality factor is Q . Assume that the width of the molecular levels Γ is always smaller than that of the cavity line, viz. $\Gamma < \omega/Q$.

6.9. A laser cavity with a ruby crystal has one mirror with 100% reflection and another with a transmission $\tau = 0.1$ at the wavelength of laser generation mode. The crystal length is $L = 12$ cm. It is known that the absorption index of light in nonexcited ruby crystal at the maximum of the working line is $\alpha = 0.4$ cm⁻¹. Find what fraction of the chromium atoms must be excited to start laser action. The scattering of light in the crystal should be neglected.

VII. Solid State

7.1. An X-ray with frequency 1.1×10^{18} Hz is incident on a cubic crystal in the [100] direction and experiences strong Bragg scattering in the [122] direction. Assuming that the

crystal is a single face-centered Bravais lattice and the observed scattering is due to first order interference, find the minimum interatomic distance in the crystal.

7.2. Find the dispersion law for longitudinal phonons in an endless chain whose elementary cell contains two atoms with masses M_1 and M_2 . The distance between two adjacent atoms is a , and the rigidity of the interatomic links is γ . Construct the plot of the dependence obtained. Indicate the limiting conversion to a monatomic chain at $M_1/M_2 \rightarrow 1$.

7.3. Find the polarizability (per cell) for a chain whose elementary cell contains two single-charged ions with charges of opposite sign. The remaining conditions are the same as those in 7.2. The electric vector of an exciting electromagnetic wave of frequency ω is oriented along the chain.

7.4. Using the conservation laws of energy and momentum, consider the inelastic single-phonon scattering of neutrons in an ideal crystal. Discuss the possibility of reconstructing the phonon dispersion law from the neutron scattering.

7.5. Derive the valuable formula $\hbar\omega \sim (m/M)^{1/2} E_{at}$ for the oscillation frequencies of molecules and crystals. Here m and M are the masses of the electron and the nuclei, respectively, and E_{at} has the atomic order of magnitude. Obtain the estimate $x_0/a \sim (m/M)^{1/4}$ for the ratio of the zero oscillation amplitude x_0 to the interatomic distance a .

7.6. Using an analogy between photons and long-wave phonons, express the low temperature lattice heat capacity of crystals in terms of the velocities of transverse and longitudinal sound.

7.7. Find the temperature dependence of the lattice heat capacity of one and two-dimensional crystals in the low temperature range.

7.8. Starting from a model of a diatomic molecule with a weak cubic anharmonicity, express the temperature coefficient of expansion α in the direction of the molecular axis in terms of the parameters of the model's potential and estimate this coefficient. Assume the motion of nuclei to be classical.

7.9. Find the low temperature entropy of crystalline helium-3. The helium-3 nucleus has a spin $I = 1/2$. Assume that the temperature $T \ll \Theta$ (Θ is the Debye temperature) is such that practically all the oscillatory degrees of freedom are "frozen-out", but the nuclear spins remain totally disordered. Under the same assumptions find the entropy of an argon-37 crystal (nuclear spin $I = 3/2 \hbar$).

7.10. Electron gas pressure is one of the principal factors controlling the compressibility of metals. Find the coefficient of the uniform compression of electron gas for copper at $T = 0$ K, if the electron concentration is $n = 8.5 \times 10^{23} \text{ cm}^{-3}$. Assume the effective mass is equal to the free electron mass.

7.11. Find the Fermi velocity v_F of the electrons in a metal with one electron per elementary cell and with a "one-dimensional" dispersion law $E = E_0 \cos(k_1 a)$ for $E_0 = 0.5$ eV and $a = 3 \text{ \AA}$ ($k = p/\hbar$ is the wave vector).

7.12. In a metal whose crystalline lattice has z-axis symmetry of order not lower than the third, the simplest dispersion law for electrons can be given as $E(p) = 1/2 m (p_x^2 + p_y^2)$. Assume that each elementary cell contains one conduction electron, the lattice period along the z-axis is $a = 3 \text{ \AA}$, the elementary cell volume is $V = 0.85 a^3$, and m is equal to the free electron mass. Calculate the Fermi velocity of the electrons and determine their frequency of rotation in a magnetic field H in terms of the angle θ between the H vector and the z-axis.

7.13. With no electric current an external static electric field only penetrates a thin surface layer of a metal. Derive the equation for the decay of the field into the metal assuming that the total potential drop is $V \ll E_F/e$ (E_F is the electron Fermi energy). Estimate the depth of field penetration (the Thomas-Fermi screening length l_{TF}) for a normal metal ($n_0 \approx 10^{23} \text{ cm}^{-3}$, $E_F \approx 5$ eV, permittivity $\epsilon \approx 1$) and for a Bi-type semimetal ($n_0 \approx 3 \times 10^{17} \text{ cm}^{-3}$, $E_F \approx 2 \times 10^{-3}$ eV, $\epsilon \approx 100$) at a $T = 0$ K. The permittivity ϵ is controlled by the polarizability of the internal electrons which do not participate in the electrical conduction.

7.14. For typical values of the parameters of metals, estimate the temperature at which electron and lattice heat capacities become equal by expressing this temperature in terms of the masses of the electrons and the nuclei, and in terms of the energy E_A , having an atomic order of magnitude. Make a numerical estimate of this temperature.

7.15. Since the mean free path in a thin wire is limited by the wire's diameter, the mean free paths of electrons and phonons are practically the same. Under these conditions, estimate the temperature at which the electron and lattice thermal conductivities of metals become equal.

7.16. Find the frequency of the natural long-wave oscillations of electron gas in conductors which induce no magnetic field (the plasma frequency ω_p). Neglect any "friction" encountered by the motion of the electrons in the crystalline lattice and assume that the electrons' dynamics is described by an effective mass m^* . Estimate the frequency ω_p and quantum $\hbar\omega_p$ for metals.

7.17. Find the frequency dependence of the complex permittivity $\epsilon(\omega)$ for a conductor. Explain the transparency of metals observed in the ultraviolet region of spectrum. Assume that the conduction electrons are described by an

effective mass m^* and a mean free time τ .

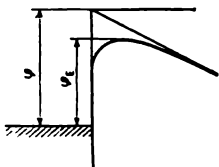


Fig. 52

7.18. Determine the depth of penetration of a high-frequency electromagnetic field with a frequency ω into a metal whose conductivity is σ . Assume that $\omega\tau \ll 1$.

7.19. Calculate the current produced by electrons moving in crossed electric and magnetic fields ($\mathbf{E} \perp \mathbf{H}$). The effective

electron mass is m^* , and the mean free time τ . Apply the results to the calculation of resistance in terms of H for the following cases: (a) a current runs through an infinite plate with $\mathbf{E}$ and $\mathbf{H}$ lying in plane of the plate; (b) a current runs through a Corbino disc (in this disc electrodes are concentric circles, while a magnetic field is applied perpendicular to the plane of the disc). Comment on the results obtained.

7.20. Thermoelectrons emitted from a metallic cathode move in a vacuum due to an external field $\mathbf{E}$ and mirror image forces. The combined action of these forces results in the work function at the metal-vacuum interface being lowered. Find the dependence of the work function φ_E (Fig. 52) and the thermoemission current J on the field E .

7.21. Derive the equation for the dissipation of space charge in conductors and the characteristic time of this process, viz. the Maxwellian time of relaxation τ_M . Determine τ_M for a crystal of pure germanium at room temperature ($\sigma = 0.014 \text{ S/cm}$, $\epsilon = 16$).

7.22. In both semiconductors and metals (cf. Problem 7.13), an external field is screened by conduction electrons. The

difference between the two is that in semiconductors the electron gas is generally nondegenerate, viz. this gas follows a Boltzmann distribution. Under these conditions, assuming the external field is weak, derive the law of the field falling off deep into a nondegenerate semiconductor. Estimate the field penetration depth l_D (Debye screening length) for a semiconductor with permittivity $\epsilon \approx 15$ and electron concentration $n \approx 10^{14} \text{ cm}^{-3}$.

7.23. Calculate the electron rotation frequency ω_c (cyclotron frequency) in a stationary, uniform, random-oriented field H whose dispersion law is

$$E(p) = \frac{p_x^2}{2m_x} + \frac{p_y^2}{2m_y} + \frac{p_z^2}{2m_z}.$$

7.24. Calculate the shape of the cyclotron resonance curve (viz. the frequency dependence of the conductivity) for electrons with an isotropic effective mass m^* and a mean free time τ . The electromagnetic wave is circularly polarized in a plane perpendicular to a stationary magnetic field.

7.25. An electron whose dispersion law is $\mathcal{E} = \mathcal{E}_0 \cos(k_x a)$ moves in a stationary uniform electric field E directed along the x -axis. Solve the equations of motion and give a physical interpretation of the result.

7.26. Solve the previous problem, given a frictional force proportional to the velocity ($F = -\gamma v$). Investigate the possible modes of the electron's motion and calculate the conductivity of such a semiconductor.

7.27. Consider the scattering of a slow electron in a semiconductor with absorption and emission of a long-wave acoustic phonon by using the laws of conservation of energy and momentum. Find the angle between the phonon wave vector q and initial electron momentum p in terms of p and q . Show that for $v < s$ (v is the electron velocity, s is the velocity of sound) the electron cannot radiate phonons, while for $v \gg s$ the electrons are scattered almost elastically, viz. little energy is changed during the scattering. Assume that the electron energy is $E = p^2/2m$, and the phonon energy is $\hbar\omega(q) = \hbar sq$.

7.28. In some semiconductors the electron mean free path appears to be of the order of the interatomic distances. In such a situation the electrons can be considered to move in random "jumps". Estimate the conductivity of such a semiconductor for $T \approx 300 \text{ K}$, if the electron concentration is $n_e \approx 10^{18} \text{ cm}^{-3}$, the average rate of the jumps is

- $v \approx 10^{13} \text{ s}^{-1}$, and the interatomic distance is $d \approx 3 \text{ \AA}$.
- 7.29. A medium whose conductivity depends on an electric field E , such that $\sigma(E) = \sigma_0 / [1 + (E/E_0)^2]$, is placed in an external homogeneous field $E \parallel x$. The space charge density $\rho(x)$ then begins to fluctuate in this medium. Find how the charge distribution along the x -axis will be changed in time. Assume the fluctuation to be small.
- 7.30. Derive a formula for the concentration of band electrons in a nondegenerate semiconductor, given the chemical potential ζ (Fermi level).
- 7.31. Find the relation between the concentrations of the electrons n and the holes p in a nondegenerate semiconductor having an arbitrary concentration of impurities, and the concentration of carriers n_i in the same semiconductor but without impurities (viz, in an intrinsic semiconductor).
- 7.32. Investigate and show schematically the temperature dependence of the concentrations of the electrons n and holes p in a semiconductor with shallow donor levels. The electron binding energy on donors is $2\Delta_1$, and the forbidden gap width $2\Delta \gg 2\Delta_1$.
- 7.33. Find the Rydberg constant, Bohr radius, and effective mass M for an exciton, viz. the hydrogen-like formation constructed from an electron and a hole. Estimate the values of these quantities assuming that the effective masses of the electron and hole are $m_n^* \sim m_p^* \sim 0.1 m_0$ (m_0 is the electron mass), and the permittivity is $\epsilon \approx 10$.
- 7.34. The holes produced by illuminating an electron semiconductor surface diffuse inside, where they recombine with conduction electrons. Determine the effective penetration depth of the holes if their lifetime is $\tau = 10^{-3} \text{ s}$ and mobility is $\mu = 2000 \text{ cm}^2/(\text{V}\cdot\text{s})$ while the sample temperature is $T = 300 \text{ K}$.
- 7.35. Assuming that the electrons in a superconductor move without resistance, calculate the depth λ to which a low frequency electromagnetic field penetrates a superconductor (London length). Estimate its value for typical parameters of metals.
- 7.36. At moderate pressures helium remains liquid down to the temperature $T = 0 \text{ K}$. Determine the temperature dependence of liquid helium's heat capacity (isotope ^3He) at low temperatures and estimate the numerical value of the coefficient in this dependence by neglecting interatomic interaction. The molar volume of liquid helium at normal pressure is $V = 37 \text{ cm}^3/\text{mol}$.

VIII. Nuclear Physics

8.1. In 1976 the Nobel prize in physics was awarded for the discovery of a new elementary particle, viz. the J -meson. The discovery was made practically simultaneously and independently in two different experiments. One used colliding beams of electrons and positrons accelerated up to the same energy E_{cms} . Determine the mass and estimate the lower bound of the J -particle lifetime.

In another experiment the J -meson decay products were detected. Find the mass of the J -meson decaying into an electron and a positron, if it is known that their energies are identical ($E_1 = E_2 = 3.1$ GeV) and the divergence angle between them is $\varphi = 60^\circ$.

8.2. The quark model of hadrons assumes that mesons are composed of a quark and an antiquark while baryons are composed of three quarks. In a simple version of the model there are four quarks and the same number of antiquarks. The quantum numbers of the quarks are given in the following table.

Quark type	Spin, h	Electric charge, Q	Baryon number, B	Strangeness, S	Charm, C
q_p	1/2	+2/3	1/3	0	0
q_n	1/2	-1/3	1/3	0	0
q_Λ	1/2	-1/3	1/3	-1	0
q_c	1/2	+2/3	1/3	0	+1

From these quarks construct a proton, Ω -hyperon, positively charged pion, kaon, and a J -particle. Also, find the orientation of the spins of the quarks constituting these particles.

8.3. In 1976 a new baryon was discovered which was interpreted as the first member of the group of charmed anti-lambda-particles. The baryon decayed according to the scheme $\Lambda_c \rightarrow \tilde{\Lambda}^0 + \pi^+ + \pi^- + \pi^-$. The baryon mass was

$Mc^2 = 2.25$ GeV. Find what fraction of the baryon rest mass is transformed into the kinetic energy T of the decay products.

8.4. The annihilation of an antiproton stopped in liquid hydrogen produces three pions: $\bar{p} + p \rightarrow \pi^+ + \pi^- + \pi^0$. Determine the energy of each if one of the pions has the maximum possible energy.

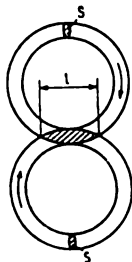


Fig. 53

8.5. In experiments with colliding beams of electrons two identical storage rings are used in which beams of ultra-relativistic particles move in opposite directions and collide with each other along an interaction length $l = 0.5$ m (Fig. 53). A system of counters surrounding the interaction region is positioned so that one in ten particle interactions (a detection efficiency $\varepsilon = 0.1$) is detected. Find the circulating current I that must be accumulated

in each ring in order for the counter system to observe not less than $K = 10$ counts per second. The cross-sectional area of the circulating beams is $S = 5$ mm² and the effective cross section of the interaction of two colliding particles is $\sigma = 10^{-25}$ barn. Assume that the number density of the particles along the orbit is constant.

8.6. Determine the kinetic energy of a proton beam which starts to emit Cherenkov radiation in nitrogen at 50 atm. At normal pressure the refractive index is $n_0 = 1.0003$.

8.7. A Cherenkov counter is installed in a beam of kaons and pions whose momentum is $pc = 500$ MeV. What must the refractive index of the counter substance n be for the counter to detect only kaons?

8.8. By virtue of the decay $\pi \rightarrow \mu + \nu_\mu$ a beam of pions with energy $E_\pi = 10$ GeV gradually transforms into a flux of muons and neutrinos. Estimate at what distance from the production region the number of muons in the beam will be double the number of pions. Neglect the escape of muons from the beam due to deviations from the pion trajectory and the decay of muons.

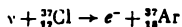
8.9. At what energy of a proton incident on a resting proton

in the reaction $p + p \rightarrow d + \pi^+$ may the kinetic energy of the pion vanish in the laboratory frame? The deuteron mass $m_d c^2 = 2m_p c^2 = 2 \times 0.94 \text{ GeV}$, and that of pion $m_{\pi} c^2 = 0.14 \text{ GeV}$.

8.10. To produce meson beams in accelerators, a thin target is installed in a beam of accelerated particles. The mesons are generated in the target. Determine how many mesons are produced per second if at the distance $L = 5 \text{ m}$ away from the target a detector records $n = 5 \times 10^2$ meson/s, and the detector area is $S = 100 \text{ cm}^2$. Perform the calculation for pions with an energy $T = 500 \text{ MeV}$, assuming that the pions are omitted from the target isotropically (viz. with the same probability at any angle).

8.11. Estimate the fraction of cosmic protons reaching the Earth without undergoing nuclear interactions in flight. Assume that the cross section of the proton-nucleus interaction is equal to the geometric cross section of nitrogen nuclei.

8.12. Nuclear reactions on the Sun can be studied by measuring the solar neutrino flux via the reaction



The average (over solar neutrino spectrum) reaction cross section is $\sigma = 1.4 \times 10^{-42} \text{ cm}^2$. Assuming that the Sun emits $N = 3 \times 10^{33}$ neutrinos per second determine how much CCl_4 (natural mixture of isotopes) is needed to produce 100 ${}^{37}_{18}\text{Ar}$ atoms per year. The natural mixture of chlorine isotopes contains 25% (by mass) of ${}^{37}_{17}\text{Cl}$ nuclei. The radius of the Earth's orbit is $R = 1.5 \times 10^8 \text{ km}$.

8.13. Determine the maximum number of pions produced in the $p + p$ reaction if the incident proton momentum $p = 5 \text{ GeV}$. Assume that prior to the reaction the target protons are at rest.

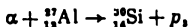
8.14. Find the lifetime of a muon in the laboratory frame if the muon was produced from the decay of a stopped kaon.

8.15. In the radioactive decay of the ${}^{60}\text{Co}$ nucleus an electron is emitted whose spin is parallel to its momentum. Estimate the angle φ through which a disc suspended by a filament will rotate if the ${}^{60}\text{Co}$ preparation is applied over one of the disc faces. The disc is thick enough to absorb all the electrons emitted towards the disc. The preparation's activity N is 10 Ci and the filament torsion modulus is $f \approx 10^{-6} \text{ dyn}\cdot\text{cm}$.

8.16. A specimen of teflon (a polymer whose chemical formula is $(CF_2)_n$, where n is an integer) of mass $M = 50$ g is magnetized in a magnetic field $H = 20$ kOe at $T = 0.05$ K. The magnetization is due to the splitting, under the action of the magnetic field, of the ground state of the ^{19}F nucleus (nuclear spin $s = 1/2$) into two sublevels. Upon switching on the field the specimen gains an angular momentum $L = 12.1 \times 10^{-6}$ erg·s (the analogue of Einstein-De Haas effect in a ferromagnetic). Determine the value of the magnetic moment μ of the fluorine nucleus.

8.17. Estimate the half-life $T_{1/2}$ of an even-even radioactive nucleus emitting 1 MeV α -particles if the ^{232}Th nucleus has $T_{1/2} = 1.4 \times 10^{10}$ years and emits 4 MeV α -particles, while a ^{210}Po preparation emits 8.8 MeV α -particles and has a $T_{1/2} = 3 \times 10^{-7}$ s.

8.18. Determine the effective cross section of the reaction



if it is known that $n_p = 8$ protons per 10^6 α -particles are produced when a thick aluminium target is irradiated by 8 MeV α -particles. The α -particle range in air under normal conditions is $R_{air} = 7.0$ cm.

8.19. In a 1 MW uranium reactor $\langle n_{\bar{\nu}} \rangle \approx 6$ antineutrinos are produced, on average, per fission event of a uranium nucleus (mainly from the decay of radioactive fission fragments). The mean antineutrino energy is $E_{\bar{\nu}} \approx 1.5$ MeV. The reactor is surrounded by a thick biological (concrete) shielding. Estimate both the antineutrino flux $\Phi_{\bar{\nu}}$ $L = 5$ m beyond the biological shielding of the reactor and the energy fraction taken from the reactor by the antineutrinos.

8.20. What is the kinetic energy T_n of the neutrons if, as a result of the neutron irradiation of a liquid hydrogen target and the reaction $n + p \rightarrow d + \pi^0$ the γ -quanta from the decay $\pi^0 \rightarrow \gamma + \gamma$ are known to be emitted in opposite directions in the laboratory frame?

8.21. In an evacuated vessel of volume $V = 1$ l ultracold neutrons are reflected from the walls with a reflection coefficient practically equal to unity. The vessel is provided with a window of area S covered with a foil that is completely transparent to the ultracold neutrons. What must the window area be if the neutrons are observed to be stored in the trap for half as long as the mean lifetime of free neutrons,

$\tau \approx 10^4$ s? Assume the velocity of all ultracold neutrons is the same and equal to $v = 5$ m/s.

8.22. Proceeding from Weizsäcker's formula, estimate the surface tension of nuclear matter.

8.23. For neutrons to enter most substances from a vacuum they must overcome an energy barrier. Therefore, sufficiently slow neutrons can be trapped in a closed cavity and may be accumulated. Determine what fraction of a flux of thermal neutrons with a Maxwellian velocity distribution will be found trapped in a copper chamber. For total internal reflection, the limiting slipping angle for neutrons moving with a mean thermal velocity will amount to $i = 10^\circ$. The collisions between the neutrons and the chamber walls may be considered as elastic.

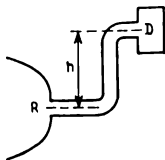


Fig. 54

8.24. After entering water, fast neutrons rapidly slow to thermal velocities and diffuse through the water unless they are caught by hydrogen nuclei (capture by oxygen atoms can be neglected). If the capture cross section is $\sigma = 0.3$ barn ($1 \text{ barn} = 10^{-24} \text{ cm}^2$) estimate the neutron lifetime in water τ .

8.25. A slow neutron is scattered from a free nucleus with spin $I = 3/2$ in the $L = 0$ state (orbital moment is zero). Determine the scattering probability in states with parallel and antiparallel orientations of the nuclear and neutron spins.

8.26. Estimate the kinetic energy of a neutron T_n , and a ${}^6\text{Li}$ nucleus T_N produced in the photodisintegration of a ${}^7\text{Li}$ nucleus induced by 15 MeV γ -quanta. The neutron is emitted forwards, viz. in the direction followed by the γ -quantum.

8.27. A well-collimated beam of γ -quanta with energy 500 MeV is incident upon a deuterium target. The secondary beam contains nucleons due to the photodisintegration of deuterium. Estimate the angle of ejection φ of the nucleons. The deuterium radius is $r_d = 2 \times 10^{-13} \text{ cm}$.

Hint. Take into account that the binding energy of the nucleons in deuterium is small compared with γ -quanta energy.

8.28. In reactors operating with thermal neutrons very slow

ultracold neutrons are produced. The feature of ultracold neutrons is that for velocities $v < v_b$ (the usual boundary velocity is $v_b \approx 10$ m/s) the neutrons will be elastically reflected from the walls for any angle of incidence. The ultracold neutrons are extracted from the reactor using hollow tubes, viz. neutron guides. Figure 54 shows a reactor R and a shaped neutron guide with a neutron detector D at its end. Assuming that the neutron velocity spectrum in the reactor is Maxwellian ($T \approx 400$ K) find how the neutron flux reaching the detector depends on its elevation h ? Estimate the height at which the flux vanishes. The effect of the detector on the neutron distribution should be neglected.

ANSWERS AND SOLUTIONS

I. Mechanics

1.1. The Earth acts upon a sample weight of mass m with

$$F = \left(\gamma m \frac{4\pi}{3} \rho_E R_E^3 \right) / R_E^2 = \frac{4\pi}{3} \gamma \rho_E R_E m.$$

The change in the force due to a lead ball of radius R_1 is

$$\Delta F = \frac{4\pi}{3} \gamma \rho_1 R_1^3 m,$$

where $\rho_1 = 11\,000 \text{ kg/m}^3$ is the lead density. To measure the change with the given accuracy requires

$$\frac{\Delta F}{F} \geq \frac{\eta}{\delta} = 10^{-6}, \quad \text{or} \quad R_1 \geq 10^{-6} \frac{\rho_E}{\rho_1} R_E \approx 3 \text{ m},$$

viz. the mass of the lead ball would exceed 10^3 ton.

1.2. Let m_1 and m_2 be the masses of the components of a double star, l_1 and l_2 the distances from their centres to the centre of mass, and v_1 and v_2 their velocities, respectively. Then, assuming for simplicity that they move in circular orbits, we obtain for the frequency of revolution

$$\omega = \frac{v_1}{l_1} = \frac{v_2}{l_2} = \frac{v_1 + v_2}{L}, \quad L = l_1 + l_2, \quad \omega = \frac{2\pi}{T}.$$

The consideration of the forces acting upon m_1 and m_2 results in

$$\frac{m_1 v_1^2}{l_1} = \frac{m_2 v_2^2}{l_2} = \gamma \frac{m_1 m_2}{L^2}, \quad \frac{v_1^2}{l_1} + \frac{v_2^2}{l_2} = \gamma \frac{m_1 + m_2}{L^2}.$$

Further, it can be easily obtained that $\gamma (m_1 + m_2) = 4\pi^2 L^3 T^{-2}$. Similarly, for the Sun-Earth system (the mass of the Earth M_E is negligible compared with the solar mass M_S), we obtain $\gamma M_S = 4\pi^2 R_E^3 T_E^{-2}$, where $T_E = 1$ year, while R_E is by definition equal to an astronomical unit. Thus,

$$M = m_1 + m_2 = M_S \left(\frac{T_E}{T} \right)^2 \left(\frac{L}{R_E} \right)^3 = \frac{8}{9} M_S.$$

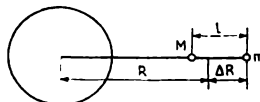
1.3. If an astronaut and a satellite move in the same orbit then the cord tension force is $F = 0$. The force is greatest when the astronaut, satellite and the Earth's centre lie on the same straight line (Fig. 55). The period of revolution T of the satellite-astronaut system near the Earth is related to the radius R of the orbit of the system's centre of mass by

$$\frac{4\pi^2}{T^2} R = \frac{\gamma M_E}{R^2},$$

where M_E is the mass of the Earth. Accordingly, to secure the astronaut on an orbit of radius $R + \Delta R$ needs an additional force F conditioned by the relation

$$\frac{4\pi^2}{T^2} (R + \Delta R) = \frac{\gamma M_E}{(R + \Delta R)^2} + \frac{F}{m}.$$

Whence, with an accuracy up to the terms of the order of $(\Delta R/R)^2$ we have



$$F \approx \frac{3\gamma M_E m}{R^3} \frac{\Delta R}{R} \approx \frac{3mg\Delta R}{R} \\ \approx \frac{3mgL}{R} \approx 2 \times 10^{-2} \text{ N},$$

Fig. 55

(here we took into account that the satellite and astronaut masses, M and m , are much less than M_E , and since $M \gg m$, $\Delta R \approx L$). The same force will be needed to secure an astronaut between the satellite and the Earth. To summarize, we may say that the gravitational force attracting the astronaut to the satellite is six orders of magnitude less than F .

1.4. The third Kepler's law suggests that the ratio between the Mars and Earth masses is

$$\frac{M_M}{M_E} = \left(\frac{L_M}{L_E} \right)^3 \left(\frac{T_E}{T_M} \right)^2 \approx 0.11,$$

where T_E is the period of revolution of the Earth satellite found from the relationship $m\omega^2 R_E \approx mg$ (m is the satellite's mass), $L_M = 1/2 (D_M + a + b)$.

1.5. $p(r) = \frac{2\pi}{3} \rho^2 \gamma (R_E^2 - r^2)$, $p(0) \approx 2 \cdot 10^{11} \text{ Pa}$.

1.6. $R_c - R_p \approx 10 \text{ km}$.

1.7. $v = (v_E + \sqrt{v_0^2 - 2gR_E})^2 - 2v_E$, where v_E and R_E are the orbital velocity of the Earth and its radius, respectively.

1.8. The potential energy of the apparatus U can be written in terms of its distance x from the Moon's centre

$$U(x) = -\frac{\gamma m M_E}{L-x} - \gamma \frac{M_M m}{x},$$

where $M_E = 6 \times 10^{24}$ kg, $M_M = 7.3 \times 10^{22}$ kg and m are the masses of the Earth, Moon, and apparatus, respectively, while $L = 3.8 \times 10^8$ m is the distance between the centres of the Earth and the Moon. The function $U(x)$ has a maximum ($dU/dx = 0$) at $x = \frac{L}{1 + (M_E/M_M)^{1/2}}$ being equal to

$U_{\max} = -\gamma m \frac{1}{L(M_M^{1/2} + M_E^{1/2})^2}$. If the kinetic energy T of the apparatus at this point vanishes then it is $T =$

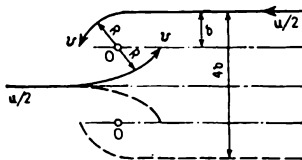


Fig. 56

$U_{\max} - U(R_M)$ near the lunar surface, where $R_M = 1.7 \times 10^6$ m is the lunar radius. The velocity of the apparatus near the lunar surface is

$$\begin{aligned} v = v_{\min} &= \left[\frac{2}{m} (U_{\max} - U(R_M)) \right]^{1/2} \\ &= \left\{ 2\gamma \left[\frac{M_M}{R_M} + \frac{M_E}{L-R_M} - \frac{(M_M^{1/2} + M_E^{1/2})^2}{L} \right] \right\}^{1/2} \\ &\approx \left(2\gamma \frac{M_M}{R_M} \right)^{1/2} \left[1 - \left(\frac{M_E}{M_M} \right)^{1/2} \frac{R_M}{L} \right] \approx 7.5 \text{ km/s.} \end{aligned}$$

1.9. A collision is a process during which the centres of two stars pass each other closer than the sum of their radii (in our case, closer than $2R_S$).

To determine the collision cross section, let us turn to the centre of mass system of the colliding stars. Let b be the maximum impact parameter (relative to the centre of mass) of this collision. Then, as it is seen from Fig. 56, the collision cross section is $\sigma = \pi (2b)^2$. It follows from the

laws of conservation of energy and momentum that

$$2 \frac{M_s}{2} \left(\frac{u}{2} \right)^2 = 2 \frac{M_s v^2}{2} - \frac{\gamma M_s^2}{2 R_s}, \quad M_s b \frac{u}{2} = M_s R_s v,$$

where v is the velocity of each star relative to a fixed centre O when these stars are at their closest. Eliminating v , we obtain

$$\frac{b^2}{R_s^2} = 1 + \frac{2\gamma M_s}{R_s u^2} \approx \frac{2\gamma M_s}{R_s u^2} \approx 10^2.$$

Hence, the collision cross section is

$$\sigma = 4\pi b^2 = 8\pi \frac{R_s \gamma M_s}{u^2}.$$

Since the mean free path length $L = 1/(n\sigma)$, where $n = N/(cT)^3$ (c is the velocity of light, $T = 1$ year $= 3.2 \times 10^7$ s), then the time interval between collisions for one star is

$$\tau = \frac{1}{n\sigma u} = \frac{u (cT)^3}{8\pi N R_s \gamma M_s} = 2.2 \times 10^{21} \text{ s} = 7 \times 10^{13} \text{ years},$$

viz. one collision per cubic light year occurs once every

$$t = \frac{\tau}{N/2} = 1.4 \times 10^{13} \text{ years}.$$

1.10. For motion in a gravitational field, the following relations are valid:

$$T^2 b^{-3} = \text{const}, \quad bE = \text{const},$$

where T is the period of revolution, b the major semiaxis of the orbit, and E the total energy. Hence we have $T^2 E^3 = \text{const}$, or for small variations of the period and velocity v we obtain

$$\frac{\Delta T}{T} = -\frac{3}{2} \frac{\Delta E}{E} = \frac{3}{2} \frac{m v \Delta v}{E},$$

where m is the satellite mass. Therefore, because E is negative, to decrease the period of revolution we must also decrease the satellite velocity ($\Delta v < 0$). As a result, the satellite will pass into an elliptical orbit with a smaller value of b and in n turns it will travel from its initial position on the circular orbit by a distance

$$L = -n v \Delta T = \frac{3}{2} n \frac{m v^3}{E} T \Delta v = -3n T \Delta v$$

(Here we have taken into account that for circular orbits $E = -mv^2/2$.) By increasing the satellite velocity by $-\Delta v$ it will again be possible to transfer the satellite back to its initial circular orbit so that the relative velocity of the satellites will vanish. Determine n and Δv at $L = 45$ km and $T = 1.5$ h:

$$\Delta v = -\frac{L}{3\pi T} = -\frac{10}{n} \text{ km/h.}$$

At $n = 1$, $|\Delta v| > 8$ km/h, and at $n = 2$, $|\Delta v| = 5$ km/h. Thus, $n_{\min} = 2$. To dock the satellites, the velocity of the satellite that lags behind B must be decreased by 5 km/h and two turns later restored to the initial value.

1.11. The thermal velocity of atmospheric molecules is much smaller than the satellite velocity, $v = 8$ km/s. Under such conditions, the collisions between molecules and the satellite are mainly inelastic. It might be, therefore, thought that after such a collision a molecule gains a velocity v . Accordingly, the atmospheric molecules that collide with the satellite acquire a momentum $S\rho_A v^2$ per second, viz., the frictional force acting on the satellite is

$$F = S\rho_A v^2 = 5 \times 10^{-4} \text{ N.}$$

The total energy of the satellite on a circular orbit of radius R is

$$E = -\frac{mv^2}{2} = -\frac{\gamma m M_E}{2R},$$

where M_E and m are the masses of the Earth and satellite, respectively. Whence, upon changing the energy by ΔE , the orbit radius changes by $\Delta R = 2R^2 \Delta E / (\gamma m M_E)$, whereas the velocity is changed by $\Delta v = -\Delta E / mv$. The satellite energy will change due to the frictional forces by $\Delta E = -2\pi R F$ per revolution. Thereby the orbit radius will be decreased by

$$\Delta R = -\frac{4\pi R^2}{\gamma m M_E} F \approx -\frac{3F}{\gamma m \rho} \approx 0.4 \text{ km.}$$

Here $\rho = 5500 \text{ kg/m}^3$ is the mean density of the Earth. The satellite velocity is incremented by

$$\Delta v = \frac{2\pi R}{m v} F \approx 0.5 \text{ m/s.}$$

1.12. The projectile velocity v can be obtained from the law of conservation of angular momentum:

$$I\omega = \frac{mvR_E}{\sqrt{1-v^2/c^2}},$$

where $\omega = 7.3 \times 10^{-5} \text{ s}^{-1}$ is the angular velocity of the Earth's rotation. Whence

$$c - v = \frac{m^2 v^2 c^2 R_E^2}{(c + v) I^2 \omega^2}.$$

Since v is only slightly different from c , a good approximation is

$$c - v \approx \frac{m^2 c^3 R_E^2}{2 I^2 \omega^2} = \frac{9}{2} \left(\frac{mc}{M_E R_E \omega} \right)^2 c = 1.56 \times 10^{-17} \text{ m/s}.$$

Since the projectile moves with an ultrarelativistic velocity, its kinetic energy T (which coincides with the total energy E) can be calculated from the formula

$$T \approx \frac{mc^2}{\sqrt{1-v^2/c^2}} \approx \frac{I\omega c}{R_E}.$$

The ratio between this quantity and the kinetic energy of the Earth's rotation, $T_E = I\omega^2/2$ is

$$\frac{T}{T_E} = \frac{2c}{\omega R_E} = 1.3 \times 10^8.$$

1.13. The change in the electromagnetic field momentum per second is equal to the thrust $F = aM$. Accordingly, the power consumed,

$$N = Fc = aMc = 3 \times 10^{13} \text{ W}$$

(c is the velocity of light) is 670 times greater than that of the Bratsk hydroelectric power station.

1.14. $h = \frac{m^2 p}{\rho g M^2}$, where ρ is the density of water.

1.15. See the solution of Problem 1.2. For a neutron star,

$$T = (3\pi/\gamma\rho)^{1/2} = 1.2 \times 10^{-3} \text{ s}.$$

1.16. On a circular orbit of radius R , the total (E), potential (U), and kinetic (T) energies are related through the relation

$$E = \frac{U}{2} = -T = -\frac{\gamma mM}{2R}.$$

After the explosion, the total energy E' will be

$$E' = T + U(1 - q) = -\frac{\gamma m M}{R} \left(\frac{1}{2} - q \right).$$

At $q > 1/2$ this energy exceeds zero and the satellite leaves the star via a hyperbolic orbit. At $q = 1/2$, $E = 0$ and the satellite's orbit will be a parabola. At $q < 1/2$ the satellite will follow an elliptical orbit, one of whose foci will contain the remainder of the central star, while the major semiaxis may be found from the well-known condition:

$$2E'b = -\gamma m M (1 - q).$$

Taking into account the expression for E' we obtain

$$b = R \frac{1 - q}{1 - 2q}.$$

1.17. In the process of braking, the spaceship velocity changes from $v_2 = 2.4$ km/s (the escape velocity of the Moon) to $v_1 = v_2/\sqrt{2} = 1.7$ km/s (the orbital velocity of the Moon). As a result, the spaceship mass should be decreased from its initial value M_0 to M

$$M = M_0 \exp \left(-\frac{v_2 - v_1}{u} \right) \approx 0.84 M_0$$

(see the solution of Problem 1.22). Here u is the gas exhaust velocity. Thus, the burnt out fuel should amount to 16% of the spaceship mass. The combustion temperature T can be estimated from the formula for the exhaust of gas velocity into a vacuum, i.e. $u = \sqrt{2c_p T}$, whence

$$T = u^2/2c_p \approx 3600 \text{ K}.$$

1.18. The equations for the vertical (v_z) and horizontal (v_x) components of a rocket velocity have the following form:

$$M\dot{v}_z = -Mu \sin \varphi - g_M M, \quad M\dot{v}_x = -Mu \cos \varphi,$$

where M is the rocket mass at the instant t . According to the statement of the Problem $v_z = 0$, therefore, we have from the first equation

$$\dot{M} = -\frac{M g_M}{u \sin \varphi} = -\frac{M}{T},$$

where T denotes a time-dimensional constant equal to $T = (u \sin \varphi)/g_M \approx 235$ s. Whence

$$M = M_0 \exp \left(-\frac{t}{T} \right).$$

Substituting $\dot{M}$ and M into the second equation, we have

$$\dot{v}_x = g_M \cot \varphi \approx g_M / \varphi \quad \text{or} \quad v_x = g_M t \cot \varphi \approx \frac{g_M}{\varphi} t.$$

The time τ needed to gain orbital velocity $v_1 = \sqrt{R_M g_M} = 1.7 \text{ km/s}$ is

$$\tau = \frac{v_1}{g_M \cot \varphi} \approx \frac{v_1 \varphi}{g_M} \approx 100 \text{ s}.$$

This expression for τ is also valid as $\varphi \rightarrow 0$. This results from the condition $v_x = 0$. As $\varphi \rightarrow 0$ the mass consumption rises infinitely and hence the time needed to boost up to the orbital velocity tends to zero.

The relative change in the rocket mass due to the burnt-out fuel is

$$\frac{M(0) - M(\tau)}{M(0)} = 1 - \exp\left(-\frac{\tau}{T}\right) \approx 0.35.$$

The overload of astronauts is

$$g_{\text{ov}} = \sqrt{g_M^2 + \dot{v}_x^2} \approx \dot{v}_x \approx \frac{g_M}{\varphi} = 17 \text{ m/s}^2.$$

1.19. $a - b \approx R_E \left(\frac{\Delta E}{E} \right) \approx 3 \text{ mm}$, where R_E is the Earth radius, and E the total satellite energy.

1.20. $\varphi \approx 1.3 \times 10^{-17} \text{ rad}$.

1.21. The conservation laws result in $\tan \varphi = v_t / v_r = 1/2\sqrt{3}$, where v_t and v_r are the tangential and radial components of velocity of the launching-pad.

1.22. The equation of motion of a rocket in the gravitational field of the Moon has the form

$$m \left(\frac{dv}{dt} \right) = m g_M + \frac{dm}{dt} u.$$

Its integration over time t yields (g_M is the free-fall acceleration on the Moon)

$$m(\tau) = m_0 \exp\left(-\frac{v_0 + g_M \tau}{u}\right),$$

where $m(\tau)$ is the rocket mass after the engine has fired for a time interval τ . Thus, the fuel supply should be

$$m_{\text{fuel}} = m_0 \left[1 - \exp\left(-\frac{v_0 + g_M \tau}{u}\right) \right].$$

1.23. Since the oscillation period of a pendulum is $T \sim (1/\sqrt{g})$, then $\frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta g}{g}$, where Δg is the magnitude of the

gravitational field strength due to the debris extracted from the tunnel. Assuming the tunnel is sufficiently long, one can easily find Δg from Gauss's theorem. As a result,

$$\Delta g = \frac{2\pi\gamma\rho R^2}{h},$$

where γ is the gravitational constant. A numerical evaluation leads to the result: $\Delta T/T \sim 10^{-7}$.

1.24. $\varphi = 8/\Delta/25h \approx 0.06$ rad.

1.25. $\Delta/\Delta_0 \approx 5$ times, where Δ is the net sag for a fall from a height h .

1.26. The laws of conservation of energy and momentum suggest that

$$\frac{mv_0^2}{2} = \frac{mv^2}{2} + \frac{Mu_1^2}{2} + \frac{Mu_1^2}{2}; \quad mv_0 = Mu_1, \quad mv = Mu_1,$$

where u_1 and u_1 are the transverse and longitudinal velocity components. Whence

$$M > m, \quad v = v_0 \sqrt{\frac{M-m}{M+m}},$$

$$u = \frac{mv_0}{M} \sqrt{\frac{2M}{M+m}}, \quad \tan \varphi = \sqrt{\frac{M-m}{M+m}}.$$

1.27. According to the law of conservation of momentum in the case of γ -decay we have

$$Mv_2 = \frac{E_\gamma}{c}.$$

Here $M = 226 \times 1.66 \times 10^{-27} \text{ kg} = 3.75 \times 10^{-25} \text{ kg}$ is the mass of a ^{226}Ra nucleus, E_γ the γ -quantum energy practically coincident with the total energy E_2 released in the γ -decay, and c the velocity of light in vacuum. Whence for the kinetic energy T_2 of the recoil nucleus we have

$$T_2 = \frac{E_2^2}{2Mc^2} = 0.095 \text{ eV}.$$

In the case of α -decay the laws of conservation of energy and momentum yield

$$(M - M_\alpha)v_1 = M_\alpha v_\alpha, \quad \frac{M - M_\alpha}{2}v_1^2 + \frac{M_\alpha}{2}v_\alpha^2 = E_1,$$

where v_α and v_1 are the velocities of the α -particle and recoil nucleus, respectively, and M_α the α -particle mass. Whence, by eliminating v_α , for the kinetic energy T_1 of the

recoil nucleus we obtain the following

$$T_1 = \frac{M_\alpha}{M} E_1 = 87 \text{ keV}.$$

Finally, the ratio of the kinetic energies is

$$\frac{T_1}{T_2} = \frac{2M_\alpha c^2 E_1}{E_1^2} = 9 \times 10^5.$$

1.28. The maximum energy which may be spent on the ionization corresponds to the kinetic energy lost in an absolutely inelastic head-on collision. (This result can be obtained in the centre of mass frame.) The lost energy is

$$\Delta T_{\max} = \frac{M_{133}}{M_{133} + M_{16}} T_0,$$

where M_{133} and M_{16} are the masses of the cesium-133 and oxygen-16 atoms, respectively. $\Delta T_{\max} = 3.56 \text{ eV} < 3.9 \text{ eV}$, viz. ionization is impossible.

1.29. $T = \frac{M_{14} + M_\alpha}{M_{14}} U > 2.97 \text{ MeV}$, where M_{14} and M_α are the masses of the nitrogen atom and α -particle, respectively.

1.30. $\frac{\Delta T}{T} \approx -\frac{I}{2I_0} \approx -10^{-3}$, viz. the watch runs faster.

1.31. $v \approx 4 \times 10^{12} \text{ cm/s} \gg c$ (v is the velocity on the "surface" of the ball).

1.32. $L = I \cdot \omega \approx 10^8 \text{ g} \cdot \text{cm}^2/\text{s}$, where I is the moment of inertia of the wheel.

1.33. Let the mass of the man be m and the swing velocity, as it passes the equilibrium position be v_0 , which is related to the initial amplitude φ_0 by the relation $v_0^2 \approx gl\varphi_0^2$ (assuming that the angles of deviation of the swing from the equilibrium position are small). The total energy of the swing at this instant is $E_0 = mv_0^2/2$. After the man has risen, the total energy is

$$E_1 = \frac{mv_1^2}{2} + mgh \approx \frac{mg(l-h)\varphi_1^2}{2} + mgh,$$

where φ_1 is the new amplitude. The change in the total energy has occurred due to the work performed by the man against the centrifugal and gravity forces: $E_1 - E_0 \approx mgh + (mv_0^2/l)h$. Comparing this expression with the differ-

once $E_1 - E_0$ found earlier we get

$$\varphi_1^2 \left(1 - \frac{h}{l}\right) = \varphi_0^2 \left(1 + 2 \frac{h}{l}\right) \quad \text{or} \quad \varphi_1 - \varphi_0 \approx \frac{3}{2} \varphi_0 \frac{h}{l},$$

which represents the increment of angular amplitude per half period. Hence the sought increment is (per period)

$$\Delta\varphi \approx 3\varphi_0 \frac{h}{l}.$$

1.34. Let the coordinate x -axis be directed along the normal to the surface of the racket while the y -axis be parallel to the surface (in the plane of incidence) (Fig. 57). In the coordinate system related to the Earth let the ball velocity

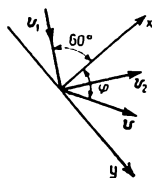


Fig. 57

be v_1 ($v_{1x} = v_1/2$, $v_{1y} = \sqrt{3} v_1/2$) and the racket velocity be v . The racket is assumed to move translationally. Then in the coordinate system related to the racket, the velocity v'_1 will have the following components

$$v'_{1x} = v_{1x} - v_x, \quad v'_{1y} = v_{1y} - v_y.$$

After the collision, the ball velocity in the moving coordinate system will be

$$v'_{2x} = -v'_{1x}, \quad v'_{2y} = v'_{1y}$$

(mirror reflection). Returning again to the rest system the expressions for the components of the ball velocity v_2 after the reflection from the racket are

$$v_{2x} = -v_{1x} + 2v_x, \quad v_{2y} = v_{1y}.$$

Under the statement of the problem, $(v_1 v_2) = 0$, viz.

$$v_{1x} (-v_{1x} + 2v_x) + v_{1y}^2 = 0.$$

The solution of this equation has the form of

$$v_x = -v_{1x}, \text{ where } v_y \text{ is arbitrary.}$$

It should be stressed, however, that when the racket hits the ball the condition of mirror reflection is strictly fulfilled only in the case of a normal fall (in the coordinate system related to the racket). This is realized at $v_y = v_{1y}$.

therefore the final answer has to be written in the form

$$v = v_1, \quad \varphi = 60^\circ.$$

1.35. Consider the coordinate system associated with the rotating bar. In this system, equilibrium can be written down as (Fig. 58)

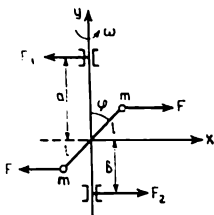


Fig. 58

$$F_1 - F_2 = 0,$$

$$F_1 a + F_2 b = 2Fl \cos \varphi.$$

Here $F = ml \sin \varphi \cdot \omega^2$ is the centrifugal force of inertia, F_1 and F_2 are the reaction forces of the ball bearings. From these equations we find

$$F_1 = F_2 = \frac{ml^2 \omega^2 \sin 2\varphi}{a+b}.$$

1.36. The moment of the elastic forces is $M_{el} = -f\varphi$, where φ is the angle of rotation of the disc. The moment of the internal friction forces in the gas is

$$M_{fr} = - \int_0^R \eta \frac{v(r)}{h} r \cdot 2\pi r dr,$$

where $v(r) = \dot{\varphi}r$ is the velocity of a point in the disc at a distance r from the centre. Elementary integration yields

$$M_{fr} = -2k\dot{\varphi}, \quad \text{where } k = \frac{\pi\eta R^4}{4h}.$$

As h increases, the moment M_{fr} falls off rapidly. This allows us, in the first approximation, to neglect the friction on the upper disc surface.

The equation of motion is

$$I\ddot{\varphi} + 2k\dot{\varphi} + f\varphi = 0.$$

Here $I = mR^2/2$ is the moment of inertia of the disc. Let us seek for a solution of this equation in the form

$$\varphi = \varphi_0 e^{-\delta t} e^{\pm i\omega t}.$$

Then we obtain

$$\delta = \frac{k}{I}, \quad \omega = \frac{2\pi}{T} = \sqrt{\frac{f}{I} - \delta^2} \approx \sqrt{\frac{f}{I}},$$

where T is the oscillation period. The logarithmic decrement is

$$d = \delta T \approx \frac{2\pi k}{\sqrt{I}} = \frac{\pi^2 \eta R^3}{h \sqrt{2fm}},$$

hence

$$\eta = \frac{dh \sqrt{2fm}}{\pi^2 R^3}.$$

$$1.37. \omega_{0 \min} = \frac{\nu_0}{R}, \quad \nu = \frac{\omega_0 R - \nu_0}{2}.$$

$$1.38. T_{\text{trans}} = \frac{p^3}{2m}, \quad T_{\text{rot}} = \frac{5p^3 k^2}{4m}, \quad k = 2/5.$$

1.39. $x \approx h/f$, where h is the height the stool centre of gravity is lifted and f is the friction coefficient.

$$1.40. \varphi = \arctan \frac{2\nu^3}{gR}.$$

1.41. The force moment is $M = Fl = \Omega L$, where $\Omega = \frac{2\pi}{T} \varphi$ and the angular momentum is $L = I \cdot 2\pi\nu$. Whence

$$F = \frac{4\pi^2 \varphi \nu I}{Tl} \approx 1.3 \times 10^4 \text{ N}.$$

$$1.42. k = \frac{m|rl|}{l}.$$

$$1.43. h = \frac{\varphi l (1 + 3 \cos^2 \varphi)^2}{12 \cos \varphi (1 - 3 \cos^2 \varphi)}; \quad 3 \cos^2 \varphi < 1.$$

1.44. The angle of rotation φ of the pendulum satisfies the equation $\ddot{\varphi} + f\varphi = 0$, where $f = \frac{\pi G D^4}{32 l}$ is the torsion modulus of the thread. According to the law of conservation of angular momentum, the initial conditions have the form

$$\varphi(0) = 0, \quad \dot{\varphi}(0) = 2 \frac{E}{c} r,$$

where E is the energy radiated by the quantum generator. The solution of the equation of motion may be represented in the form

$$\varphi = \varphi_0 \sin \omega t, \quad \text{where } \varphi_0 = \frac{\dot{\varphi}(0)}{\omega} \text{ and } \omega = \sqrt{\frac{f}{I}}.$$

For the sensitivity $\eta = 2\varphi_0/E$ we obtain

$$\eta = \frac{4r}{\omega c I}.$$

Whence the period of oscillations,

$$T = \frac{2\pi}{\omega} = \frac{\pi c \eta l}{2r} = 19 \text{ s.}$$

Furthermore, since $T = 2\pi\sqrt{I/f}$, we have

$$D = 4 \left(\frac{\pi I l}{2G r^3} \right)^{1/4} = 22 \text{ } \mu\text{m.}$$

1.45. The velocity of elastic waves s is $\sqrt{E/\rho} = 5 \text{ km/s}$. For estimation, we assume the collision time τ to be the time it takes a wave to travel from one end of the bar to the

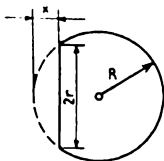


Fig. 59

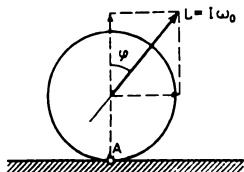


Fig. 60

other and back again, viz. $\tau \approx 2l/s = 4 \times 10^{-5} \text{ s}$. Inelastic phenomena will emerge when at some points along the bar the pressure reaches p , viz. the potential energy per unit volume reaches the value $p^2/2E$. Such points are those where the bars come into contact. At the instant of impact all the kinetic energy (per unit volume) at these points will be converted into potential energy.

Thus, $\rho \frac{v^2}{2} \approx \frac{p^2}{2E}$ and hence

$$v \approx \frac{p}{\sqrt{\rho E}} = \frac{ps}{E} = 5 \text{ m/s.}$$

1.46. For simplicity, let us neglect the rigidity of the cover and the bladder and assume that the ball deformation can be characterized by a single parameter x (Fig. 59) which is much smaller than the ball radius R (a soft kick). Then in a good approximation one may assume that the excess pressure p in the football during the kick remains constant. The force acting on the football from the wall, $F = pS$ (S is the area of the contact between the football and wall),

can be represented as

$$F = \pi p r^3 \approx 2\pi p R x.$$

Since the force F is proportional to the football deformation x , the equation of motion of the football near the wall will have the form of an oscillatory equation:

$$\ddot{x} + \omega^2 x = 0, \quad \omega^2 = \frac{4\pi^3}{\tau^2} = \frac{2\pi p R}{m},$$

where m is the football mass. The collision time τ is then half an oscillation period, viz. $\tau \approx (2\pi m/pR)^{1/2}$. Taking for estimates $m = 0.5$ kg, $p = 10^5$ Pa, and $R = 0.1$ m, we obtain $\tau \approx 2 \times 10^{-2}$ s.

1.47. The moments of the external forces (those of gravity and friction) relative to point A (Fig. 60) vanish. Hence,

$$I\omega_0 \cos \varphi = \text{const}, \quad I\omega_0 \sin \varphi = I\omega_{\text{hor}} + mvR \\ = (I + mR^2) \omega_{\text{hor}},$$

where $v = \omega_{\text{hor}} R$ is the velocity of the ball under steady-state pure rolling. Whence

$$\omega_{\text{vert}} = \omega_0 \cos \varphi, \quad \omega_{\text{hor}} = \frac{I\omega_0 \sin \varphi}{I + mR^2} = \frac{2}{7} \omega_0 \sin \varphi,$$

$$v = \frac{2}{7} \omega_0 R \sin \varphi, \quad \omega = (\omega_{\text{hor}}^2 + \omega_{\text{vert}}^2)^{1/2}.$$

1.48. In the problem at hand one should distinguish between two cases.

(1) The slipping persists during the impact time τ . Then

$$mv_{\text{vert}} = f \int_0^\tau N(t) dt,$$

where N is the reaction of the wall. On the other hand, the change in the horizontal momentum component is

$$2mv_0 = \int_0^\tau N(t) dt.$$

Hence,

$$v_{\text{vert}} = 2fv_0, \quad \tan \alpha = 2f.$$

It may be easily shown that the condition of slipping at the end of the impact, viz. $\omega(r) > v_{\text{vert}}/r$, leads to the condition $f < 1/7$.

(2) Pure rolling sets in before the end of the impact, viz. at $\tau_1 < \tau$, this taking place for $f > 1/7$. Then

$$\omega(\tau_1) = \frac{v_{\text{vert}}}{r}, \quad mv_{\text{vert}} = \int_0^{\tau_1} N(t) dt.$$

The change in the angular momentum of the ball during this period is

$$I\omega(\tau_1) - I\omega_0 = -rf \int_0^{\tau_1} N dt,$$

or

$$\frac{v_{\text{vert}}}{r} - \frac{v_0}{r} = -\frac{rfmv_{\text{vert}}}{I},$$

whence it follows that

$$v_{\text{vert}} = \frac{2}{7} v_0, \quad \text{viz.} \quad \tan \alpha = \frac{2}{7}.$$

Thus

$$\tan \alpha = 2f \quad \text{at} \quad f < 1/7,$$

$$\tan \alpha = \frac{2}{7} \quad \text{at} \quad f > 1/7.$$

$$1.49. \quad t = \frac{S}{\sigma} \left(\frac{2h}{g} \right)^{1/2}.$$

$$1.50. \quad t = \frac{S}{\sigma} \frac{h-h_0}{(2gh_0)^{1/2}}.$$

1.51. The relative change in volume is

$$\frac{\Delta V}{V} = \frac{3(1-2\mu)}{E} p,$$

where p is the pressure, $\Delta V = m \left(\frac{\rho_w - \rho}{\rho_w \rho} \right)$ (ρ_w is the density of water and m is the mass). Hence

$$\frac{\Delta V}{V} = \frac{\rho_w - \rho}{\rho_w}, \quad p = \frac{E(\rho_w - \rho)}{\rho_w \cdot 3(1-2\mu)} \approx 2 \times 10^9 \text{ Pa.}$$

1.52. For gravitational waves, the velocity c and the length l are related as $c \approx \sqrt{gl}$. Therefore the wave patterns will be similar if all the dimensions are changed in proportion to the squared velocity of motion. Hence, the velocity of the model must be $3.6 \text{ km/h} = 1 \text{ m/s}$. Notice that in the given problem the dimensionless similarity parameters are the

ratios c/v and l/L , where v is the velocity of the ship and L its linear size.

1.53. The phase velocity of surface waves depends on the liquid density ρ , the free-fall acceleration g , and the wavelength λ : $v \sim \rho^a g^m \lambda^p$. Comparing the dimensions of the left- and right-hand parts of this equation, we obtain $n = 0$, $m = 1/2$, $p = 1/2$. Thus, $v = A\sqrt{g\lambda}$, where A is a numerical dimensionless factor.

II. Thermodynamics and Molecular Physics

2.1. Using the first law of thermodynamics, the Mondoleov-Clapeyron equation, and the equation of the process (see Fig. 5), we get

$$C(V) = R \frac{\gamma - (\gamma + 1)V/V_0}{(\gamma - 1)(1 - 2V/V_0)}, \quad T_{\max} = p_0 V_0 / 4R.$$

At the point $C(V) = 0$ the straight line in Fig. 5 is tangent to an adiabatic curve, and to an isothermic curve at $C(V) = \infty$.

2.2. According to the problem $dQ = -dU$, where U is the internal energy. Therefore the heat capacity $C = dQ/dT = -C_V$. Using this relationship, the first law of thermodynamics, and the Mendelev-Clapeyron equation we find the equation of the process:

$$TV^{(\gamma-1)/2} = \text{const},$$

and the expression for the work of the external forces expended in compressing the gas from a volume V_0 to V is

$$-A = 2C_V T_0 \left[\left(\frac{V_0}{V} \right)^{(\gamma-1)/2} - 1 \right].$$

At $V = V_0/2$ the work is

$$-A = 2C_V T_0 (2^{(\gamma-1)/2} - 1).$$

2.3. With an arbitrary equation of state $p = p(V, T)$,

$$dp = \left(\frac{\partial p}{\partial V} \right)_T dV + \left(\frac{\partial p}{\partial T} \right)_V dT.$$

In an isobaric process $dp = 0$, the ratio of differentials dV/dT is equal to the partial derivative $(\partial V/\partial T)_p$ and therefore

$$\left(\frac{\partial p}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_p \left(\frac{\partial T}{\partial p} \right)_V = -1.$$

By introducing in the usual fashion

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p, \quad \gamma = \left(\frac{\partial p}{\partial T} \right)_V, \quad \alpha = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T,$$

we obtain $\beta = \gamma \cdot \alpha$.

2.4. The conventional formula for the velocity of an adiabatic jet exhausting into a vacuum is $v_0 = \sqrt{2C_p T/\mu}$, where μ is the molar mass, and C_p the molar heat capacity. Using the equation of state $p = \rho RT/\mu$ and the formula for the velocity of sound $s^2 = \gamma p/\rho$, we obtain $v = s\sqrt{2/(\gamma - 1)}$.

2.5. According to the Tsiolkovskii formula $M_0/M = e^{v_0/v_s}$, where $v_0 = \sqrt{2C_p T/\mu}$ is the exhaust velocity of the gas in a rocket system (see the previous problem). Therefore

$$\frac{M_0}{M} = \exp \left\{ v_s \sqrt{\frac{\gamma-1}{2\gamma} \frac{\mu}{RT}} \right\} \approx 11.3.$$

2.6. A direct calculation shows that the efficiency of a cycle composed of two adiabats and two isochores is $\eta = 1 - \alpha^{1-\gamma}$, where α is the compression ratio. As a result, given $\gamma = 4/3$

$$\frac{\eta_2}{\eta_1} = \frac{1 - \alpha_2^{1-\gamma}}{1 - \alpha_1^{1-\gamma}} \approx 1.29.$$

2.7. Taking a value $\Delta T \approx 50$ K as an estimate of the difference between the temperatures at the tropics and the poles we obtain from Carnot's theorem $\eta_{\max} = \Delta T/T_1 \approx 0.16$. The power N converted per second into wind energy is

$$N = \frac{\eta Q}{4\pi R_E^2} = 0.1 \frac{\eta_{\max} Q}{4\pi R_E^2} \approx 5 \times 10^6 \text{ W/km}^2,$$

where R_E is Earth's radius. Causes for the difference between η and $\eta_{\max}$ are the absorption of solar energy at high altitudes, excess thermal radiation (viz. exceeding the average level) from low altitudes, thermal conductivity, and the incomplete absorption of solar radiation by the atmosphere.

2.8. Let us assume the opposite and arrange for two polytropes A and B to go between two adjacent states (1 and 2) (Fig. 61a). We then form a closed cycle from A and B , and by integrating over the cycle and taking into account that $dQ = C dT$ on the polytropes we obtain

$$\oint dS = (C_A - C_B) \ln \frac{T_2}{T_1}, \quad A = \oint dQ = (C_A - C_B) (T_2 - T_1).$$

Here C_A and C_B are the heat capacities on the polytropes A and B . The first integral vanishes since entropy is a function of state and therefore an integral of it along a closed contour also vanishes. We can immediately see that the second integral vanishes automatically too, irrespective of what causes the first integral to vanish ($C_A = C_B$ or $T_1 = T_2$). But since the work A is equal to the area on the p - V -plane enclosed by the cycle, the curves A and B must coincide.

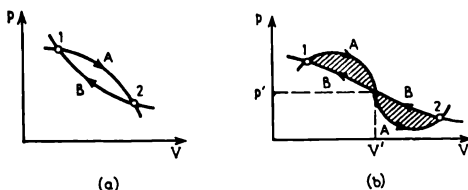


Fig. 61

If the function $T(p, V)$ is not single-valued, this proof becomes invalid since the situation shown in Fig. 61b, is possible. Here at the point (p', V') the polytropes are associated with different temperatures, while $\oint p dV$ vanishes because the two hatched regions cancel each other out. As to the contour integrals covering the hatched regions, both $\oint dS$ and $\oint dA$ are nonzero since these contours are not a closed cycle.

2.9. It can easily be seen that the first law of thermodynamics can be represented as

$$dQ = (C_p - C_v) \left(\frac{\partial T}{\partial V} \right)_p dV + C_v dT.$$

In fact, for $V = \text{const}$ the right-hand side cancels to $C_v dT$ while for $p = \text{const}$ and when $dV = (\partial V / \partial T)_p dT$, the right-hand side becomes $C_p dT$. Whence it follows that for an adiabatic process

$$\left(\frac{dT}{dV} \right)_{\text{ad}} = \left(\frac{C_p}{C_v} - 1 \right) \left(\frac{\partial T}{\partial V} \right)_p,$$

therefore

$$\frac{C_p}{C_v} \approx 1 + \beta \frac{\Delta T}{\Delta V/V} = 1.1.$$

Changing the variables in the first formula from V, T to p, T and taking into account that

$$\left(\frac{\partial T}{\partial V}\right)_p \left(\frac{\partial V}{\partial p}\right)_T \left(\frac{\partial p}{\partial T}\right)_V = -1$$

we obtain

$$dQ = C_p dT - (C_p - C_v) \left(\frac{\partial T}{\partial p}\right)_V dp$$

whence the pressure increment

$$\Delta p \approx \left(\frac{\partial p}{\partial T}\right)_V \frac{C_p/C_v}{C_p/C_v - 1} \Delta T \approx 1.1 \times 10^6 \text{ N/m}^2.$$

2.10. Using the standard expression

$$T dS = dW - V dp = \left[\left(\frac{\partial W}{\partial p}\right)_T - V \right] dp + \left(\frac{\partial W}{\partial T}\right)_p dT,$$

where W is the enthalpy, and the fact that dS is a full differential we obtain the thermodynamic identity

$$\left(\frac{\partial W}{\partial p}\right)_T - V = -T \left(\frac{\partial V}{\partial T}\right)_p.$$

For an adiabatic process ($dS = 0$) we have,

$$\left(\frac{\partial T}{\partial p}\right)_S = -\frac{T}{c_p} \left(\frac{\partial V}{\partial T}\right)_p.$$

For stretching the wire, in an analogous fashion, we find that

$$\left(\frac{\partial T}{\partial F}\right)_S = -\frac{\alpha l T}{C},$$

where $C = cpl\sigma$ (σ is the area of wire's cross section and l its length). Finally

$$\Delta T = -\frac{\alpha T F}{c p \sigma} \approx -0.03 \text{ K}.$$

2.11. The energy balance that includes the potential energy of the piston and the condition governing the piston's mechanical equilibrium will be

$$2C_p (T' - T) = \frac{mg}{\pi r^2} \Delta V, \quad 2 \frac{T'}{T} \frac{\Delta V/V}{1 - (\Delta V/V)^2} = \frac{mg}{\pi r^2 p}$$

where T' is the gas's final temperature, C_p the heat capacity of ν gas moles, $2V$ the total volume of the cylinder, and ΔV the volume increment in the upper section during a down-stroke. The equations demonstrate that, given that $mg \ll \pi r^2 p$, we will have $\Delta V \ll V$ and $(T' - T) \ll T$. Assuming

in the second equation that $T' \approx T$ and omitting $(\Delta V/V^2)$ in the denominator, we obtain

$$\frac{T' - T}{T} \approx \frac{1}{4} \frac{\nu R}{C_v} \left(\frac{mg}{\pi r^2 p} \right)^2; \quad \frac{\Delta V}{V} \approx \frac{1}{2} \frac{mg}{\pi r^2 p}.$$

Using the same approximation we get for the change in entropy:

$$\Delta S \approx \frac{\nu R}{4} \left(\frac{mg}{\pi r^2 p} \right)^2.$$

2.12. For a reversible isothermal process ($T = \text{const}$), the second law of thermodynamics yields $dA = -dF$, where $F = U - TS$ is the free energy. Therefore $A_{\text{max}} = -\Delta F$. Since for an ideal gas in an isothermal process $U = \text{const}$, we have $A_{\text{max}} = T \Delta S$. Whence it immediately follows that

$$A_{\text{max}}^{T=\text{const}} = RT \left(\nu_1 \ln \frac{V_1 + V_2}{V_1} + \nu_2 \ln \frac{V_1 + V_2}{V_2} \right) \approx 1.8 \text{ kJ}.$$

For a reversible adiabatic process ($dQ = 0$), the first law of thermodynamics yields $A_{\text{max}} = -\Delta U$. This process is isentropic and condition $S = \text{const}$ results in the following equation for the final temperature T' :

$$(\nu_1 C_{V1} + \nu_2 C_{V2}) \ln \frac{T'}{T} + \frac{A_{\text{max}}^{T=\text{const}}}{T} = 0.$$

Whence $T' \approx 225 \text{ K}$. The adiabatic work is

$$\begin{aligned} A_{\text{max}}^{S=\text{const}} &= U(T) - U(T') = (\nu_1 C_{V1} + \nu_2 C_{V2}) (T - T') \\ &= \frac{T - T'}{T \ln(T/T')} A_{\text{max}}^{T=\text{const}} \approx \frac{T + T'}{2T} A_{\text{max}}^{T=\text{const}} \approx 1.6 \text{ kJ}. \end{aligned}$$

2.13. Each piston performs work isothermally compressing the gas for which it is not "transparent" from the volume V_0 to $V' = V_0/2$; the second gas passes freely through the piston. The total work of the external forces in this process is

$$A = 2\nu RT \ln \left(\frac{V_0}{V'} \right) = 2 \ln 2 \cdot \nu RT.$$

A direct calculation of the change in the total entropy (as a sum of the changes in the entropies of both gases) shows that $\Delta S = -A/T$, according to the general theory (see the previous problem).

2.14. Let us enclose a bag of volume V_1 containing ν_1 helium moles inside a larger volume V_2 containing ν_2 heli-

um moles and v air moles at the same temperature T and total pressure p (for simplicity we do not take into account that the air itself is a gas mixture, viz. the result is independent of this fact). The entropy in the initial state, when helium is in the bag, will be

$$S_1 = Rv_1 \ln \frac{V_1}{v_1} + Rv_2 \ln \frac{V_2}{v_2} + Rv \ln \frac{V_3}{v},$$

while in the final state, when all the helium has diffused out, the entropy is

$$S_2 = R(v_1 + v_2) \ln \frac{V_1 + V_2}{v_1 + v_2} + Rv \ln \frac{V_1 + V_2}{v}.$$

The change in entropy, $\Delta S = S_2 - S_1$ should be calculated in the limit when $v_1, v_2, V_2 \rightarrow \infty$ for which V_2/v_1 and V_2/v_2 retain their final values. A simple calculation yields

$$\Delta S = Rv_1 \ln \frac{V_2/v_2}{V_1/v_1} = \frac{pV_1}{T_1} \ln \frac{n_1}{n_2} \approx 110 \text{ J/K},$$

where n_1 and n_2 are the molar concentrations of the helium in the bag and that in the surroundings, respectively. The minimum isothermal work needed to collect the same amount of helium together again is (see problem 2.12):

$$|A|_{\min} = T \Delta S = pV_1 \ln \frac{n_1}{n_2} \approx 32 \text{ kJ}.$$

2.15. If $dQ = 0$ and $p = \text{const}$, enthalpy will be preserved. Before relaxation $C_p = 7/2R$, and after it $C'_p = 9/2R$, therefore $T_2 = 7/9 T_1$. The change in entropy ΔS is the sum of ΔS_{ext} and ΔS_{int} , the entropies due to the external and internal degrees of freedom respectively. In an isobaric process

$$\Delta S_{\text{ext}} = C_p \ln \frac{T_2}{T_1}, \quad \Delta S_{\text{int}} = \int_{T_0}^{T_2} C_0(T) \frac{dT}{T},$$

where $C_0(T)$ is the vibrational specific heat. However, the specific heat of a harmonic oscillator for quantum region (viz. for $kT \ll \hbar\omega$) is exponentially small and for this reason the main contribution to the integral comes from the integration region $T \gtrsim \hbar\omega/k$, where $C_0 \approx R$. Hence, $\Delta S_{\text{int}} \approx R \ln(kT_2/\hbar\omega) \gg \Delta S_{\text{ext}}$ and because of this $\Delta S \approx R \ln(kT_1/\hbar\omega)$, with the same accuracy. Physically, the predominance of the vibrational contribution to ΔS means that during the relaxation a maximum relative temperature change occurs in the vibrational subsystem.

2.16. The derivation of the Clausius-Clapoyron equation shows that the heat of vaporization λ is $T(s_v - s_l)$, where s is the specific entropy. Whence

$$\Delta S = -\frac{\lambda m}{T} = -\frac{\lambda \mu p \Delta V}{RT^2} \approx -0.67 \text{ J/K}.$$

Here the Mendeleev-Clapoyron equation was used and m is the vapour's mass, and μ the molar mass. Since the condition for the liquid and vapour to be in equilibrium is the invariability of the potential $\Phi = U + pV - TS = F + pV$ during a phase transition ($\Phi_v = \Phi_l$), we have

$$\Delta F = -p \Delta V \approx -10 \text{ J}$$

2.17. First of all, it must be demonstrated that adiabats are always steeper than isotherms. Along an adiabat $dS = 0$ and

$$C_v dT + \left[\left(\frac{\partial U}{\partial V} \right)_T + p \right] dV = 0.$$

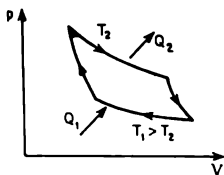


Fig. 62

According to a well-known thermodynamical identity, the bracketed expression is equal to $T(\partial p/\partial T)_v$. Substituting in the full differential of the function $T = T(V, p)$, taking into account the validity of the above relation along an adiabat $S = \text{const}$, and using the identity from Problem 2.3 we obtain

$$\left(\frac{\partial p}{\partial V} \right)_S = \left(\frac{\partial p}{\partial V} \right)_T - \frac{T}{C_v} \left(\frac{\partial p}{\partial T} \right)^2 < \left(\frac{\partial p}{\partial V} \right)_T,$$

viz. an adiabat is steeper than an isotherm for an arbitrary sign of $(\partial p/\partial T)_v$. Hence, as a whole, the cycle has a conventional form (Fig. 62). The (clockwise) direction in which it is followed is dictated by the sign of the work, $A > 0$. From $(\partial p/\partial T)_v < 0$ it follows that the "hot" isotherm T_1 lies below the "cold" one T_2 . Along an isotherm $dS = (\partial p/\partial T)_v dV$ and this is why the elementary heat change leftwards along the lower isotherm ($dV < 0$) is $dQ = T_1 dS > 0$, viz. the system receives heat. When moving along the upper ("cold") isotherm, the system releases heat.

2.18. Using the Van der Waals gas equation we find that p is $5p_{cr}$, and the isothermal compressibility κ is $1/(6p_{cr})$. The

equation for the oscillations of a piston of mass M is

$$\omega^2 = 6S^2 p_{cr} / MV.$$

Since $V = V_{cr} = 3b$ for a Van der Waals gas and the piston equilibrium condition is $Mg = 5p_{cr}S$ we find that

$$\omega^2 = 2gS/5b.$$

2.19. The adiabatic (Laplacian) velocity of sound is $s^2 = (\partial p / \partial \rho)_s$. Differentiating the Van der Waals equation and substituting the critical parameter values ($T_{cr} = 8a/27bR$, the molar volume $V_{cr} = 3b$) for coefficients, we assure ourselves that the coefficient of dV vanishes and $dp = (R/2b) dT$. That this coefficient vanishes means that $(\partial p / \partial \rho)_T = 0$, viz. at the critical point the isothermal (Newtonian) velocity of sound is zero. Differentiating the entropy at the critical point and then assuming $dS = 0$ (an adiabatic process) we find that

$$C_V dT + \frac{4a}{27b^2} dV = 0.$$

Since the density ρ is μ/V , where μ is the molar mass, we finally obtain for the adiabatic velocity of sound

$$s = \sqrt{2aR/3b\mu C_V}.$$

2.20. Let us consider a flat layer in the liquid of thickness $2x$ lying symmetric to the central plane. For stationary motion the flux of the momentum through side faces must equal the difference in the forces acting on the ends, i.e.

$$-2dl\eta \frac{dv}{dx} = 2xlpgh.$$

Integrating this equation with a boundary condition $v(\delta/2) = 0$ we obtain

$$v(x) = \frac{\rho gh}{2d\eta} \left[\left(\frac{\delta}{2} \right)^2 - x^2 \right], \quad v_{\max} = \frac{\rho gh\delta^2}{8d\eta} \approx 2.3 \text{ cm/s.}$$

By integrating over the cross section we determine the liquid flow rate

$$m = \frac{2}{3} \delta l \rho v_{\max} \approx 1.2 \times 10^{-4} \text{ g/s.}$$

2.21. We take a cylindrical layer of liquid with arbitrary inner and outer radii and whose axis is that of the tube.

Since the motion of the liquid in this layer is stationary while the pressures on both ends are the same, the total fluxes of momentum through its inner and outer surfaces are also the same. Therefore the flow of momentum $\Pi(r)$ through a cylindrical surface of radius r , if taken per unit length,

$$\Pi = -2\pi r \eta \frac{dv}{dr} = \text{const}$$

which is independent of r . Here $v(r)$ is the velocity of the liquid. By integrating this relationship we find

$$v + \left(\frac{\Pi}{2\pi\eta} \right) \ln r = \text{const}$$

and using the boundary conditions $v(R_1) = v_0$, $v(R_2) = 0$, we finally obtain

$$v(r) = v_0 \frac{\ln(R_2/r)}{\ln(R_2/R_1)}, \quad F = \Pi = \frac{2\pi\eta v_0}{\ln(R_2/R_1)} = 0.27 \text{ dyn/cm.}$$

2.22. We take a cylinder of liquid of radius r whose axis coincides with that of the tube. The motion of the liquid inside the cylinder is stationary if the momentum flux through the cylinder's surface equals the difference in the forces acting on its ends, i.e. if

$$-2\pi r \eta \frac{dv}{dr} = \frac{\pi r^2 p}{l},$$

where p is the pressure differential between the ends of the tube, $\eta = \nu\rho$ is the dynamic viscosity, and $v(r)$ is the flux velocity. By integrating this equation with a boundary condition $v(R) = 0$ we obtain

$$v(r) = \frac{p}{4\eta l} (R^2 - r^2).$$

By integrating $\rho v(r)$ over the tube's cross section we find the mass flux (Poiseuille's formula)

$$- \frac{dm}{dt} = \frac{p}{8\nu h} \pi R^4,$$

where $h(t)$ is the height of the liquid column, and $p = \rho gh$ is the pressure differential. Integrating this equation yields

$$m(t) = m_0 - \frac{\rho g}{8\nu} \pi R^4 t,$$

which, in conjunction with an initial condition $m_0 = \pi R^2 l \rho$, leads to $\tau = 8\nu l / g R^2$.

2.23. Suppose that a narrow layer of thickness Δz between two planes is filled with a liquid whose dynamic viscosity is η , and that the planes themselves move towards the x -axis with velocities v_x and $v_x + \Delta v_x$ (Fig. 63). Then the viscous friction forces acting upon the planes will be $\eta \Delta v_x / \Delta z$ and $-\eta \Delta v_x / \Delta z$ (per unit area), respectively. The total power dissipated in the layer is

$$\begin{aligned}
 & -\eta \frac{\Delta v_x}{\Delta z} v_x + \eta \frac{\Delta v_x}{\Delta z} (v_x + \Delta v_x) \\
 & = \eta \left(\frac{\Delta v_x}{\Delta z} \right)^2 \Delta z \approx \eta \left(\frac{dv_x}{dz} \right)^2 \Delta z.
 \end{aligned}$$

Since the pressure difference between the ends of the capillary is $p = \rho gh$, the velocity distribution in it (see the previous problem) is

$$v(r) = \frac{\rho gh}{4\eta l} (R^2 - r^2).$$

By taking a thin cylindrical layer of liquid and applying the above formula for dissipation we obtain the following for the total power dissipated in the capillary

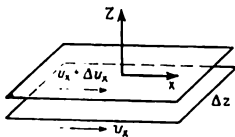


Fig. 63

$$\begin{aligned}
 N &= \eta l \int_0^R \left(\frac{dv_x}{dr} \right)^2 2\pi r dr \\
 &= \frac{\pi}{8} \frac{(\rho gh R^2)^2}{\eta l} \approx 10^{-3} \text{ W.}
 \end{aligned}$$

2.24. When the engine is running, the spatial gas distribution in the cabin is heterogeneous because of the inertia in the coordinate system fixed with respect to the rocket. After the engine is switched off, the distribution becomes homogeneous.

(1) During a quasistatic transition to the homogeneous distribution the gas performs the work against the inertial forces. The work due to a displacement of the gas under an infinitely small acceleration change Δa is

$$\Delta A = - \sum_i F_i \Delta x_i = \sum_i m_i a \Delta x_i = ma \Delta X = ma \frac{dX}{da} \Delta a,$$

where the summation over i is over all elementary volumes and the m_i are their masses and the Δx_i their displacements; $X(a)$ is the coordinate of the centre of gravity of the gas.

For $\mu ah/RT_0 \ll 1$ we can easily see that

$$X \approx \frac{h}{2} \left[1 - \frac{1}{6} \frac{\mu ah}{RT_0} \right].$$

Using the same approximation, when calculating dX/da we can neglect temperature changes, which only yield very small corrections. Therefore

$$\frac{dX}{da} = -\frac{\mu h^3}{12RT_0}.$$

The total work is

$$A = m \int_a^0 a \frac{dX}{da} da = \frac{m\mu h^3 a^2}{24RT_0}$$

which, given thermal insulation ($Q = 0$), governs the change in the internal energy, $\nu C_V (T_1 - T_0) + A = 0$, where ν is the number of moles. Finally,

$$T_1 - T_0 = -\frac{1}{24} \frac{RT_0}{C_V} \left(\frac{\mu ah}{RT_0} \right)^2.$$

Since the process is reversible, the entropy is unchanged: $S_1 = S_0$.

(2) Given a rapid change in acceleration, the gas does not move at the instant the engine is switched off and therefore does not perform the work against external forces. As a result, $Q = A = 0$ and hence the internal energy and the temperature also do not change. The entropy may be calculated from the general formula

$$S = R \sum_i \nu_i \ln \frac{\Delta V_i}{\Delta v_i} = R \int \frac{dV}{v(r)} \ln v(r),$$

where $v(r)$ is the molar volume at the point r . Yet it may be simpler to use the fact that in both cases the initial states of the gas are the same while the final ones differ in temperature by $T_0 - T_1$. Therefore the entropy change is

$$S_2 - S_0 = \frac{\nu C_V}{T_0} (T_0 - T_1) = \frac{1}{24} \frac{mR}{\mu} \left(\frac{\mu ah}{RT_0} \right)^2.$$

2.25. $\Delta S = -\frac{R}{24} \left(\frac{\mu \omega^2 r^2}{2RT} \right)^2$ (see the previous problem).

2.26. Applying Stokes' formula for viscous drag we obtain

$$R^{1/2} \sim \frac{2\eta}{g(\rho - \rho_0)} \sqrt{\frac{kT}{\rho}} \quad \text{and} \quad R \sim 10^{-3} \text{ cm.}$$

The particles precipitated are those for which the average column height evaluated using the barometric formula is $h \sim kT/mg \ll R$ (m is the mass of the particles), which is equivalent to

$$R \gtrsim \left(\frac{kT}{4\rho g} \right)^{1/4} \sim 0.5 \times 10^{-4} \text{ cm.}$$

Hence a sediment does form in the paint.

2.27. The hole is so small that it is possible to assume its presence does not influence the distribution of the gas atoms (in terms of volume or velocity) and that the flow is a molecular effusion. Therefore the flow of particles N is $S\bar{n}\bar{v}/4$, where n is the concentration and $\bar{v}$ the average thermal velocity. The mean energy carried away by each escaping atom is

$$\bar{E}_{\text{esc}} = \frac{\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} dv_x dv_y \int_0^{\infty} dv_z (mv^2/2) v_z e^{-mv^2/2kT}}{\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} dv_x dv_y \int_0^{\infty} dv_z v_z e^{-mv^2/2kT}} = 2kT$$

and exceeds the mean thermal energy $\bar{E} = 3kT/2$ by $kT/2$. Therefore to maintain a constant gas temperature a heat flux Q of $(kT/2)N$ must be supplied. The concentration decrease over time is governed by the equation

$$\frac{d}{dt}(nV) = -N = -\frac{S\bar{n}\bar{v}}{4},$$

whence

$$n(t) = n_0 e^{-t/\tau}, \quad \tau = \frac{4V}{S\bar{v}}, \quad Q = \frac{kT}{8} S\bar{n}_0 e^{-t/\tau}.$$

2.28. This problem is similar to the previous one. The laws for the conservation of the number of particles and energy are

$$\frac{d}{dt}(nV) = -N, \quad \frac{d}{dt} \left(\frac{3}{2} kTnV \right) = -2kTN.$$

We eliminate n and dn/dt and so for $T(t)$ obtain the equation

$$\frac{dT}{dt} = -\frac{1}{12} \frac{S}{V} T\bar{v}(T),$$

which, with the initial condition $T(0) = T_0$, yields

$$T = \frac{T_0}{[1 + (S/6V)\sqrt{kT_0/2\pi m}]^2},$$

where m is the atomic mass.

2.29. Since the problem states that the vapour near the surface is saturated, the evaporation rate will be limited by the vapour's diffusion rate through the tube. The equation controlling the mass balance in evaporation and diffusion has the form

$$-\frac{dh}{dt} = \frac{\rho_{\text{sat}}}{2\rho} \frac{D}{l-h},$$

where $h(t)$ is the height of the liquid column in the tube, and ρ the water density. Integrating this equation with the initial condition $h(0) = l/2$ and using $D \approx 1/3 \bar{v}\Lambda$ we find the following for the evaporation time

$$t = \frac{9RT\rho l^3}{4v\Lambda p_{\text{sat}}\mu} \approx 160 \text{ days},$$

where μ is the molar mass of water.

2.30. The drift velocity v_{dr} is μE , where μ is the mobility related to the diffusion coefficient $D \approx 1/3 \bar{v}\Lambda$ by the Einstein equation $eD = kT\mu$. Therefore

$$v_{dr} \approx \frac{e\bar{v}\Lambda}{m\bar{v}^2} E \approx 15 \text{ m/s}.$$

2.31. If $\Lambda \gg d$, then for the gas flowing in a tube, the role of the diameter d will be the particle range and therefore the effective diffusion coefficient $D_{\text{eff}} \sim \bar{v}d$. The flux through the tube will be

$$N = -\frac{\pi d^3}{4} D_{\text{eff}} \frac{dn}{dx} \propto d^3,$$

hence, with the d value halved, the gas concentration in the vessel will be a factor of $2^3 = 8$ times greater.

2.32. At a pressure of 10^{-6} mm Hg the free path is $\Lambda \sim 1 \text{ m} \gg R$, viz. the flow will be Knudsenian. Under these conditions, the molecular flow is

$$N = \frac{2}{3} \pi R^3 \bar{v} \frac{n - n_1}{l},$$

where n is the concentration in the vessel, n_1 the concentration at the pipeline outlet, and $\bar{v}$ the mean thermal velocity. By definition, the pumping rate N is $n_1 V_1$. The rate V at which the flow will be the same if pumped directly from the vessel is $N = nV$. By eliminating n and n_1 from these

three equations we find that

$$V = \frac{V_1}{1 + \frac{3}{2\pi} \frac{V_1}{R^2 v}} \approx 330 \text{ hp.}$$

2.33. The time needed for the molecules to diffuse a distance of the order of πR_E , where R_E is the Earth radius, is estimated to be $t_{\text{diff}} \sim (\pi R_E)^2 / \bar{v} \Lambda$. Given $R_E \sim 6 \times 10^8 \text{ cm}$, $\Lambda \sim 10^{-3} \text{ cm}$, $\bar{v} \sim 3 \times 10^4 \text{ cm/s}$, the formula yields $t_{\text{diff}} \sim 4 \times 10^{11} \text{ years}$, viz. $t_{\text{diff}} \sim 10^{11} t_{\text{univ}}$.

2.34. Since at the pressure given in the problem the free path (viz. the hole size) is $\Lambda \gg S^{1/2}$, the flow will be molecular. Thus the exit of each individual molecule will be independent and for this reason N obeys a Poisson distribution. Hence,

$$\frac{\sqrt{(\Delta N)^2}}{\bar{N}} = \frac{1}{\sqrt{\bar{N}}} = \frac{2}{\sqrt{n \bar{v} S t}} \approx 10^{-6},$$

where $\bar{v}$ is the mean velocity and n the concentration of the helium atoms in the vessel. Given a Poisson distribution, the probability of fixing N particles in one run is

$$w(N) = \frac{\bar{N}^N}{N!} e^{-\bar{N}}.$$

The probability of $N = 0$ is $w(0) \approx \exp(-10^{12})$.

2.35. Using the definition of mean values and the Boltzmann distribution we obtain in case (a)

$$\left[\frac{(\Delta U)^2}{\bar{U}^2} \right]^{1/2} = \frac{\left[\int_{-\infty}^{+\infty} (U(x) - \bar{U})^2 e^{-U(x)/kT} dx \right]^{1/2}}{\int_{-\infty}^{+\infty} U(x) e^{-U(x)/kT} dx} = \sqrt{2},$$

where $U(x) = Kx^2/2$. In case (b) the fluctuation is equal to $\left(\frac{2}{3N-5} \right)^{1/2}$ and $\left(\frac{2}{3(N-1)} \right)^{1/2}$ for molecules with linearly and nonlinearly arranged atoms, respectively. In case (c) the fluctuation is $\sqrt{2/N_A}$.

2.36. Since the temperature remains constant, the pressure fluctuation is due entirely to a fluctuation in the number of particles. Let us use the relation between probability and

entropy. The entropy of the initial and final states is

$$S_0 = 2Rv \ln \frac{V}{v},$$

$$S = R(v + \Delta v) \ln \frac{V}{v + \Delta v} + R(v - \Delta v) \ln \frac{V}{v - \Delta v},$$

where v is the number of moles of gas in each vessel, and $\Delta v = \frac{V}{RT} \Delta p$ is the number of moles transferred from one vessel to the other during the fluctuation period. For $|\Delta v| \ll v$ we obtain

$$\Delta S = -R \frac{(\Delta v)^2}{v} \quad \text{and} \quad \frac{w}{w_0} = \exp \left[-nV \left(\frac{\Delta p}{p} \right)^2 \right],$$

where n is the equilibrium concentration of the gas molecules. Finally

$$V = \frac{kT}{p} \left(\frac{p}{\Delta p} \right)^2 \ln \frac{w_0}{w} \approx 4 \text{ cm}^3.$$

2.37. When considering the temperature fluctuations we assume that the number of moles inside and outside the volume v (v_1 and v_2 , respectively) are fixed. Initially we assume that the vessel is completely isolated so that $Q = A = 0$ and the invariance of the internal energy is expressed thus

$$\Delta U = v_1 C_V \Delta T_1 + v_2 C_V \Delta T_2 = 0.$$

The change in entropy due to the temperature fluctuation is

$$\Delta S = v_1 C_V \ln \frac{T + \Delta T_1}{T} + v_2 C_V \ln \frac{T + \Delta T_2}{T}$$

and for $v_1 \ll v_2$ in the region of small fluctuations this reduces to

$$\Delta S = -\frac{1}{2} v_1 C_V \left(\frac{\Delta T_1}{T} \right)^2.$$

Thus, the probability distribution is Gaussian:

$$w(\Delta T_1) \propto \exp \left[-\frac{1}{2} v_1 \frac{C_V}{k} \left(\frac{\Delta T_1}{T} \right)^2 \right].$$

But if a quantity x has a Gaussian distribution, viz. $w(x) \propto e^{-x^2/2\lambda^2}$, then the mean-square fluctuation must be

$$\overline{x^2} = \frac{\int_{-\infty}^{+\infty} x^2 e^{-x^2/2\lambda^2} dx}{\int_{-\infty}^{+\infty} e^{-x^2/2\lambda^2} dx} = \lambda^2.$$

Therefore, as applied to our problem, $\overline{(\Delta T_1)^2} = kT^2/v_1 C_V$. Since additionally $p v = v_1 R T$, we get

$$v = \frac{R}{C_V} \frac{kT}{p} \left(\frac{T}{\Delta T} \right)^2 \approx 2.5 \times 10^{-8} \text{ cm}^3.$$

The size of the vessel drops out of the final formulas, therefore we can see that it is irrelevant whether the vessel is isolated: processes occurring far away cannot influence the magnitude of the local fluctuations.

2.38. The ratio of the lengths of time the system stays in the homogeneous and separated states (t_0 and t_1) is related to the ratio between the corresponding probabilities, which in turn can be expressed via the difference between the entropies of the states, $S_0 - S_1 = 2Nk \ln 2$, i.e.

$$\frac{t_0}{t_1} = \frac{w_0}{w_1} = \exp \frac{S_0 - S_1}{k} = 2^{2N}.$$

The time t_1 it takes for the separated state to disintegrate is comparable to the time needed for the molecules to fly apart, i.e. $t_1 \sim d/\bar{v} \sim 10^{-4}$ s ($d \sim 10$ cm is the vessel size). The time t_0 is the expected time it takes for a separated state to form: $t_0 \sim 10^{10}$ years $\sim 10^{17}$ s. Whence

$$N = \frac{\log(t_0/t_1)}{2 \log 2} \approx 35.$$

2.39. The energy associated with an isothermal fluctuation volume is given by the work done against the quasistatic force, $(\partial p / \partial V)_T \Delta V$. Therefore, the energy is

$$E(\Delta V) = \frac{1}{2} \left| \left(\frac{\partial p}{\partial V} \right)_T \right| (\Delta V)^2,$$

and the probability of the fluctuation, given the Boltzmann distribution, is

$$w(\Delta V) \propto \exp \left[-\frac{1}{2kT} \left| \left(\frac{\partial p}{\partial V} \right)_T \right| (\Delta V)^2 \right],$$

viz. the resulting distribution is Gaussian. The root-mean-square fluctuation (see Problem 2.37) is

$$\overline{(\Delta V)^2} = kT \left| \left(\frac{\partial V}{\partial p} \right)_T \right|.$$

By calculating $(\partial p / \partial V)_T$ for a Van der Waals gas and assuming (in the derivative) that the molar volume is equal

to the critical one we find that

$$\frac{(\Delta V)^2}{V^2} = \frac{4b}{3N_A V} \frac{T}{T - T_{cr}},$$

where N_A is Avogadro's number. It can be seen that the fluctuations build up rapidly near the critical point. If $T \approx T_{cr}$, then $N_A V \approx 3bN$, where N is the number of gas molecules per unit volume. Then

$$\frac{(\Delta V)^2}{V^2} \approx \frac{4}{9N} \frac{T}{T - T_{cr}}.$$

2.40. The well-known formula for the temperature dependence of saturated gas pressure is

$$p(T) = p_0 \exp \left\{ \frac{\Lambda}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right\},$$

where $\Lambda = \lambda \mu$ is the molar heat of evaporation. The condition for the valve to be opened is $F = \frac{\pi D^2}{4} (p - p_0)$.

Finally, $F \approx 18$ N.

2.41. The pressure in the subterranean reservoir p is $p_0 + \rho gh$, where p_0 is the atmospheric pressure. The boiling sets in at $p(T) = p$, when the saturated vapour pressure is

$$p(T) = p_0 \exp \left\{ \frac{\Lambda}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right\}.$$

The boiling will stop when the water temperature drop due to the boiling reaches T_0 . This will take place when the mass of water that boils away is

$$\Delta M \approx \frac{MC}{\Lambda} (T - T_0),$$

where C is the molar heat capacity of water. By obtaining T from the first equation we find from the second one that $\Delta M/M \approx 0.14$.

2.42. From the Clausius-Clapeyron equation, along the melting curve,

$$\frac{dp}{dT} = \frac{q}{T \Delta V}, \quad q = T(S_1 - S_2).$$

The pressure minimum $dp/dT = 0$ corresponds to $q = 0$. It follows from the data on the entropy of both phases that the corresponding temperature $T_{min} = \Theta \ln 2 \approx 0.15$ K. The melting heat $q = RT^3/\Theta - RT \ln 2$ is a parabola that in-

tersects zero at $T_{\min}$. At lower temperatures $q < 0$. This change in the melting heat is a unique fact called the Pomeranchuk effect. The falling section of the $p(T)$ curve corresponds to the negative q . We stress that $\Delta V > 0$ for helium-3 (as opposed to the general situation when $\Delta V < 0$, as in the case of the water-ice system). Integrating the Clausius-Clapeyron equation yields

$$p(T) = p_{\min} + \frac{R\theta}{2\Delta V} \left(\frac{T}{\theta} - \ln 2 \right)^2.$$

Whence $p(0) \approx 32.5$ atm.

2.43. The equation of mechanical equilibrium governing the pressure distribution inside the core is

$$\frac{dp}{dr} = -\rho g(r) = -\frac{4}{3}\pi\gamma\rho^2r,$$

where γ is the gravitational constant and $g(r)$ the acceleration due to gravity. From the Clausius-Clapeyron equation for the phase equilibrium curve,

$$\frac{dp}{dT} = \frac{q}{T\Delta V} = -\frac{\rho^2q}{T\Delta\rho},$$

where q is the specific melting heat. Dividing these equations into each other (at $r = R$, where they are both valid) we find that

$$\frac{dT}{dR} = \frac{4\pi}{3} \frac{\gamma\Delta\rho}{q} RT.$$

This equation relates the radius R of the solid state part to the temperature T . Since

$$\frac{dR}{dT} = \frac{dR/dt}{dT/dt} = -\frac{1}{u} \frac{dR}{dt},$$

we find that

$$\Delta R = -\frac{3}{4\pi} \frac{qut}{\gamma\Delta\rho RT} \approx -35 \text{ km.}$$

2.44. The internal energy of the system is

$$U = m_v u_v + m_l u_l,$$

where u_v and u_l are the specific internal energies of the vapour and liquid. Differentiating U with respect to T and allowing for the conservation of the total mass ($m_v + m_l = \text{const}$) we get

$$(m_v + m_l)c = (u_v - u_l) \frac{dm_v}{dT} + \left(\frac{3R}{\mu} m_v + c_0 m_l \right).$$

In the first term on the right-hand side $u_v - u_l = \lambda - kT/\mu$ ($\lambda = \Lambda/\mu$ is the latent heat of vaporization, viz. the difference between enthalpies per unit mass) and according to the Clausius-Clapeyron equation,

$$\frac{dm_v}{dT} = \frac{\mu}{RT^2} \left(\lambda - \frac{RT}{\mu} \right) m_v.$$

Finally

$$c = \frac{1}{1+\beta} \left\{ c_0 + \beta \left[\frac{3R}{\mu} + \frac{\mu}{RT^2} \left(\lambda - \frac{RT}{\mu} \right)^2 \right] \right\} \approx 10^4 \text{ J/(K} \cdot \text{kg)}.$$

2.45. The answer follows from the solution of the previous problem at the limit as $\beta \rightarrow \infty$, viz.

$$c = \frac{3R}{\mu} + \frac{\mu}{RT^2} \left(\frac{\Lambda}{\mu} - \frac{RT}{\mu} \right)^2 \approx 7 \times 10^4 \text{ J/(K} \cdot \text{kg)}.$$

2.46. The total entropy change is the sum of the changes in the volume and surface entropies. The volume entropy is

$$(\Delta S)_{vol} = \nu R \ln \frac{V'}{V} = \frac{p_0}{T} \cdot \frac{4}{3} \pi r^3 \ln 8 = 4\pi \ln 2 \frac{p_0 r^3}{T},$$

since we can approximate the pressure inside the bubble to p_0 . The specific surface entropy s is q/T , where q is the heat of the isothermal formation of a unit of surface, as can be seen from the conventional definition $dQ = T dS$. Therefore

$$(\Delta S)_{surf} = 2 \cdot 4\pi [(r')^2 - r^2] \frac{q}{T} = 24\pi r^2 \frac{q}{T}.$$

Finally,

$$\Delta S = \frac{24\pi r^2}{T} \left(q + \frac{\ln 2}{6} p_0 r \right).$$

III. Electricity and Magnetism

3.1. An electron emitted from a negatively charged plate is acted upon by the force due to the electric field in a capacitor, $F_1 = eE$, and the force of electric image, $F_2 = -e^2/(4\pi\epsilon_0 \cdot 4x^2)$. The total force acting upon the electron is

$$F = eE - \frac{e^2}{16\pi\epsilon_0 x^2}.$$

It follows from this expression that the electron slows down unless $x < \sqrt{e/(16\pi\epsilon_0 E)}$. The electron velocity is at a minimum at

$$x = x_0 = \frac{1}{4} \sqrt{\frac{e}{\pi\epsilon_0 E}} \approx 10^{-6} \text{ m}.$$

Notice that this solution does not allow for the space charge that may form near the negatively charged plate.

3.2. In an electric field gas molecules acquire a dipole moment $p = \epsilon_0 \alpha E$ and an energy (see Problem 3.16) $W = -pE/2 = -\epsilon_0 \alpha E^2/2$. The molecules in the force field have the Boltzmann distribution

$$n = n_0 \exp \left(-\frac{W}{kT} \right) = n_0 \exp \frac{\epsilon_0 \alpha E^2}{2kT},$$

where n_0 is the concentration of the molecules outside the capacitor. Substituting in numerical values shows that the exponent is small compared to unity. Therefore it is possible to write

$$\frac{\Delta n}{n_0} \approx \frac{1}{2} \frac{\epsilon_0 \alpha E^2}{kT} \approx 4 \times 10^{-8}.$$

The concentration of the molecules (and hence the pressure) in the capacitor is higher than in the remainder of the vessel. The physical reason for the higher pressure in the capacitor is that the molecules with a dipole moment are pulled into the region of the strong field.

3.3. It can easily be shown using Gauss's theorem that the strength of the electric field of a uniformly charged ball at $r \leq R$ is

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} r = \frac{1}{3\epsilon_0} \rho r.$$

Here r is the radius-vector from the ball's centre to the point of observation, Q the total charge on the ball, R its radius, and ρ the volume charge density. The field in the cavity may be treated as a superposition of the fields of two uniformly charged balls:

$$E = E_1 + E_2 = \frac{1}{3\epsilon_0} \rho r_1 - \frac{1}{3\epsilon_0} \rho r_2 = \frac{1}{3\epsilon_0} \rho (r_1 - r_2) = \frac{1}{3\epsilon_0} \rho a.$$

The notation is evident from Fig. 64. Thus, the electric field in the cavity is uniform. This conclusion remains valid irrespective of the ratio between the radii of the balls and the distance between their centres. In particular, the field in an uncharged spherical cavity inside a uniformly charged ball of large radius is also uniform.

3.4. Using the solution of Problem 3.3, consider two identical balls of radii R with charge uniformly distributed within their volumes and charge densities ρ and $-\rho$. Assume that the balls intersect so that their centres are a distance

$a \ll R$ apart. Within the intersection region the charges will cancel each other out. It follows from Problem 3.3 that the field in this region will also be uniform, i.e.

$$E = \frac{1}{3\epsilon_0} \rho a.$$

Here a is the vector connecting the centres of the balls. As shown in Fig. 65, the space charge is only nonzero in a thin surface layer. As $a \rightarrow 0$ (given the condition that $\rho a = \text{const}$) we arrive at an idea of a surface charge on a sphere.

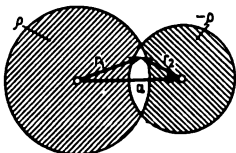


Fig. 64

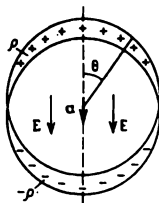


Fig. 65

It can be seen that the thickness of the charged layer at the point defined by the angle θ is $a \cos \theta$ and hence the charge per unit area is

$$\sigma(\theta) = \rho a \cos \theta = 3\epsilon_0 E \cos \theta.$$

Since the field of a uniformly charged ball is the same for $r > R$ as that of a point charge located at the centre of the ball, we can conclude that in our case the field outside the sphere is that of a point dipole whose dipole moment is

$$p = -Qa = -\frac{4}{3} \pi R^3 \rho a = -4\pi\epsilon_0 R^3 E.$$

It is possible to check that the uniform field inside the sphere and the field of a dipole outside it both satisfy the boundary conditions on the surface of the sphere, thus indicating that this solution is correct.

3.5. It follows from the solution of Problem 3.4 that in an external uniform field E a conducting ball of radius R acquires a dipole moment

$$p = 4\pi\epsilon_0 R^3 E.$$

In fact, the electric field inside the conducting ball should vanish. The charge on the ball should be redistributed so that the field of the charge inside the ball becomes uniform and cancels the external field. Thus outside the ball the field of the charge inside it will be the same as that of a dipole.

The electric moment P per unit volume of an "ideal gas" composed of conducting balls will in an external field be

$$P = np = 4\pi e_0 n R^3 E.$$

Now we can write

$$D = e_0 E + P = e_0 (1 + 4\pi n R^3) E = e_0 \epsilon E.$$

Whence it follows that

$$\epsilon = 1 + 4\pi n R^3.$$

Because the ball concentration is small we can ignore the interaction of the balls when determining the dipole moment.

3.6. The electric field outside a spherical dust speck will be composed of a uniform external field E and the field E_d of a dipole:

$$E_d = \frac{1}{4\pi e_0} \left[\frac{(pr)r}{r^3} - \frac{p}{r^3} \right],$$

whose dipole moment (see Problem 3.4) is

$$p = 4\pi e_0 R^3 E.$$

At points C and D the field $E_d = -E$, while at points A and B the field $E_d = 2E$. Whence it follows that the total field at points C and D will vanish while at points A and B it will be trebled.

3.7. The dipole moment of an argon atom in an external field E can be written as

$$p = Zxe = e_0 \alpha E,$$

where x is the displacement of the electron shell, Z the number of electrons in the atom, and α the atomic polarizability:

$$\alpha = Zxe/e_0 E.$$

A dielectric permittivity ϵ is related to its atomic polarizability α thus

$$\epsilon = 1 + n_0 \alpha,$$

where $n_0 = 2.7 \times 10^{25} \text{ m}^{-3}$ is the concentration of the atoms under normal conditions (Loschmidt number). Hence

$$x = (e - 1) e_0 E / n_0 Z e \approx 2 \times 10^{-18} \text{ m} = 2 \times 10^{-16} \text{ cm}.$$

Notice that the atoms are of the order of 10^{-8} cm in size, which significantly exceeds the amount by which they are deformed in electric fields.

$$\begin{aligned} 3.8. \quad \rho(r) &= \frac{3e}{4\pi R^3} = \text{const}, \quad \nu = \frac{1}{2\pi} \sqrt{\frac{e^2}{4\pi\epsilon_0 R^3 m}} \\ &= 2.5 \times 10^{13} \text{ Hz}. \end{aligned}$$

This value of the frequency ν is within the visible spectrum.

3.9. In the coordinate reference frame of the rocket, the electrons in the matter are acted upon by the force of inertia ma . In equilibrium, this force must be counter-balanced by the force eE due to the electric field. Hence, at the ends of a conductor of length l a potential difference $U = El = malle$ emerges during acceleration. Thus, the acceleration might be detected by measuring the value of U . For $a = 10 g$ and $l = 10 \text{ m}$ the potential difference is $U \approx 6 \times 10^{-8} \text{ V}$. Modern equipment can, in principle, measure such small potential differences. Note, however, that in this case the standard method of measuring using wires connected to the end-points of the conductor cannot be used because an analogous potential is set up in the connecting wires themselves. However, the potential could be determined indirectly, e.g. by measuring the amplitude of the current in a conductor whose orientation is periodically changed by 180° .

3.10. The charge is uniformly distributed on an external surface:

$$\sigma = q/4\pi R^2.$$

The field outside the shell is the same as that of a point charge q at its centre:

$$\Phi = \frac{1}{4\pi\epsilon_0} \frac{q}{R}.$$

In order to determine σ_1 and σ_2 we need to know the electric field inside the shell. Using the method of electric images it is possible to demonstrate that the electric field inside the shell coincides with that of two point charges q and q' (Fig. 66). The charge-image q' has the value

$$q' = -q \frac{r}{a}$$

and is located a distance

$$b = \frac{r^2}{a}$$

from the shell's centre.

The value and position of the charge-image are chosen so that the potential of the inner surface of the shell is every-

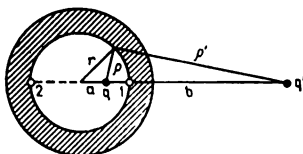


Fig. 66

where zero in the field of the charges q and q' , i.e.

$$\varphi' = \frac{q}{4\pi\epsilon_0\rho} + \frac{q'}{4\pi\epsilon_0\rho'} = 0.$$

The field strength at points 1 and 2 is

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2} - \frac{1}{4\pi\epsilon_0} \frac{q'}{(b-r)^2} = \frac{q}{4\pi\epsilon_0} \frac{1+a/r}{(r-a)^2},$$

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2} + \frac{1}{4\pi\epsilon_0} \frac{q'}{(b+r)^2} = \frac{q}{4\pi\epsilon_0} \frac{1-a/r}{(r+a)^2}.$$

Now it is possible to determine the surface densities σ_1 and σ_2 using the boundary condition on the conductor surface, $E_n = \sigma/\epsilon_0$. The sign in this relation corresponds to the choice of the external normal. In our case $E_{1n} = -E_1$ and $E_{2n} = -E_2$, therefore

$$\sigma_1 = -\frac{q}{4\pi} \frac{1+a/r}{(r-a)^2}, \quad \sigma_2 = -\frac{q}{4\pi} \frac{1-a/r}{(r+a)^2}.$$

3.11. From the condition of continuity of the normal components of the induction vector $D = \epsilon_0 E + P$ we have

$$E_1 = E_3 = E = E_2 + \frac{P}{\epsilon_0}.$$

Here E_1 and E_3 are the fields in the air gaps, and E_2 the field in the plate. The potential difference between the capac-

itor plates is zero, therefore

$$E(d-h) + E_2 h = 0.$$

Whence

$$E_2 = -\frac{P}{\epsilon_0} \left(1 - \frac{h}{d}\right), \quad D_2 = \epsilon_0 E_2 + P = P \frac{h}{d}.$$

$$3.12. E_A = \frac{1}{2\epsilon_0} P, \quad \frac{E_A}{E_B} = \left(\frac{l}{r}\right)^2.$$

3.13. The field in the plate is

$$E_1 = -\frac{P}{\epsilon_0} = -\frac{P_0}{\epsilon_0} \left[1 - \frac{x^2}{d^2}\right].$$

The field outside the plate is

$$E_2 = 0.$$

The potential difference between the sides of the plate is

$$U = -\int_{-d}^d E_1 dx = \frac{P_0}{\epsilon_0} \int_{-d}^d \left(1 - \frac{x^2}{d^2}\right) dx = \frac{4}{3\epsilon_0} P_0 d.$$

3.14. If all the energy of the capacitor is consumed to heat the gas the atoms will all be hot enough to ionize. If N is the number of hydrogen molecules in a given volume, the law of conservation of energy is

$$\frac{CU^2}{2} = 4N \frac{3}{2} kT - N \frac{3}{2} kT_0.$$

The factor 4 in the first term on the right-hand side takes into account the processes of dissociation and ionization. We can neglect the second term on the right-hand side ($T \gg T_0$) of this relationship. Assuming $N = pV/kT_0$ we find that

$$T = T_0 \frac{CU^2}{12pV} \approx 3 \times 10^7 \text{ K}.$$

A possible reason why the temperature might be lower is that the capacitor plates get hot. If all the discharge energy is consumed by heating the plates, the law of conservation of energy would be of the form

$$\frac{CU^2}{2} = cm \Delta T,$$

whence $\Delta T \approx 200 \text{ K}$. Thus, only a relatively small rise in the temperature of the capacitor plates results in a lowering of the gas temperature by a few orders of magnitude.

3.15. Let us apply the method of electrical images. As shown in Fig. 67, a dipole image has a dipole moment $-p$. The electric field strength of this dipole on its axis (the basic dipole position) is

$$E = \frac{1}{4\pi\epsilon_0} \left[-\frac{3(pr)r}{r^5} + \frac{p}{r^3} \right] = \frac{p}{2\pi\epsilon_0 r^3}.$$

Here r is the distance to the image and equal in our case to $2L$. The force acting upon a dipole with moment p in an inhomogeneous electric field E is

$$F = (p \nabla) E$$

whence it follows that

$$F = p \frac{\partial E}{\partial r} = \frac{3p^2}{2\pi\epsilon_0 r^4} = \frac{3p^2}{32\pi\epsilon_0 L^4} \approx 5.4 \times 10^{-6} \text{ N} = 0.54 \text{ dyn.}$$

As can be seen $F > 0$, which means that the dipole is repelled from the conducting plane.

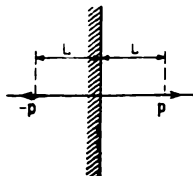


Fig. 67

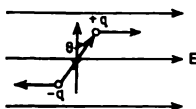


Fig. 68

Now let us find out the work done by the external forces to displace the dipole to a new position:

$$\begin{aligned} A &= - \int_L^{L_1} F dx = - \int_L^{L_1} \frac{3p^2}{32\pi\epsilon_0 x^4} dx = \frac{p^2}{32\pi\epsilon_0 L^3} \left(\frac{L^3}{L_1^3} - 1 \right) \\ &= 1.6 \times 10^{-6} \text{ J} = 0.16 \text{ erg.} \end{aligned}$$

3.16. A rigid electric dipole can be moved from infinity to any point in a field without any work being done by moving it along one of the lines of force provided that all the time $p \perp E$. Therefore the energy of a rigid dipole in an external field can be defined as the work done by the external forces to rotate the dipole (Fig. 68):

$$W = A = -2q \frac{l}{2} \sin \theta \cdot E = pE \sin \theta = -(pE).$$

Here l is the length of the dipole and q its charge.

In order to determine the energy of an elastic dipole let us consider a quasistatic change in the dipole moment when the external field is changed. Assuming that the dipole's length is changed by dl , electric forces will perform the work

$$qE \, dl = \alpha \epsilon_0 E \, dE.$$

Expressing the energy of the dipole through the work of the external forces we find that

$$W = -\frac{\alpha \epsilon_0 E^2}{2} = -\frac{pE}{2} = -\frac{(pE)}{2}.$$

3.17. $U = \frac{mv_e^2}{2\epsilon} \frac{l}{d} \approx 14 \, \mu\text{V}$, where m and e are the mass and charge of an electron.

3.18. The solution to this problem can be found by the method of electric images. The field of the point charge q and that of the charge-image should satisfy the boundary conditions at the boundary of the dielectric. As a result, we obtain

$$\sigma(\rho) = -\frac{1}{2\pi} \frac{\epsilon-1}{\epsilon+1} \frac{qL}{(\rho^2 + L^2)^{3/2}}.$$

Here ρ is the distance from the perpendicular projection of the point where the charge q is on the dielectric surface. The total charge Q induced on the dielectric surface is

$$Q = -\frac{\epsilon-1}{\epsilon+1} q.$$

The field of the surface charge in the half-space above the dielectric is the same as that of a point charge Q located inside the dielectric at a distance L from the boundary.

The force acting upon the charge q is given by the expression

$$F = \frac{1}{4\pi\epsilon_0} \frac{qQ}{4L^2} = -\frac{1}{16\pi\epsilon_0} \frac{\epsilon-1}{\epsilon+1} \frac{q^2}{L^2}.$$

The minus sign means that the charge q is attracted to the dielectric.

3.19. We apply the law of conservation of energy to the process of recharging the capacitor after bringing its plates together:

$$Q = A_{\text{bat}} - \Delta W_e.$$

Here Q is the Joule heat liberated in the resistor, A_{bat} the work done by the extraneous forces in the battery, and ΔW_e

the change in the electric energy of the capacitor. If Δq was the change in the capacitor charge due to its having been recharged we can write

$$\Delta q - \Delta C \mathcal{E} = (2C - C) \mathcal{E} \quad C \mathcal{E}, \quad A_{\text{net}} = \Delta q \mathcal{E} = C \mathcal{E}^2.$$

To obtain an expression for ΔW_e we must take into account that when the plates are brought together the capacitor capacitance becomes equal to $2C$ and hence the potential difference, given the charge remains the same, is $\mathcal{E}/2$. When the recharging process is complete the potential difference will be restored to $\mathcal{E}$. Therefore

$$\Delta W_e = \frac{2C \mathcal{E}^2}{2} - \frac{2C (\mathcal{E}/2)^2}{2} = \frac{3}{4} C \mathcal{E}^2.$$

Thus, $Q = \frac{1}{4} C \mathcal{E}^2$.

In order for the capacitor's charge to remain practically unchanged during the time interval Δt when the plates were brought together, it is necessary that

$$\Delta t \ll \tau = RC.$$

The constant $RC = \tau$ determines the rate of the process in a circuit composed of a resistor and a capacitor. Substituting the numerical values we obtain

$$R \gg \frac{\Delta t}{C} = 10^8 \, \Omega.$$

3.20. To estimate the quantities we assume that the field of a needle tip is the same as that of a charged ball. In an electric field of strength E an atom acquires a dipole moment $p = \alpha \epsilon_0 E$. The inhomogeneous field acts upon the dipole with a force $F = p \frac{\partial E}{\partial r}$. As a result we get

$$F = \frac{2\alpha\epsilon_0}{r^3} \frac{\eta^2}{(4\pi\epsilon_0)^2} = \frac{2\alpha\epsilon_0 R^2 \Phi^2}{r^3} \approx 10^{-11} \text{ N} = 10^{-6} \text{ dyn}.$$

In this expression $r = R + l = 300 \text{ \AA} = 3 \times 10^{-8} \text{ m}$ is the distance between the atom and the tip centre of curvature, and the polarizability $\alpha \approx a^3$, where $a = 10^{-10} \text{ m}$ is the atom's size.

3.21. Since $\lambda_1 \gg \lambda$, we can assume, in a good approximation, that the cylindrical conductors have constant potentials along their whole length and that to determine the field we need only consider an electrostatic problem. Suppose the linear charge density on the conductors is $\pm \rho$. Applying Gauss's

theorem to one of the rods we find that the field strength of the rod a distance r from its axis is

$$E = \pm \rho / (2\pi\epsilon_0 r).$$

The potential difference between the rods is found by integrating over the field strength:

$$U = \frac{\rho}{2\pi\epsilon_0} \int_{r_0}^{b-r_0} \left(\frac{1}{r} + \frac{1}{b-r} \right) dr = \frac{\rho}{\pi\epsilon_0} \ln \frac{b-r_0}{r_0} \approx \frac{\rho}{\pi\epsilon_0} \ln \frac{b}{r_0}.$$

The total current J can be easily found if we consider that the field near each rod is independent of the charge on another rod ($b \gg r_0$). Assuming that the current density is constant across the sheet thickness we obtain the following for the total current flowing out of a cylindrical conductor:

$$J = 2\pi r_0 a j = 2\pi r_0 a \lambda E = \frac{a \lambda \rho}{\epsilon_0}.$$

Hence

$$R = \frac{U}{J} \approx \frac{1}{\pi \lambda a} \ln \frac{b}{r_0}.$$

3.22. In the first case, the current will remain constant as the voltage increases; therefore the power liberated at the anode will double. In a stationary state this power, when anode temperature is high enough, is dissipated mainly as radiation so that $W \propto T^4$, where T is the absolute temperature (in our estimates we shall neglect the heat released through conductors and the other structural elements of the kenotron). Thus, the absolute temperature will be increased $2^{1/4} \approx 1.2$ times, viz. it will reach about 1014 °C. In the second case the current will be increased $2^{3/2}$ times (three-halves power law) so that the power will be increased $2 \times 2^{3/2} = 2^{5/2}$ times and the temperature will rise $(2^{5/2})^{1/4} = 2^{5/8}$ times, viz. it will reach some 1400 °C.

3.23. The conductivity λ of an ionized gas is proportional to the ion concentration which in turn is proportional to the flux of the fast electrons moving between the spheres. Thus, $\lambda \propto j$. It follows from Ohm's law ($j_{\text{cond}} = \lambda E$) that throughout the space between the spheres the electric field is constant. Hence,

$$E = \frac{\phi}{R_2 - R_1}.$$

This sort of field can only emerge in a spherical capacitor when a space charge is present.

3.24. $\Lambda = \frac{4\pi\lambda}{1/R_1 - 1/R_2}$, where λ is the specific conductivity of the dielectric, and R_1 and R_2 are the radii of the spheres.

3.25. $\lambda \propto 1/r^4$.

3.26. In a stationary mode of operation the load current is equal to the current carried by the belt:

$$j = \sigma lv.$$

The maximum current value depends on the electric strength of the gas:

$$j_{\max} = 2e_0 E_{st}, \quad j_{\max} = \sigma_{\max} lv = 2e_0 E_{st} lv \approx 10^{-3} \text{ A.}$$

The maximum potential of the sphere is also controlled by the electric strength of the gas. For a spherical electrode,

$$\varphi = RE,$$

where E is the field strength near the electrode. Therefore

$$\varphi_{\max} = RE_{st} = 4.5 \times 10^6 \text{ V.}$$

$$3.27. \quad \varepsilon = \frac{C}{4\pi e_0} \frac{R_2 - R_1}{R_1 R_2}, \quad \rho = \frac{4\pi R_1 R_2}{C(R_2 - R_1)} \frac{t}{\ln 2}.$$

3.28. The magnetic moment of a uniformly charged ball of radius R rotating with frequency ω is

$$p_m = \frac{QR^2\omega}{5},$$

where Q is the total charge on the ball. In fact, the magnetic moment of a ring with current j is jS , where S is the ring area. Changing to spherical coordinates r, θ, φ we find that the magnetic moment of the ring at r, θ is

$$dp_m = \pi \rho \omega r^4 \sin^3 \theta \, dr \, d\theta,$$

where ρ is the space charge density. Integrating over r from 0 to R and over θ from 0 to π we arrive at the above formula for p_m .

Let us introduce the quantity $y = (q_p - q_e)/q_p$ and write the charge on one atom as $Z(q_p - q_e) = Zy q_p$. If the Earth is composed of atoms with an atomic number Z and a relative mass A , then the ratio between its charge Q and mass M_E is

$$\frac{Q}{M_E} = \frac{yZq_p}{m_e A},$$

where m_0 is the mass of a nucleon. Using this relationship the expression for p_m can be rewritten as

$$p_m = y \frac{Zq\nu}{m_0 A} \frac{4\pi}{15} \omega R_E^2 \rho,$$

where ρ is the Earth's density. If the Earth possessed this magnetic moment, then the magnetic field induction at the Pole would be given by

$$B_E = \frac{2\mu_0 p_m}{4\pi R_E^3}.$$

Whence

$$y = \frac{A}{Z} \frac{15m_0 B_E}{2\mu_0 \rho q \nu \omega R_E^3} \approx 2.4 \times 10^{-10}.$$

Thus, the hypothesis in the problem about the origin of the Earth's magnetic field contradicts experimental data.

3.29. The tensile stress in the walls of a rotating tube can be determined by equating the elastic forces and the centrifugal forces of inertia because they are in equilibrium. The stress proves to be $f = \rho\nu^2$, where ρ is the density of tube's material, and ν the speed of rotation. Its limiting value is $\nu_{\max} = \sqrt{f_{\max}/\rho} = 4.7 \times 10^3$ m/s. As the tube rotates, a transfer current emerges whose surface density j is given by the formula $j = \sigma\nu$, where σ is the surface charge density. From Gauss's theorem, the electric field strength on the surface E is σ/ϵ_0 so that for the current density we have $j = \epsilon_0 E \nu$. As to the shape of the currents, the rotating tube is like a solenoid, therefore the magnetic field outside the tube is zero while inside it is $B = \mu_0 j$. Substituting $j = \epsilon_0 E \nu$ and $\epsilon_0 \mu_0 = c^{-2}$ we get

$$\frac{w_m}{w_e} = \frac{B^2/2\mu_0}{\epsilon_0 E^2/2} = \frac{\nu^2}{c^2} \approx 2.4 \times 10^{-12}.$$

3.30. $\sigma_{\max} = 2\epsilon_0 E = 3.5 \times 10^{-5}$ C/m²,

$$B_{\max} = \frac{\mu_0}{2} \sigma_{\max} \nu = \epsilon_0 \mu_0 E \nu \approx 3.3 \times 10^{-10} \text{ T} \approx 3.3 \times 10^{-6} \text{ G}.$$

3.31. First we find the total force acting upon an electron a distance r from the beam's axis. Gauss's theorem yields $E = -en_0 r/2\epsilon_0$ for the electric field, while the magnetic field acting upon the electron is determined from the magnetic field circulation theorem:

$$H = -\frac{en_0 \nu r}{2}.$$

The total force F acting upon the electron is thus

$$F = \frac{n_0 e^2 r}{2\epsilon_0} - \frac{\mu_0 n_0 e^2 v^2 r}{2}.$$

The first term represents Coulomb repulsion, and the second magnetic compression. This expression can be rewritten as

$$F = \frac{n_0 e^2 r}{2\epsilon_0} \left(1 - \frac{v^2}{c^2} \right).$$

If the concentration of positive ions is n , then the equilibrium condition for the beam electrons is

$$\frac{n_0 e^2 r}{2\epsilon_0} \left(1 - \frac{v^2}{c^2} \right) = \frac{n e^2 r}{2\epsilon_0},$$

whence

$$\frac{n}{n_0} = 1 - \frac{v^2}{c^2}.$$

3.32. The induction B of the magnetic field is zero inside a cylinder and $B = \mu_0 \mathcal{I} / 2\pi R$ near its outer surface. The pressure on a plasma cylinder in a magnetic field is

$$p_m = B^2 / 2\mu_0 = \frac{\mu_0 \mathcal{I}^2}{8\pi^2 R^2}.$$

Substituting numerical values we find $p_m \approx 6.4 \times 10^4 \text{ N/m}^2$. In our case $p_m < p$ and hence the plasma cylinder will expand. It can be easily calculated that the condition $p = p_m$ will be fulfilled at the current $\mathcal{I} = 1.25 \times 10^8 \text{ A}$.

3.33. Under equilibrium the gas-dynamic pressure in a plasma is matched by the magnetic pressure, i.e.

$$nkT = \frac{B^2}{2\mu_0}.$$

whence

$$B = \sqrt{2\mu_0 nkT} \approx 6 \text{ T} = 6 \times 10^4 \text{ G}.$$

$$3.34. \quad n(r) = \frac{1}{v\epsilon\mu_0} \frac{dB}{dr} = \frac{q}{R\mu_0\epsilon} \sqrt{\frac{m}{2eU}} B_0 \left(\frac{r}{R} \right)^{q-1},$$

$$E(r) = \frac{1}{\epsilon_0\mu_0 v} B(r) = c^2 \sqrt{\frac{m}{2eU}} B_0 \left(\frac{r}{R} \right)^q.$$

Here $v = \sqrt{2eU/m}$ is the velocity of the electrons in the beam, and c the velocity of light.

3.35. The work performed by magnetic forces in displacing a contour with current $\mathcal{I}$ in a magnetic field is

$$A = \mathcal{I} \Delta\Phi,$$

where $\Delta\Phi$ is the change in the magnetic flux. In our case $\Delta\Phi = \Delta\Phi_{21}$ is the change in the flux traversing the d.c. frame. According to reciprocity theorem, $\Delta\Phi_{21} = \Delta\Phi_{12}$, if $\mathcal{I}_1 = \mathcal{I}_2$. Since $\Delta\Phi_{12} = -2\Phi_{12}$, the problem boils down to determining the magnetic flux in the frame in a d.c. field:

$$\Phi_{12} = - \int_a^b \frac{\mu_0 \mathcal{I}_1}{2\pi r} a dr = - \frac{\mu_0 \mathcal{I}_1 a}{2\pi} \ln \left(1 + \frac{a}{b} \right).$$

The minus sign depends on the choice of the positive direction of the currents in Fig. 16. Finally we get

$$A = \frac{\mu_0 \mathcal{I}_1 \mathcal{I}_2 a}{\pi} \ln \left(1 + \frac{a}{b} \right) = 2.72 \times 10^{-7} \text{ J} = 2.72 \text{ erg}.$$

$$3.36. F = \left(\frac{\partial W_m}{\partial x} \right)_{\mathcal{I}=\text{const}} = - \left(\frac{\mu_0 r N \mathcal{I}}{4} \right)^2 \frac{1}{[(\pi R/\mu) + x]^3},$$

where W_m is the magnetic energy in the coil. The minus sign indicates that both halves of the toroidal coil attract each other. For $x = 0$ we get

$$F = - \left(\frac{\mu_0 \mu r N \mathcal{I}}{4\pi R} \right)^2.$$

3.37. The force on a sample in heterogeneous magnetic field is

$$F = p_m \frac{dB}{dr},$$

where p_m is the magnetic moment of the sample and equal to κBV . This force reaches a maximum at $r_{\max} = 1/2\sqrt{\alpha} = 5 \text{ cm}$. The maximum value of the force is

$$F_{\max} = \kappa V B_0^2 \sqrt{\alpha} e^{-1/2} \approx 10^{-1} \text{ dyn}.$$

3.38. As the tube is compressed, the magnetic flux will be conserved:

$$\pi R_0^2 B_0 = \pi R^2 B, \quad R = R_0 \sqrt{B_0/B}.$$

The increase in the magnetic induction in the tube is due to the appearance of a current in its walls. Thus, the tube behaves like a solenoid. The magnetic field pressure is

$$p_m = \frac{B^2}{2\mu_0}.$$

Substituting in numerical values we get

$$B = \sqrt{2\mu_0 p_m} = 5 \times 10^2 \text{ T} \quad \text{and} \quad R = R_0 \sqrt{B_0/B} = 5 \times 10^{-3} \text{ m}.$$

3.39. When a conducting liquid moves in a magnetic field, an e.m.f. of vBd is induced in the circuit. This expression can be easily obtained by considering the action of the Lorentz force on the free charge carriers (electrons and ions). A current of

$$J = \frac{\mathcal{E}^{\text{ind}}}{r+R}, \quad \text{where } r = \frac{1}{\lambda} \frac{d}{S},$$

will run through the circuit. Thus,

$$P = J^2 R = \frac{(\mathcal{E}^{\text{ind}})^2 R}{(r+R)^2} = \frac{v^2 B^2 d^2 R}{\left(\frac{1}{\lambda} \frac{d}{S} + R\right)^2}.$$

3.40. The magnetic induction due to the current is $B = \mu_0 J / 2\pi R$. The force F acting on the needle is $F = p_m (dB/dR)$, where p_m is the magnetic moment of the needle and is equal to $B_0 V / \mu_0$ (V is the needle volume). Thus, $F = B_0 V J / 2\pi R^2$. In order to lift the needle off its peg we must have $F > \rho g V$, where ρ is the steel density. Whence

$$J > \frac{2\pi \rho g R^2}{B_0} = 2.4 \times 10^3 \text{ A}.$$

3.41. The equation for small oscillations of a bar has the form

$$I \ddot{\varphi} = -p_m B_E \varphi,$$

where I is the moment of inertia of the bar and is equal to $\frac{1}{12} \rho V$ (V is the bar volume), and p_m is the magnetic moment equal to $B_0 V / \mu_0$. Finally, the oscillation equation will be

$$\ddot{\varphi} + \frac{12 B_0 B_E}{\mu_0 \rho l^2} \varphi = 0.$$

The oscillation frequency is determined by the relationship

$$\omega_0^2 = \frac{12 B_0 B_E}{\mu_0 \rho l^2}.$$

Whence

$$T_0 = \frac{2\pi}{\omega_0} = 2\pi l \sqrt{\frac{\mu_0 \rho}{12 B_0 B_E}} \approx 4.1 \text{ s}.$$

3.42. We want to find the residual magnetization of the ball. From the definition of the demagnetization factor we have

$$H_0 - H = \beta I,$$

where H_0 is the external magnetic field, H the field inside the ball, and I the magnetization (viz. the magnetic moment per unit volume). In our case the external field is absent, therefore

$$H = -\beta I = \frac{1}{3} I.$$

In this relationship H is the magnetic field strength inside the ball due to the residual magnetization I . The magnetic induction inside the ball is

$$B = \mu_0 (H + I) = \frac{2}{3} \mu_0 I.$$

According to the problem, $B = B_0 (1 + H/H_0)$. To determine I we have the equation

$$\frac{2}{3} \mu_0 I = B_0 \left(1 - \frac{1}{3} \frac{I}{H_0} \right),$$

whence

$$I = \frac{3B_0 H_0}{2\mu_0 H_0 + B_0} \approx 1.2 \times 10^4 \text{ A/m}^2.$$

3.43. If we apply the magnetic field circulation theorem we have

$$H(L - l) + H'l = N\mathcal{I}.$$

where H is the field strength in the core, and H' the field strength in the gap. For a linear case (the inclined part of the magnetization curve) the boundary condition is $\mu H = H'$. Thus,

$$H(L - l) + \mu l H = N\mathcal{I} \text{ or } H[L + l(\mu - 1)] = N\mathcal{I}.$$

Whence

$$\mathcal{I}_0 = \frac{H_0}{N} [L + l(\mu - 1)].$$

Allowing for $(\mu - 1) = \kappa = I_0/H_0$ we find finally that

$$\mathcal{I}_0 = \frac{1}{N} [LH_0 + lI_0].$$

In the case of saturation (at $\mathcal{I} > \mathcal{I}_0$) the boundary condition should be

$$H' = H + I_0, \text{ viz. } B' = B.$$

From the magnetic field circulation theorem we get

$$(H' - I_0)(L - l) + H'l = N\mathcal{I}, \quad H' = \frac{1}{L} [N\mathcal{I} + I_0(L - l)].$$

Hence

$$B' = \mu_0 H' = \frac{\mu_0}{L} [N\gamma + I_0(L-l)] \\ = \mu_0 \left[H_0 + I_0 + \frac{N}{L} (\gamma - \gamma_0) \right].$$

$$3.44. \quad \omega = \frac{3}{2} \frac{\pi N B Q}{m} \approx 0.1 \text{ s}^{-1}.$$

$$3.45. \quad \gamma = l/n.$$

$$3.46. \quad B_A = \frac{\mu_0 I}{2}, \quad \frac{B_A}{B_{A'}} = \left(\frac{l}{r} \right)^2.$$

3.47. The equation governing a proton's radial motion in the field of an electron beam can (neglecting the magnetic field) be written

$$m\ddot{r} = eE_r = 0.$$

Here E_r is the absolute value of the radial component of the electric field of the beam as determined from Gauss's theorem: $2\pi r e_0 E_r = Q$, where $Q = en\pi r^2$ is the charge per unit length of a tube of radius r (n is the electron concentration in the beam). Thus, $E_r = ner/2e_0$. Since $\gamma = en v \pi R^2$, where for a relativistic beam $v \approx c$, the expression for E_r can be written as

$$E_r = \frac{\gamma}{2\pi e_0 c R^2} r.$$

The equation of motion for the proton takes the form

$$m\ddot{r} + \frac{e\gamma}{2\pi e_0 c R^2} r = 0,$$

whence for the frequency of the natural oscillations ω we find

$$\omega^2 = \frac{e\gamma}{2\pi e_0 c R^2 m}, \quad \text{or} \quad \omega = \frac{1}{R} \sqrt{\frac{e\gamma}{2\pi e_0 m c}}.$$

Substituting in numerical values we arrive at $\omega = 3.9 \times 10^8 \text{ Hz}$.

3.48. It follows from the solution of problem 3.31 that for $v \ll c$ we can neglect the magnetic component of the Lorentz force acting on the beam charge, as compared to the electric component. The electric field strength at the beam's boundary is

$$E = \frac{ne}{2e_0} = \frac{\gamma}{2e_0 v},$$

where n is the number of electrons per unit area of a sheet flow, and v the electron velocity. The time needed for the beam's width to double is

$$t = \sqrt{\frac{2d}{a}} = \sqrt{\frac{4e_0 d v}{j} \frac{m}{e}}.$$

Here a is the transverse acceleration of the electrons under the action of the Coulomb forces. Finally we find that

$$x = vt = \left(\frac{2W}{m}\right)^{3/4} \left(\frac{4e_0 d}{j} \frac{m}{e}\right)^{1/2} \approx 17 \text{ cm}.$$

3.49. The ion will continue to spin so long as the frequency of the a.c. electric field coincides with that of the ions' rotation in the magnetic field (the synchronicity condition):

$$\nu_0 = \frac{eB}{2\pi m} \approx 170 \text{ kHz}.$$

In this expression $m = 28 \times 1.67 \times 10^{-27} \text{ kg}$ is the mass of an N_0^+ ion.

To estimate the frequency change we must first find the spiral pitch. The velocity change for adjacent turns can be estimated from Newton's second law, i.e.

$$m\Delta v \approx Ee \frac{T}{2},$$

where T is the period of revolution. Since $R = \frac{T}{2\pi} v$, we get

$$\Delta R = \frac{T}{2\pi} \Delta v \approx \frac{T^2}{4\pi} \frac{eE}{m}.$$

The synchronicity condition will be violated if

$$\frac{\Delta v}{v_0} \gg \frac{\Delta R}{R}.$$

Whence it follows that

$$\Delta v \approx \frac{E}{2RB} \approx 15 \text{ kHz}.$$

3.50. Each charge on a dipole is acted upon in the direction $[\mathbf{vB}]$ by the force qvB . The equation governing the small oscillations of the dipole due to this force is

$$I_0 \ddot{\varphi} + p\nu B \varphi = 0,$$

whence for the oscillation frequency we get

$$\omega_0^2 = \frac{p\nu B}{I_0}.$$

$$3.51. B(t) = -\frac{B_{av}(t)}{2} + \text{const.}$$

$$3.52. t = \frac{R^2}{2} \frac{\ln(R/r)}{\mu^* U} \approx 5 \times 10^{-4} \text{ s.}$$

$$3.53. j_{\max} \approx \frac{4e_0 \delta m}{e l^2} \left(\frac{2W}{m} \right)^{3/2} \approx 40 \text{ A/m}^2 = 4 \text{ nA/cm}^2.$$

$$3.54. B \leq \frac{(A_2 - A_1) m}{e \Delta} \sqrt{\frac{2W}{A_1 m}} \approx 1.3 \times 10^{-2} \text{ T} = 130 \text{ G},$$

where A_1 and A_2 are the atomic numbers of the uranium isotopes ^{235}U and ^{238}U , and m and e are the proton's mass and charge, respectively. The time required is

$$t = \frac{Me}{mA_2 \gamma} = 8 \times 10^7 \text{ s} \approx 2.5 \text{ years.}$$

$$3.55. B \leq \frac{3b}{e l^2} \sqrt{\frac{mV}{2}} \approx 10^{-3} \text{ T} = 10 \text{ G.}$$

3.56. The estimate is easy for particles leaving the column with a root mean-square velocity $v = \sqrt{3kT/m}$ directed normally to column surface. The magnetic field near the column surface is

$$B = \frac{\mu_0 J}{2\pi R}.$$

In this field a particle that has escaped from the column will move along a circle whose radius is given by

$$\frac{mv^2}{r} = evB.$$

As a result we get

$$J = \frac{2\pi R}{\mu_0 e r} \sqrt{3kTm} \approx 2 \times 10^5 \text{ A.}$$

3.57. There is a restoring force (a quasielastic one) for the particles which leave the horizontal plane, the total force depending on the magnetic field component B_r . Since $B \sim H$ and $\text{rot } H = 0$ in the magnet gap, we get

$$\frac{\partial B_r}{\partial z} = \frac{\partial B_z}{\partial r},$$

whence for small z 's

$$B_r = \frac{\partial B_z}{\partial r} z = -\frac{n}{r_0} B_z z.$$

The Lorentz force component is

$$F_z = evB_r = -e\omega_0 n B_z z = -\omega_0^2 \mu m z,$$

where v is the linear velocity of the particle on an orbit and $\omega_0 = eB_z/m$ is the angular frequency (cyclotron frequency). It follows from the equation of vertical oscillations, i.e.

$$m\ddot{z} = -n\omega_0^2 m z,$$

that

$$\omega_z = \omega_0 \sqrt{n}.$$

3.58. An equilibrium orbit is stable to small radial deviations of a particle if the Lorentz force changes with radius r slower than the centrifugal force (Fig. 69). For small deviations $\rho = r - r_0$, so it is possible to write

$$B_z(r) = B_z(r_0) + \frac{\partial B_z}{\partial r} (r - r_0) = B_z(r_0) - \frac{n}{r_0} B_z(r_0) \rho.$$

The radial component of the Lorentz force is

$$F_L = evB_z(r) = e\omega r B_z(r) = e\omega_0 \frac{r^2}{r} B_z(r_0) \left(1 - \frac{n}{r_0} \rho\right)$$

$$= e\omega_0 r_0 B_z(r_0) \left(1 - \frac{\rho}{r_0}\right) \left(1 - \frac{n}{r_0} \rho\right) \approx F_L(r_0) - m\omega_0^2 (1 + n) \rho.$$

This expression also includes the law of the conservation of angular momentum: $\omega_0 r_0^2 = \omega r^2$, where $\omega_0 = eB_z(r_0)/m$ is the angular frequency of particle rotation on an equilibrium orbit (cyclotron frequency). We now turn to the expression for the centrifugal force:

$$F_{ct} = m\omega^2 r = \frac{m\omega_0^2 r^4}{r^3} \approx m\omega_0^2 r_0 \left(1 - 3 \frac{\rho}{r_0}\right).$$

The equation for the radial oscillations has the form

$$m\ddot{\rho} = -m\omega_0^2 (2 - n) \rho.$$

It can be seen that an oscillating mode (an equilibrium of a stationary orbit) is possible for $n < 2$. In this case

$$\omega_\rho = \omega_0 \sqrt{2 - n}.$$

3.59. It follows from symmetry that the ring currents perpendicular to an external magnetic field will cross the ball's surface. It can be easily checked that the field of the currents outside the ball will coincide with the field of a dipole placed in its centre. The field of a magnetic dipole with di-

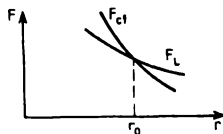


Fig. 69

pole moment p_m is

$$B = \frac{\mu_0}{4\pi} \left[\frac{3(p_m r) r}{r^5} - \frac{p_m}{r^3} \right].$$

This field along with a homogeneous external magnetic field B_0 satisfies the boundary condition on the surface of a superconducting ball: $B_{0n} + B_n = 0$ (the condition of continuity of normal components of induction vector). In fact, for any point A on the ball's surface it is possible to write (Fig. 70):

$$B_{0n} = B_0 \cos \theta, \quad B_n = \frac{\mu_0}{4\pi} \left[\frac{3p_m \cos \theta}{R^3} - \frac{p_m \cos \theta}{R^3} \right] = \frac{\mu_0 p_m \cos \theta}{2\pi R^3}.$$

At $p_m = -\frac{2\pi}{\mu_0} R^3 B_0$ the boundary condition is by definition fulfilled at any point of the ball surface. Thus,

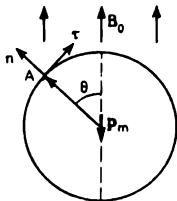


Fig. 70

$$B_0 + B = \left(1 + \frac{R^3}{2r^3} \right) B_0 - \frac{3R^3 (B_0 r) r}{2r^5}.$$

The current surface density i will be found from the boundary condition for the tangential components of magnet-

ic field strength: $H_{0\tau} + H_\tau = i$. Finally,

$$i(\theta) = \frac{3B_0 \sin \theta}{2\mu_0}.$$

3.60. In a magnetic field B_0 each ball acquires a magnetic moment

$$p_m = -\frac{2\pi}{\mu_0} R^3 B_0 = -2\pi R^3 H_0$$

(see Problem 3.59). The magnetic moment of a unit volume of a gas composed of the superconducting balls is

$$I = n p_m = -2\pi n R^3 H_0.$$

Whence it follows that the magnetic susceptibility χ defined in SI units by the expression $I = \chi \mu_0 H$ is

$$\chi = -\frac{2\pi}{\mu_0} n R^3.$$

Thus,

$$\mu = 1 - \chi = 1 - \frac{2\pi}{\mu_0} n R^3 < 1.$$

An "ideal gas" composed of superconducting balls possesses diamagnetic properties. The condition $nR^3 \ll 1$ means that the balls are spaced far apart and so their interaction may be neglected.

3.61. In our estimates, we shall assume that the ball is in a non-uniform field that changes from B_{cr} to zero at a distance of the order of a ball diameter. The magnetic moment may thus be estimated from

$$p_m \approx -\frac{2\pi}{\mu_0} R^3 B_{av},$$

where $B_{av} = \frac{B_{cr}}{2}$ (see Problem 3.59). The force acting upon the ball is

$$F_x = p_m \frac{\partial B_x}{\partial x} \approx -\frac{\pi}{2\mu_0} R^2 B_{cr}^2.$$

The superconducting ball is pushed out from the region of the strong magnetic field. By equating this force to the ball's weight we find

$$R_{max} \approx \frac{3B_{cr}^2}{8\mu_0 \rho g} \approx 1.7 \text{ cm.}$$

3.62. A dipole in a magnetic field is acted upon by a force

$$F_x = p_m \frac{\partial B_x}{\partial x} = -\frac{2\pi B R^3}{\mu_0} \frac{\partial B_x}{\partial x}$$

(see Problem 3.59).

The work against this force,

$$A = - \int_{-\infty}^0 F dx = \frac{2\pi R^3}{\mu_0} \int_0^{\mu_0} B dB = \frac{\pi B_0^2 R^3}{\mu_0},$$

is equal to the initial kinetic energy of the ball:

$$\frac{mv_0^2}{2} = \frac{\pi R^3 B_0^2}{\mu_0},$$

whence

$$v_0 = B_0 \sqrt{\frac{2\pi R^3}{m\mu_0}} = 20 \text{ m/s.}$$

Thus, at the initial velocity $v_0 \geq 20 \text{ m/s}$ the ball will fly into the solenoid.

3.63. When a superconducting solenoid is deformed, the magnetic flux is conserved: $BSN = \text{const}$, where S is the solenoid's crosssection, and N the number of turns. The magnetic pressure on the solenoid walls is

$$p_m = -\frac{H^2}{2\mu_0}.$$

Whence,

$$p_m S^2 = \text{const.}$$

3.64. From the magnetic field circulation theorem we have

$$n\gamma_0 = H_1 L + H_2 l,$$

where n is the number of turns, H_1 the field in the steel and H_2 the field in the gap. The boundary conditions yield $\mu H_1 = H_2$. Whence

$$n\gamma_0 = H_2 (L/\mu + l).$$

Since the winding is superconducting, $d\Phi/dt = 0$, viz. the magnetic flux is conserved irrespective of any change in the gap. Hence, as the gap is decreased, the field in it is unchanged. Whence

$$n\gamma = H_2 \left(\frac{L}{\mu} + \frac{l}{2} \right),$$

where γ is the new current in the winding. Thus,

$$\gamma = \gamma_0 \frac{1 + \frac{\mu}{2} \frac{l}{L}}{1 + \mu \frac{l}{L}}.$$

3.65. Inside a superconductor the magnetic field vanishes. It follows from boundary conditions that the normal component of the magnetic field induction vanishes on its surface, too. To determine the magnetic field produced by the plane we can use the method of images, viz. we imagine there is a direct current at the same height over the plane, but running in the opposite direction. The force acting upon the current per unit length due to the image is $F = \gamma B$, where B is the magnetic induction of the field produced by the image. This force is directed upwards. For the conductor to hover freely over the plane a distance h above it will be

$$F = \rho g, \quad \text{or} \quad \frac{\mu_0 \gamma^2}{2\pi \cdot 2h} = \rho g.$$

$$h = \frac{\mu_0 \gamma^2}{4\pi\rho g} \approx 2 \times 10^{-3} \text{ m.}$$

Note that the solution can easily be generalized to the case of a flat curve of arbitrary shape, if the radius of curvature of the curve significantly exceeds h in all points.

3.66. Let us assume that the cavity's dimensions are small compared with a resonance wavelength. Then the resonance frequency can be determined from the formula for a quasi-stationary circuit: $\omega_0 = 1/\sqrt{LC}$, where C is the capacitance of a parallel-plate capacitor in the central part of the cavity, while L is the inductance of the toroidal part:

$$C = \frac{\epsilon_0 \pi a^2}{4d} \approx 7 \times 10^{-12} \text{ F.}$$

When doing the calculation we use the formula $\Phi = L\mathcal{I}$. The induction B of the magnetic field in the cavity is $\mu_0 \mathcal{I}/2\pi R$, where R is the distance from the cavity's centre:

$$\Phi = \int B dS = \frac{\mu_0 a}{2\pi} \mathcal{I} \int_{a/2}^{3a/2} \frac{dr}{r} = \frac{\mu_0 a \ln 3}{2\pi} \mathcal{I} = L\mathcal{I}.$$

Hence,

$$L = \frac{\mu_0 a \ln 3}{2\pi} \approx 2.2 \times 10^{-8} \text{ H.}$$

For the resonance frequency $\omega_0 = 1/\sqrt{LC}$ we find that $\omega_0 = 2.5 \times 10^9 \text{ s}^{-1}$. The wavelength $\lambda = 2\pi c/\omega_0 \approx 1 \text{ m} \gg a$, viz. the assumption that the resonance wavelength is much larger than the cavity dimensions was valid.

$$3.67. \frac{\Delta L}{L} = \frac{1}{Q} \sqrt{\frac{2\Delta V}{U}} \approx 4.5 \times 10^{-6}.$$

$$3.68. \Delta d = \frac{d}{Q} \sqrt{\frac{2\Delta V}{U}} \approx 4.5 \times 10^{-8} \text{ cm.}$$

$$3.69. \mathcal{I} = \frac{\mathcal{E}_0}{2R} \cos 2\Omega t, \quad \mathcal{I}_1 = \frac{\mathcal{E}_0}{2(R+R_1)}.$$

$$3.70. \mathcal{I} = \frac{\mathcal{E}_0}{2R}, \quad \mathcal{I}_1 = \frac{\mathcal{E}_0 \cos(2\Omega t + \varphi)}{2\sqrt{(R_1+R)^2 + (1/2\Omega C_1)^2}},$$

$$\tan \varphi = \frac{1}{(R_1+R)2\Omega C_1}.$$

$$3.71. \Omega RC = 1, \quad \frac{U_{\text{out}}}{U_{\text{in}}} = \frac{1}{3}.$$

$$3.72. \quad q = \frac{q_0}{R} \sqrt{\frac{L}{C}}.$$

$$3.73. \quad \omega_{\text{res}} = \frac{1}{\sqrt{L_{\text{equiv}} C}}, \quad \text{where} \quad L_{\text{equiv}} = L_1 - M^2/L_2 \quad \text{is}$$

the equivalent inductance of the oscillatory circuit, allowing for the effect of the short-circuited coil.

IV. Optics

$$4.1. \quad D_c = D \frac{b_1}{F_1}, \quad E = \frac{\pi}{4} \left(\frac{D}{E_2} \right)^2 B.$$

$$4.2. \quad I = 2I_0 \left[1 - \frac{\sin 2\pi\psi(h/\lambda)}{2\pi\psi(h/\lambda)} \cos \frac{4\pi h\alpha}{\lambda} \right].$$

$$\text{At } \psi = \frac{\lambda}{2h} m \quad (m = 1, 2, 3, \dots) \quad I = 2I_0.$$

4.3. (1) The interference order $m \approx 400$. (2) The width of interference fringes $l = 2.8 \text{ cm}$. (3) The greatest source size $b \approx l = 2.8 \text{ cm}$. (4) The source's nonmonochromaticity $\Delta\lambda \leq 14 \text{ \AA}$.

4.4. (1) $m_{\text{max}} \approx 1000$, $m_{\text{min}} \approx 720$; (2) $\Delta\lambda \leq 5 \text{ \AA}$; (3) the source may have any dimensions.

$$4.5. \quad A = 1 + \frac{Dl}{Fd} = 5.$$

$$4.6. \quad \frac{\Delta l}{l} \approx \frac{2d\alpha \sqrt{n^2 - 1}}{\lambda} \approx 0.75.$$

$$4.7. \quad \Delta f \approx \frac{c(1-\rho)}{2\pi L} \approx 15 \text{ MHz}, \quad \Delta\nu = \frac{c}{2L} = 1500 \text{ MHz}.$$

4.8. The radiation from a white light source has a wide spectrum. The intensity of the interference maxima will yield the spectral components such that $\Delta = m\lambda$, where m is an integer.

Analogously, the spectral components for which $\Delta = (m + 1/2)\lambda$ will yield the intensity minima. Thus, as a result of interference, the spectral instrument will receive a radiation whose spectrum contains alternating maxima and minima. The distance between adjacent spectral maxima can be found from the relationship

$$(m+1)\lambda_{m+1} = m\lambda_m, \quad \text{or} \quad m(\lambda_m - \lambda_{m+1}) = \lambda_{m+1},$$

$$\Delta\lambda = \lambda_m - \lambda_{m+1} = \lambda/m = \lambda^2/\Delta.$$

The spectral instrument should possess a resolving power sufficient to detect two maxima spaced $\Delta\lambda$ apart:

$$R \geq \lambda/\Delta\lambda = m = \Delta/\lambda \approx 5 \times 10^5.$$

4.9. $\Delta = c/\Delta\nu = 30 \text{ m.}$

4.10. An electron excites identical perturbations in a lattice which follow one another every time interval $\tau = d/v$. An interval of time between adjacent perturbations at the point of observation, which is located at an angle θ , is equivalent to the path difference

$$\Delta = \tau c - d \sin \theta = d \left(\frac{c}{v} - \sin \theta \right).$$

In order for the perturbations to amplify one another in the θ direction we must have

$$d \left(\frac{c}{v} - \sin \theta \right) = m\lambda,$$

where m is a positive integer number ($m > 0$, since $c/v > 1$). The possible m and λ values should satisfy the relationship

$$|\sin \theta| = \left| \frac{c}{v} - m \frac{\lambda}{d} \right| \leq 1.$$

4.11. It is necessary that at the point of observation the Fraunhofer diffraction be fulfilled. The mirror size should be much smaller than the dimensions of the first Fresnel zone, i.e.

$$\frac{d}{2} \ll r_1 = \sqrt{\lambda \frac{ab}{a+b}}.$$

Assuming that the source of the radiation is at the focus of a parabolic mirror (viz. $a = \infty$) we get

$$\frac{d}{2} \ll \sqrt{\lambda b}, \quad \text{or} \quad b \gg \frac{d^2}{4\lambda} \approx 10 \text{ m.}$$

4.12. $I = I_0/4$.

4.13. $\Delta\nu = \frac{\Omega l}{c} v$; the effect is due to the differences between the optical path lengths for the two sets of waves.

4.14. $\tau \approx \frac{1}{2\pi\nu} \left(\frac{\lambda}{b} + \frac{b}{2L} \right) \approx 10^{-6} \text{ s.}$

4.15. $\frac{I}{I_0} \approx \left(\frac{D^2}{2\lambda F} \right)^2 \approx 1,5 \times 10^7$, $d \approx \frac{2\lambda F}{D} \approx 10^{-3} \text{ cm.}$

4.16. In the first case the telescope objective receives radiation whose power is

$$P_1 = P_0 \left(\frac{D_1}{L_1} \right)^2.$$

This power is distributed over an area S_1 of diffraction spot ($S_1 \approx \frac{\lambda^2}{D_1^2} F_1^2$) and produces an illumination

$$E_1 \approx \frac{P_0}{L_1^2} \frac{D_1^4}{\lambda^2 F_1^2}.$$

The background illumination in the focal plane is proportional to the lens aperture:

$$E_1^0 = a \left(\frac{D_1}{F_1} \right)^2.$$

Thus, the image contrast is

$$\gamma_1 = \frac{E_1 + E_1^0}{E_1^0} = 1 + \frac{P_0 D_1^4 F_1^2}{L_1^2 \lambda^2 F_1^2 a D_1^2} = 1 + \frac{P_0}{a} \frac{D_1^2}{L_1^2 \lambda^2}.$$

When making observations through a telescope with an objective D_2 ,

$$\gamma_2 = \frac{E_2 + E_2^0}{E_2^0} = 1 + \frac{P_0}{a} \frac{D_2^2}{L_2^2 \lambda^2}.$$

Hence, $\gamma_1 = \gamma_2$ provided that

$$\frac{L_1^2}{D_1^2} = \frac{L_2^2}{D_2^2} \quad \text{whence} \quad L_2 = L_1 \frac{D_2}{D_1} = 5 \times 10^4 \text{ km.}$$

4.17. The angular divergence of a light beam is λ/D , therefore the radiative power scattered by the satellite is

$$P' = \rho P \frac{d^2}{4 (\lambda L/D)^2} = \frac{\rho P}{4} \left(\frac{dD}{\lambda L} \right)^2.$$

Here L is the distance to the satellite. Since the satellite uniformly scatters the optical power P' throughout a solid angle 2π , the received power will be

$$P' \frac{\pi (D^2/4)}{2\pi L^2} = \frac{P'}{8} \left(\frac{D}{L} \right)^2.$$

The condition for detecting a scattered signal is now that

$$\frac{P'}{8} \left(\frac{D}{L} \right)^2 = \frac{\rho P}{32} \left(\frac{d}{\lambda} \right)^2 \left(\frac{D}{L} \right)^4 \geq P_{\text{thr}},$$

$$L \leq \frac{D}{2} \sqrt[4]{\frac{\rho P d^2}{2 P_{\text{thr}} \lambda^2}} \approx 70 \text{ km.}$$

4.18. $L \leq \frac{dD}{2\lambda} \sqrt{\frac{P}{h\nu}} \approx 5 \times 10^7 \text{ km.}$ Here h is Planck's constant.

4.19. $l_{\text{min}} \approx 1 \text{ m.}$

4.20. The resolving power of the objective is fully utilized for a normal magnification of

$$\gamma_n = \frac{D}{d} = \frac{F_1}{F_2}.$$

Whence

$$F_2 = \frac{d}{D} F_1 = 6 \text{ cm.}$$

The angular resolution of a telescope when being used for visual observations is γ_n times higher than that of the naked eye. The distance at which a text can be read using a telescope is governed by the condition

$$\frac{l}{L} \geq \frac{\lambda}{D}.$$

Whence it follows that

$$L \leq \frac{D}{\lambda} l \approx 600 \text{ m.}$$

4.21. The limiting resolving power of a five-meter telescope corresponds to a lunar surface area of $S_0 \approx \pi (\lambda R/D)^2$, where R is the distance to the Moon. This means that the illumination of any point in the Moon's image in the focal plane of the telescope objective is due to the radiation arising from every element on the Moon's surface within an area of S_0 . If part of this area is screened by a black (nonreflecting) canvass, the illumination of the moon's image at the telescope's focus will be changed in proportion to $S_0 - S$, where S is the circle area. Given the data in the problem

$$\frac{S_0 - S}{S_0} \leq 0.99.$$

Whence

$$r^2 \geq 0.01 \left(\frac{\lambda}{D} R \right)^2, \text{ or } r \geq 4 \text{ m.}$$

$$4.22. D \approx \sqrt{\frac{k\rho_1}{\rho_2}} \frac{2\lambda L}{d} \approx 6.5 \text{ m.}$$

4.23. A microscope resolves an object of size,

$$l = \frac{0.61\lambda}{n \sin u},$$

where λ is the wavelength, n the refractive index, and u the angular aperture. An electron wavelength is

$$\lambda_e = \frac{h}{mv} = \frac{12.24}{\sqrt{U}} \text{ \AA}.$$

where h is Planck's constant, and U the accelerating voltage expressed in volts.

The resolving powers of the optical and electron microscopes are equal when

$$\frac{0.61\lambda_{\text{opt}}}{\sin u_{\text{opt}}} = \frac{0.61\lambda_e}{\sin u_e} = \frac{0.61 \times 12.24}{\sqrt{U} \cdot \sin u_e}.$$

Whence

$$U = \frac{12.24 \sin u_{\text{opt}}}{\lambda_{\text{opt}} \sin u_e} \approx 400 \text{ V}.$$

4.24. The order of magnitude of the Doppler broadening can be found from the r.m.s. value of the longitudinal velocity of the gas molecules, i.e.

$$\frac{\Delta\lambda_D}{\lambda} \approx 2 \sqrt{\frac{kT}{m}}.$$

To investigate the Doppler profile a spectral instrument should have the resolving power of

$$R > \frac{\lambda}{\Delta\lambda_D} \approx \frac{c}{2\sqrt{\frac{kT}{m}}} \approx 4 \cdot 10^5.$$

When working in first order $R = N = \frac{L}{d}$, where L is the size of the diffraction grating. Thus

$$L = Rd > 40 \text{ cm}.$$

4.25. In order for a spaceship's motion to be detected by a Doppler shift in spectral lines, the shift must exceed the spectral line width caused by the thermal motion of the molecules on the Sun's surface (viz. the spaceship linear velocity should exceed the root mean-square velocity of the hydrogen molecules, which is $v_T \approx 10^4$ m/s at $T = 6000$ K). Whence it follows that $v \gtrsim v_T \approx 10^4$ m/s. The minimum number of diffraction grating line rulings is determined from the relation

$$R = mN \gtrsim \frac{\lambda}{\delta\lambda} \approx \frac{c}{v}.$$

For $m = 2$ we get $N \gtrsim 1.5 \times 10^4$.

4.26. For a split line $\delta\lambda$ to be detected in the emission spectrum of a double star due to the Doppler effect, the split must exceed the spectral line width caused by the thermal

motion of molecules. The condition is fulfilled if the linear velocity of the stars exceeds the root mean-square velocity of the molecules on their surface. For $\tau = 10$ days this condition is fulfilled if

$$R = mN \geq \frac{\lambda}{\delta\lambda} \approx \frac{c}{2v},$$

$$N \geq 10^3.$$

For $\tau = 10$ years the splitting of the lines is much less than their width.

4.27. An interference reflection of light occurs only from those sound waves for which the Wulf relation is fulfilled (Fig. 71). The Doppler shift of the scattered light is

$$\Delta\nu = 2v \frac{\nu}{c/n} \sin \frac{\theta}{2}.$$

For reliable resolution we must have

$$R = mN \geq \frac{\nu}{\Delta\nu}.$$

Whence for $m = 1$

$$N \geq \frac{c}{2vn \sin(\theta/2)} \approx 1.2 \times 10^5.$$

4.28. $N \geq \frac{\alpha}{4} \left(\frac{c}{v} \right)^2 \approx 2.5 \times 10^6.$

4.29. The resolving power R of a Fabry-Perot interferometer can be estimated by considering the following. The minimum path-length difference between rays forming an interference pattern in the interferometer is $2L$. Hence, the interference order m is $2L/\lambda$. The effective number of rays leaving the interferometer is $N \approx 1/(1-\rho)$. By analogy with a diffraction grating we can write

$$R = \frac{\omega}{\Delta\omega} = mN \approx \frac{2L}{\lambda} \frac{1}{1-\rho} = \frac{\omega L}{c(1-\rho)}.$$

Whence

$$\Delta\omega \approx \frac{c(1-\rho)}{L}.$$

Since $\omega = \omega_{IR} + \omega_L$, where $\omega_L = \text{const}$ is the laser radiation frequency, and ω_{IR} the frequency of the investigated source in the infrared range, we can take $\Delta\omega = \Delta\omega_{IR}$.

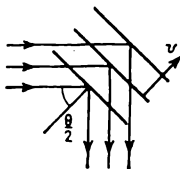


Fig. 71

Hence,

$$R_{IR} = \frac{\omega_{IR}}{\Delta\omega_{IR}} = \frac{\omega_{IR}}{\Delta\omega} = \frac{\omega_{IR}}{\omega} R \approx \frac{\omega_{IR}L}{c(1-\rho)} = \frac{2\pi L}{\lambda_{IR}(1-\rho)} \\ \approx 2 \cdot 10^8.$$

4.30. Linearly polarized light does not transfer angular momentum. If one of the major directions of the quarter-wave plate coincides with that of the incident light's polarization, the transmitted light will also be linearly polarized and the plate will not experience a torque. As the major directions of the plate are rotated through $\pm 45^\circ$ relative to the polarization plane of the incident light, the transmitted light will be circularly polarized, either clockwise or anticlockwise. This light will transfer angular momentum at a rate $L = \pm \hbar n$ per unit time, where n is the number of photons passing through the plate per second. The plate will experience a recoil angular momentum of

$$M = -L = \pm \hbar n = \pm \hbar \frac{P}{\hbar \nu} = \pm \frac{P}{\nu},$$

where P is the power of the transmitted light and ν its frequency. Substituting the numerical P and ν values into the expression we get

$$M = 6.28 \times 10^{-16} \text{ N}\cdot\text{m}.$$

4.31. z is the optical axis. $n_o = \sqrt{\frac{\epsilon_1 d_o + \epsilon d}{d + d_o}},$

$$n_e = \sqrt{\epsilon \epsilon_1 \frac{d + d_o}{d_o \epsilon + d \epsilon_1}}.$$

4.32. $A = \frac{A_0}{2^N} \left| \frac{\sin(2^N \varphi/2)}{\sin(\varphi/2)} \right|$, where $\varphi = \frac{2\pi}{\lambda} d(n_e - n_o)$.

Such a system is a polarized interference filter acting like a diffraction grating.

4.33. The gas's refractive index is

$$n = \sqrt{1 + 4\pi\alpha N},$$

where α is the polarizability of the gas molecules (in the Gaussian system) and N their concentration. Since

$$N(h) = \frac{p_0}{kT} \exp \left[-\frac{mgh}{kT} \right],$$

where p_0/kT is the molecular concentration at $h = 0$, we get

$$n = \sqrt{1 + 4\pi\alpha \frac{p_0}{kT} \exp\left[-\frac{mgvh}{kT}\right]} \approx 1 + 2\pi\alpha \frac{p_0}{kT} \left(1 - \frac{mgvh}{kT}\right).$$

The radius of curvature of a beam that starts horizontally near planet's surface is

$$R = \frac{n}{dn/dh} \approx -\frac{(kT)^2}{2\pi\alpha p_0 mgv} \approx -10^2 \text{ km.}$$

Since $R < 0$, the centre of curvature is below the surface. Thus, horizontal and near-horizontal rays cannot escape from Venus's atmosphere ($|R| < R_V$). Circular refraction is thus possible in Venus's atmosphere, if a light beam envelopes the planet at some height.

4.34. The refraction of a horizontal beam is described by the formula

$$\frac{1}{R} = \frac{1}{n} \frac{dn}{dh},$$

where R is the radius of curvature of the beam and n the gas's refractive index as defined by the expression

$$n - 1 = (n_0 - 1) \exp\left[-\frac{mgh}{kT}\right] \approx (n_0 - 1) \left(1 - \frac{mgh}{kT}\right).$$

The last expression is valid at small values of the height ($mgh/kT \ll 1$). Allowing for $(n - 1) \ll 1$ we get

$$\frac{1}{R} = -(n_0 - 1) \frac{mg}{kT}, \quad R = -\frac{1}{n_0 - 1} \frac{kT}{mg} \approx -2.9 \times 10^4 \text{ km.}$$

The negative R value indicates that the beam's centre of curvature is at $h < 0$.

Given circular refraction the beam's radius of curvature is equal to the Earth's radius, i.e. $R_E = 6.4 \times 10^3 \text{ km}$. Since

$R \sim \frac{1}{n_0 - 1} \sim \frac{1}{p_0}$, the atmospheric pressure (and density, too) should be increased by $|R|/R_E = 4.5$ times.

4.35. The lens and hence the ball receive a radiation power of $\beta \frac{\pi D^2}{4}$, where D is the objective's diameter. The ball's radius is equal to that of the Sun's image, $\alpha F/2$, where F is the lens's focal distance. The ball radiates a power of $\sigma T^4 \cdot 4\pi \frac{\alpha^2 F^2}{4}$ ($\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2\text{K}^4)$ is the Stephan-Boltzmann constant).

The steady-state temperature T can be found from the energy balance:

$$\beta \frac{\pi D^2}{4} = \sigma T^4 \cdot 4\pi \frac{\alpha^2 F^2}{4}.$$

Whence

$$T = \sqrt[4]{\frac{\beta}{4\sigma\alpha^2} \left(\frac{D}{F}\right)^2} \approx 1.7 \times 10^3 \text{ K}.$$

4.36. An electric breakdown occurs when an electron affected by the electric field of a light wave gains an energy of $10 \text{ eV} = 1.6 \times 10^{-18} \text{ J}$. This energy is much less than the electron's rest energy of $m_0 c^2 = 10^{-13} \text{ J}$, viz. the electrons' velocity during the breakdown is much less than c . In one oscillation period the electron moves a distance of order of c/v , where v is the frequency of light. This is much less than the light's wavelength, $\lambda \approx 5 \times 10^{-5} \text{ cm}$, and electron free path $\Lambda = 10^{-4} \text{ cm}$.

Thus, we can consider the action of a light wave upon a free electron

$$m\ddot{x} = -eE_0 \cos \omega t,$$

where E_0 is the light wave's amplitude. By integrating we get an expression for the electron velocity: $v = -\frac{eE_0}{m\omega} \sin \omega t$.

The maximum electron kinetic energy is

$$\frac{mv^2}{2} = \frac{m}{2} \left(\frac{eE_0}{m\omega} \right)^2 = eU.$$

Whence

$$E_0^2 = \frac{2U m \omega^2}{e}.$$

The radiation power flux density in the beam is

$$\frac{P}{S} = ce_0 E_{av}^2,$$

where $S = \frac{\pi}{4} D^2$, $E_{av}^2 = E_0^2/2$. Finally,

$$P = ce_0 S \frac{E_0^2}{2} = \frac{ce_0 S U m \omega^2}{e} \approx 2 \times 10^{12} \text{ W}.$$

4.37. Since at sliding angles of incidence, $\varphi \approx 4\alpha$, we find from the condition of total internal reflection that

$$\cos \alpha = \cos \frac{\varphi}{4} = n = 1 - \frac{\varphi_2}{32}.$$

The refractive index is

$$n = \sqrt{\epsilon} = \sqrt{1 - \frac{e^2 N}{\epsilon_0 m \omega^2}} \approx 1 - \frac{e^2 N}{2 \epsilon_0 m \omega^2}.$$

For Be: $Z = 4$, $A = 9$, $N = Z N_A \rho / A$ (N_A is Avogadro's number). Whence

$$\varphi \approx \sqrt{32 \frac{\lambda^2 e^2 Z N_A \rho}{\epsilon_0 m c^2 8 \pi^2 A}} \approx 5 \cdot 10^{-2} \text{ rad} \approx 2.9^\circ.$$

4.38. A photon with energy 5 eV corresponds to the wavelength $\lambda = 3 \times 10^{-5}$ cm. The transparency boundary is given by the equation

$$\epsilon = 1 - \frac{e^2 N}{\epsilon_0 m \omega^2} = 0.$$

Whence

$$N = \frac{\epsilon_0 m \omega^2}{e^2} = \frac{4 \pi^2 \epsilon_0 c^2 m}{\lambda^2 e^2} \approx 10^{24} \text{ m}^{-3}.$$

The concentration of atoms in silver is

$$n = \frac{N_A \rho}{A} \approx 6 \times 10^{28} \text{ m}^{-3},$$

where N_A is Avogadro's number. Thus,

$$\frac{N}{n} \approx \frac{1}{6}.$$

4.39. Plasma's permittivity ϵ

$$\epsilon = n^2 = 1 - \frac{e^2 N}{\epsilon_0 m \omega^2},$$

where e and m are the electron's charge and mass, respectively, and N the electron concentration. Whence

$$N = \frac{(1 - n^2) m \omega^2 \epsilon_0}{e^2} \approx 2.4 \times 10^{11} \text{ m}^{-3}.$$

Now we find the radiowave's phase velocity, i.e.

$$v = c/n \approx 3.32 \times 10^8 \text{ m/s}.$$

To determine the group velocity u we need to know the function $v(\lambda)$. This can be easily established from the expression for n^2 , if we take $n = c/v$ and $\omega = 2\pi v/\lambda$:

$$v = \sqrt{c^2 + \frac{N e^2 \lambda^2}{4 \pi^2 \epsilon_0 m}}.$$

Using the Rayleigh relationship

$$u = v - \lambda \frac{dv}{d\lambda},$$

we get

$$u = \frac{c^2}{v} = cn \approx 2.7 \times 10^8 \text{ m/s.}$$

4.40. The focal spot's diameter is

$$\frac{\lambda}{D} F \approx \lambda.$$

The acceleration of the electron is due to the electric field E . The final electron velocity is given by the relation

$$\frac{m_0 c^2}{\sqrt{1-\beta^2}} = 2m_0 c^2, \quad \text{whence } \beta^2 = \left(\frac{v}{c}\right)^2 = \frac{3}{4}.$$

Thus, the average electron velocity is close to $c/2$. Since the acceleration only occurs for an interval of about $T/4$, the distance travelled will be about $\frac{c}{2} \frac{T}{4} = \frac{\lambda}{8}$, viz. smaller than the focal spot size. Now we estimate the accelerating field strength E :

$$\frac{4c}{T} = \frac{4c^2}{\lambda} \approx \frac{eE}{m_0}, \quad E \approx \frac{4m_0 c^2}{\lambda e}.$$

The laser's power should be

$$P \approx (\epsilon_0 E^2) c \lambda^2 = \frac{16 \epsilon_0 m_0^2 c^5}{e^2} \approx 10^{10} \text{ W.}$$

4.41. Using a system of coordinates in which the satellite is at rest, we find the frequency of the signal received by the satellite, i.e.

$$\omega_1 = \omega_0 \frac{\sqrt{1-v^2/c^2}}{1 + \frac{v}{c} \cos \alpha}.$$

The reflected signal will have the same frequency in the satellite's system, but in the Earth-bound system we detect, at point B , a signal whose frequency is

$$\omega_2 = \omega_1 \frac{\sqrt{1-v^2/c^2}}{1 - \frac{v}{c} \cos \theta} = \omega_0 \frac{1 - v^2/c^2}{\left(1 + \frac{v}{c} \cos \alpha\right) \left(1 - \frac{v}{c} \cos \theta\right)}.$$

The relativistic correction to the frequency is

$$\left(\frac{\Delta\omega}{\omega_0}\right)_{\text{rel}} \approx \frac{v^2}{c^2} \approx 10^{-9},$$

whence it follows that the resolving power of the spectral device must be

$$R \gg \left(\frac{\omega_0}{\Delta\omega}\right)_{\text{rel}} \approx 10^9.$$

4.42. The corrections due to relativity account for a $(v/c)^2$ fraction of measured quantities. The plasma's refractive index $n = \sqrt{\epsilon}$ differs from unity by

$$\Delta n = \sqrt{\epsilon} - 1 \approx \frac{e^2 N}{2\epsilon_0 m \omega^2},$$

where e and m are an electron's charge and mass, respectively. Since the ionosphere (its electron density) changes in a random fashion, the orbital parameters must be measured with errors less than Δn in order to detect reliably the effects predicted by the theory of relativity. Thus,

$$\left(\frac{v}{c}\right)^2 \geq \Delta n \approx \frac{e^2 N}{2\epsilon_0 m \omega^2}.$$

Whence

$$v = \frac{\omega}{2\pi} \geq \frac{ec}{2\pi\nu} \sqrt{\frac{N}{2\epsilon_0 m}} \approx 0.8 \times 10^{11} \text{ Hz}.$$

4.43. $v = \frac{3}{5} c.$

4.44. $\Delta t = 3 \times 10^{-8} \text{ s}, \quad \delta(\Delta t)/\Delta t \approx 0.06\%.$

V. Atomic Physics

5.1. The laws of conservation of energy and momentum for the Compton scattering of a γ -quantum through the angle of 180° yield

$$\hbar\omega + mc^2 = \hbar\omega' + \sqrt{p^2 c^2 + m^2 c^4}, \quad \frac{\hbar\omega}{c} = -\frac{\hbar\omega'}{c} + p,$$

where p is the electron momentum while $\hbar\omega$ and $\hbar\omega'$ are the γ -quantum energies before and after the scattering, respectively. Given that $E \gg mc^2$ the momentum $p \gg mc$. It follows that $\hbar\omega' \approx mc^2/2 = 0.25 \text{ MeV}.$

5.2. If the total energy, momentum, and angle of ejection of a recoil electron are E , p , and φ , respectively, the laws of conservation for the Compton scattering, given data of the

problem, are

$$\hbar\omega + mc^2 = \frac{\hbar\omega}{2} + E, \quad \frac{\hbar\omega}{c} = p \cos \varphi, \quad \frac{\hbar\omega}{2c} = p \sin \varphi,$$

and we also have

$$E^2 = p^2 c^2 + m^2 c^4.$$

Solving this system of equations we get $\hbar\omega = mc^2 = 0.5$ MeV, whence $\lambda \approx 2.5 \times 10^{-10}$ cm.

5.3. The resolving power of the spectrograph is

$$R = \frac{\lambda}{\delta\lambda} = Nm,$$

where N is the number of reflecting layers, m the interference order. According to the Bragg relation:

$$2d \sin \theta = m\lambda, \quad m = \frac{2d \sin \theta}{\lambda}.$$

Here d is the diffraction grating constant. The change in the wavelength is governed by the Compton formula,

$$\delta\lambda = 2\pi\Lambda (1 - \cos \varphi),$$

where $\Lambda = \hbar/mc = 3.9 \times 10^{-11}$ cm is the Compton length. Thus,

$$R = Nm = \frac{D}{d} \frac{2d \sin \theta}{\lambda} = \frac{\lambda}{2\pi\Lambda (1 - \cos \varphi)}.$$

Whence

$$D = \frac{\lambda^2}{4\pi\Lambda \sin \theta (1 - \cos \varphi)}.$$

Substituting in the numerical values we get $D = 20$ Å.
5.4. If the initial photon energy is $\hbar\omega$, the final one $\hbar\omega'$, the photon scattering angle θ , the electron kinetic energy E_k , its momentum p , and the electron ejection angle φ , the law of conservation of energy and momentum is

$$\hbar\omega = \hbar\omega' + E_k, \quad \frac{\hbar\omega'}{c} \sin \theta = p \sin \varphi, \quad \frac{\hbar\omega}{c} = \frac{\hbar\omega'}{c} \cos \theta + p \cos \varphi.$$

Whence

$$\left(\frac{\hbar\omega'}{c} \right)^2 = \left(\frac{\hbar\omega}{c} \right)^2 + p^2 - \frac{2\hbar\omega}{c} p \cos \varphi$$

and

$$\hbar\omega' = \frac{p^2 c^2 + E_k^2 - 2pcE_k \cos \varphi}{2(pc \cos \varphi - E_k)}, \quad \text{if } (pc \cos \varphi - E_k) > 0.$$

Finally,

$$\cos \varphi > \left(\frac{E_k}{pc} \right) = \sqrt{1 + \left(\frac{mc^2}{pc} \right)^2} - \frac{mc^2}{pc}.$$

5.5. Assume a photon with momentum $p = \hbar\omega/c$ and mass $m = p/c$ flies along the z -axis (Fig. 72). The change in the photon's transverse momentum is

$$\begin{aligned} \Delta p_x &= \int_{-\infty}^{+\infty} F_x dt = \int_{-\infty}^{+\infty} \frac{\gamma M_S m}{R_S^2 + z^2} \cos \theta dt = \int_{-\infty}^{+\infty} \frac{\gamma M_S m R}{(R_S^2 + c^2 t^2)^{3/2}} dt \\ &= \frac{2\gamma M_S m}{R_S c}. \end{aligned}$$

The angle of photon's deviation can be estimated thus

$$\sin \varphi \approx \frac{\Delta p_x}{p_x} = \frac{2\gamma M_S}{R_S c^2} = 5 \times 10^{-5} \text{ rad.}$$

5.6. The m -th order reflection in the coordinate frame related to the crystal will be

$$2d \sin \varphi = m\lambda = m \frac{h}{Mv},$$

where h is Planck's constant, M a neutron's mass, and v its velocity relative to the crystal. In the first case the neutron source is at rest, and in the second case it moves with a velocity u . Therefore

$$\frac{\sin \varphi_0}{\sin \varphi} = \frac{|v+u|}{|v|}.$$

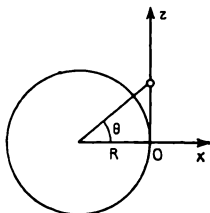


Fig. 72

Further, we assume that $u \ll v$. In this situation we have approximately

$$\frac{|u+v|}{|v|} \approx 1 + \frac{u}{v} \sin \varphi_0$$

and hence

$$\sin \varphi \approx \left(1 + \frac{u}{v} \sin \varphi_0 \right) \sin \varphi_0.$$

5.7. The neutron's De Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{ht}{mL} = \frac{h\varphi}{mL\omega} = \frac{h\varphi}{mL\nu} \approx 6 \text{ \AA} \text{ and}$$

$$\frac{\Delta \lambda}{\lambda} = \frac{b}{R\varphi} = 0.2.$$

5.8. $D = \left(\frac{hL}{\sqrt{3mkT}} \right)^{1/2} \approx 2 \text{ } \mu\text{m}$, where m is the mass of silver atoms, k the Boltzmann constant.

5.9. The energy at the n -th level is

$$E_n = \frac{\pi^2 \hbar^2}{2mb^2} n^2.$$

Each level possesses two electrons, therefore the total energy of all the electrons is

$$E = 2 \frac{\pi^2 \hbar^2}{2mb^2} \sum_{n=1}^{N/2} n^2$$

and the force

$$F = \frac{\partial E}{\partial b} = - \frac{2\pi^2 \hbar^2}{mb^3} \sum_{n=1}^{N/2} n^2.$$

5.10. Given that $\Delta p \approx p$ and $\Delta r \approx r$, the uncertainty relation yields

$$p = \frac{\hbar}{r}, \quad E_k = p^2/2m \approx 0.2 \text{ MeV}.$$

5.11. Nucleons would be expected to emit field quanta virtually within a time interval t as allowed by the uncertainty relation, $t \sim \hbar/mc^2$. During this interval the quanta succeed in travelling a distance $r \sim ct \approx \hbar/mc \approx 2 \times 10^{-13} \text{ cm}$ from the nucleon.

5.12. The angle φ between the direction of the particle's trajectory and the radiation's direction is given as

$$\cos \varphi = \frac{c}{nv}.$$

An electron, when placed in a layer of thickness d , has a momentum uncertainty of $\Delta p \approx \hbar/d$ and a velocity uncertainty of $\Delta v = \hbar/dm$, where m is an electron's mass. The related uncertainty in the angle φ is

$$\Delta \varphi \approx \frac{c\hbar}{mdnv^2 \sin \varphi}.$$

5.13. $E_n = U + n^2 \frac{\pi^2 \hbar^2}{2mb^2}$; for $n=1$ the energy $E_1 \approx 35 \text{ eV}$.

5.14. The ionization potential I is $\frac{m e^4}{2\hbar^2}$, where $m = \frac{m_e}{1+m_e/M}$ is the reduced mass. For positronium, $M = m_e$ and $I = 6.8 \text{ eV}$, for muonium $I = 13.5 \text{ eV}$.

5.15. The position of energy level is given by the formula

$$E_n = -\frac{1}{2} \mu c^2 \cdot \alpha^2 \frac{1}{n^2} = A \frac{1}{n^2}, \quad (1)$$

where the reduced mass μ is defined as

$$\frac{1}{\mu} = \frac{1}{m_p} + \frac{1}{m_\mu} \quad (2)$$

and α is the fine structure constant and is equal to $1/137$. The energy of a transition is

$$\Delta E = A \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right), \quad (3)$$

where $n_1 = 2$ and $n_2 = 3$ for the M - N -transition. Finally, from Eqs. (1)-(3) we get $\Delta E = 124$ eV.

5.16. The mean value of the potential energy of an electron at the instant the meson decays can be found from the equation

$$\langle E_p \rangle = \frac{1}{8\pi r_1^3} \int_0^\infty e^{-r/r_1} \left(1 - \frac{r}{2r_1} \right)^2 \left(-\frac{e^2}{r} \right)^2 4\pi r^2 dr = -\frac{e^2}{4r_1}.$$

The mean kinetic energy is

$$\langle E_k \rangle = E - \langle E_p \rangle = -\frac{e^2}{8r_1} + \frac{e^2}{4r_1} = \frac{e^2}{4r_1} \approx 4 \text{ eV}.$$

5.17. The probability of finding a meson a distance r from the centre of a nucleus is

$$w(r) = \frac{4\pi r^2}{\pi r_1^3} e^{-2r/r_1} dr.$$

The energy correction is

$$\Delta E = \int_0^{r_N} \frac{ze^2}{r} \frac{4r^2}{r_1^3} e^{-2r/r_1} dr = \frac{ze^2}{r_1} \left[1 - \left(1 + \frac{2r_N}{r_1} \right) e^{-2r_N/r_1} \right].$$

Since

$$r_N = 1.3 \times 10^{-13} A^{1/3} \approx 3.5 \times 10^{-13} \text{ cm},$$

$$r_1 = \frac{\hbar^2}{2m_{\mu} \alpha e^2} \approx 2.5 \times 10^{-12} \text{ cm},$$

we see that $r_1 \gg r_N$ and

$$\Delta E \approx \frac{ze^2}{r_1} \left(\frac{2r_N}{r_1} \right)^2.$$

Since

$$\langle E \rangle = 2 \langle E_p \rangle = \frac{2ze^2}{r_1},$$

where E_p is the potential energy, finally we get

$$\frac{\Delta E}{E} \approx 2 \left(\frac{r_N}{r_1} \right)^2 \approx 4 \times 10^{-2}.$$

5.18. To make an estimate, we shall assume that the electron's initial momentum p_0 was perpendicular to the photon's trajectory. The value of p_0 will be found from

$$p_0 r \sim \hbar. \quad (1)$$

The electron's longitudinal momentum p_e , after colliding with the photon will be found from the law of conservation of momentum:

$$\frac{\hbar \omega}{c} = \frac{\hbar \omega'}{c} + p_e \quad (2)$$

and $\Delta \omega$ from the Compton formula

$$\Delta \lambda = \lambda_2 - \lambda_1 = 2\pi \Lambda (1 - \cos \varphi), \quad (3)$$

where $\Lambda = 3.9 \times 10^{-11}$ cm is the Compton wavelength. In our case

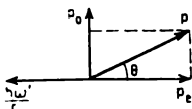


Fig. 73

$$\Delta \lambda = 4\pi \Lambda. \quad (4)$$

Using (2) and (4) we can easily see that the electron in this problem is nonrelativistic. Now we estimate the spread in

the electron's scattering angle (Fig. 73):

$$\tan \theta = p_0/p_e \approx 0.14, \text{ or } \theta \approx 10^\circ.$$

5.19. The intensity variation in time t is

$$J = J_1 e^{-t/\tau}, \quad (1)$$

where τ is the mean lifetime of an atom and $t = x/v$. Taking the natural logarithms of both sides of (1) we get

$$\ln \frac{J}{J_1} = -\frac{x}{v\tau}.$$

From the graph in Fig. 49 we find that $1/v\tau = 1 \text{ cm}^{-1}$, whence $\tau = 10^{-8}$ s. The uncertainty in the energy is governed by the natural linewidth, i.e.

$$\Delta E \approx \frac{\hbar}{\tau} = 10^{-7} \text{ eV}.$$

5.20. By emitting a γ -quantum the dust speck acquires a recoil momentum $Mv = E/c$, where E is the γ -quantum's energy. The Doppler shift in the γ -quantum's frequency due to the motion of the emitter (speck) is given by

$$\frac{\Delta\omega}{\omega} = \frac{v}{c}. \quad (2)$$

We can find the natural width of the spectral line from the uncertainty principle, viz.

$$\Delta\omega \approx \frac{\Delta E}{\hbar} = \frac{2\pi}{\tau}. \quad (3)$$

Using (2)-(3) we determine the smallest mass the speck can have for the Mössbauer effect still to be observed:

$$M = \frac{E}{vc} = \frac{E^2\tau}{2\pi c^2\hbar} \approx 1.6 \times 10^{-17} \text{ kg}.$$

The speck's radius will be approximated by assuming the speck is a sphere with a density ρ of 8 g/cm^3 , i.e.

$$r = \left(\frac{3M}{4\pi\rho} \right)^{1/3} \approx 2 \times 10^{-5} \text{ cm}.$$

5.21. The change in the γ -quantum's energy as it moves a distance H against gravity is

$$\Delta E = mgH = \frac{E}{c^2} gH.$$

Using the data in the problem, $\Delta E = 10 \Gamma = 10\hbar/\tau$, where Γ is the linewidth of the resonance absorption line. Finally,

$$H = \frac{10\hbar c^2}{\tau g E} \approx 400 \text{ m}.$$

5.22. Rotational levels have energies

$$E_l = \frac{\hbar^2}{2I} l(l+1).$$

For each of the three levels we can write

$$E_1 = \frac{\hbar^2}{2I} l(l+1), \quad E_2 = \frac{\hbar^2}{2I} (l+1)(l+2),$$

$$E_3 = \frac{\hbar^2}{2I} (l+2)(l+3),$$

whence

$$\Delta E_{1,2} = E_2 - E_1 = \frac{\hbar^2}{I} (l+1) = 10^{-4} \text{ eV},$$

$$\Delta E_{2,3} = E_3 - E_2 = \frac{\hbar^2}{I} (l+2) = 2 \times 10^{-4} \text{ eV},$$

or

$$\frac{l+2}{l+1} = 2,$$

whence it follows that $l = 0$ and $l = 0, 1$, and 2 correspond to the three levels. The value of

$$I = \frac{\hbar^2(l+1)}{\Delta E_{1,2}} = 7 \times 10^{-46} \text{ kg} \cdot \text{m}^2.$$

5.23. The splitting of a neutron beam in a homogeneous magnetic field is due to the reorientation of the neutron spins along and against the field. The change in a neutron's energy when it intersects the boundary of a magnetic field ΔE_k is $\pm \mu B$, where the "+" sign corresponds to the case when the neutron spin is parallel to the field, and the "-" sign when it is antiparallel, while $\mu = \mu_B$ is the projection of the magnetic moment onto the field direction and μ_B is the Bohr magneton. The neutron's kinetic energy is related to the De Broglie wavelength λ thus $E_k = 2\pi^2\hbar^2/m\lambda^2$. The refractive index is

$$n_{\pm} = \left\{ 1 \pm \frac{\mu B}{E_k} \right\}^{1/2} \approx 1 \pm \frac{\mu B}{2E_k}.$$

For the angles of refraction at the boundary we have

$$\frac{\sin 60^\circ}{\sin \varphi} = \frac{1}{n_{\pm}}.$$

For a small angle of splitting $\theta = \varphi_+ - \varphi_-$, finally we have

$$\theta \approx \frac{\sin 60^\circ}{\cos \varphi} (n_+ - n_-) = \frac{\mu B}{E_k} \tan 60^\circ = \frac{m\lambda^2 \mu B}{2\pi^2 \hbar^2} \sqrt{3} \approx 8 \times 10^{-6} \text{ rad}$$

(φ is the average angle of refraction, viz. $\approx 60^\circ$).

5.24. Na atoms at the temperature given are mainly in the $^2S_{1/2}$ state. The projection of the magnetic moment of a Na atom onto the field direction is

$$\mu_H = m_l g \mu_B = \mu_B,$$

since $m_l = \pm 1/2$, $g = 2$ (Lande's splitting factor); μ_B is the Bohr magneton. The force which splits a beam of atoms flying along the y -axis is

$$F_x = \pm \mu_B \frac{dB}{dx},$$

which in time $t = l/v$ imparts to the atoms a velocity

$$v_x = \pm \mu_B \frac{dB}{dx} \frac{l}{Mv}$$

and this results in the displacement of the beams across a detector, i.e.

$$x' = \pm \mu_B \frac{dB}{dx} \frac{l(L+l/2)}{Mv^2}.$$

The kinetic energy of the atoms is

$$\frac{Mv^2}{2} = \frac{3}{2} kT.$$

The distance between spots on the screen will thus be

$$\Delta = 2x' = 2\mu_B \cdot \frac{dB}{dx} \cdot \frac{l(L+l/2)}{3kT} \approx 2 \text{ cm.}$$

5.25. As in the previous problem, the force deflecting the Li atoms is

$$F = ma = \mu_B \cdot \frac{dB}{dx},$$

where a is the acceleration perpendicular to the direction of the velocity $v = \sqrt{2E_k/m}$. The linear deflection of the atoms across the screen during the time they fly through the magnet, $t_1 = L_1/v$, and through the gap between the magnet and screen, $t_2 = L_2/v$, is

$$\Delta = a \frac{t_1^2}{2} + at_1 t_2 = at_1(t_1/2 + t_2).$$

The distance between two beams on the screen is 2Δ . The size of the image of each beam is $D_{\text{im}} = D \cdot \frac{L_1 + L_2 + L/2}{L/2}$. Since $D_{\text{im}} < \Delta$ and for $L_2 \gg L$ and L_1 we finally get

$$D < \frac{\mu_B dB/dx}{2E_k} L L_1 \approx 0.9 \text{ cm.}$$

5.26. The resolving power of an interferometer is

$$R = \frac{\lambda}{\Delta\lambda} = mN,$$

where $m = 2L/\lambda$. Thus,

$$\frac{\lambda}{\Delta\lambda} = \frac{2LN}{\lambda},$$

whence

$$L = \frac{\lambda^2}{2N \Delta\lambda}.$$

If we transform from the wavelength λ to the circular frequency $\omega = \frac{2\pi c}{\lambda}$ and $\Delta\omega = -\frac{2\pi c}{\lambda^2}\Delta\lambda$, we get $L = \frac{\pi c}{N\Delta\omega}$. The Zeeman splitting is

$$\Delta\omega = \frac{eB}{2mc} = \frac{\mu_B B}{\hbar}.$$

Finally,

$$L = \frac{2\pi mc^2}{N\epsilon B} \approx 5 \cdot 10^{-2} \text{ cm.}$$

5.27. The Zeeman splitting should exceed the Doppler broadening both because of the thermal motion of the atoms

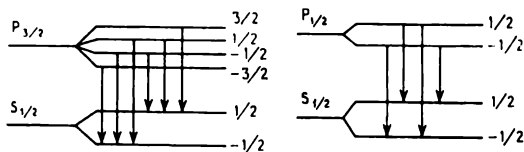


Fig. 74

(mainly hydrogen atoms) and because of the star's rotation, viz.

$$\frac{\Delta\omega}{\omega} = \frac{2\mu_B B}{\hbar\omega_0} \geq \left(\frac{\Delta\omega}{\omega}\right)_{\text{therm}} + \left(\frac{\Delta\omega}{\omega}\right)_{\text{rot}} = \frac{v_{\text{therm}}}{c} + \frac{v_{\text{rot}}}{c} \approx \frac{v_{\text{therm}}}{c}.$$

Finally,

$$B \geq \frac{\hbar\omega_0}{2\mu_B c} \sqrt{\frac{2kT}{m}} \approx 0.18 \text{ T.}$$

5.28. The splitting in a magnetic field,

$$\Delta\lambda \approx \frac{\Delta E \lambda^2}{2\pi c \hbar} \approx 0.3 \text{ \AA},$$

where $\Delta E \approx \mu_B B \approx 2 \times 10^{-4} \text{ eV}$. Since $\Delta\lambda$ is much less than $\lambda_1 - \lambda_2$, the effect is anomalous. One line of the doublet will split into six components, the other into four components (Fig. 74).

5.29. The difference in energy is due to the rotation of the electron's spin in the magnetic field produced by the orbital motion of the electron. Therefore $\Delta E = 2\mu_B B$, whence

$$B = \frac{\Delta E}{2\mu_B} = \frac{\hbar\Delta\omega}{2\mu_B} = \frac{2\pi\hbar c \Delta\lambda}{2\lambda^2 \mu_B} \approx 0.3 \text{ T.}$$

5.30. The radius of the second Bohr orbit of positronium is

$$r_2 = \frac{4\hbar^2}{me^2} = 8r_B,$$

where $m = m_e/2$ is the reduced mass for positronium, and r_B the radius of the first orbit of a hydrogen atom. The projections of the magnetic moment of electron and that of a positron are equal to the Bohr magneton μ_B . The interaction energy is approximately

$$E \approx \frac{\mu_B^2}{(8r_B)^3} \approx 10^{-8} \text{ eV}.$$

The spins of the states are 0 and 1, respectively.

5.31. The energy of hyperfine splitting is

$$\Delta E \approx \frac{\mu_p \mu_e}{r_n^3} \approx 10^{-8} \text{ eV},$$

where μ_p and μ_e are the magnetic moments of proton and electron, respectively, and $r_n = \frac{2^2 \hbar^2}{mc^2}$ is the Bohr orbit radius at $n = 2$. The uncertainty in the energy due to the natural broadening of spectral line, $\delta E \approx \hbar/\Delta t \approx 10^{-7} \text{ eV}$. Since $\Delta E \approx 0.16E$, the splitting is undetectable.

5.32. The energy change during spin reorientation is

$$\Delta E \approx 2 \frac{\mu_p \mu_e}{r_B^3} \approx 10^{-8} \text{ eV},$$

where r_B is the radius of the ground state of a hydrogen atom. Finally, $\lambda = \hbar c/\Delta E \approx 1 \text{ m}$.

5.33. The dipole electric moment is $p = 2\pi\hbar c/\lambda E$, the dipole size $l = p/e \approx 10^{-8} \text{ cm}$.

5.34. The wave function of the lowest quantum level is spherically symmetric. Writing the Schrödinger equation in spherical coordinates, viz.

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Psi}{dr} \right) = E\Psi$$

and making the substitution $\Psi(r) = u(r)/r$, we find that

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} = Eu.$$

This equation must be solved with the boundary condition $u(R) = 0$, which corresponds to an infinitely high wall. In addition, the wave function Ψ must be finite everywhere,

therefore $u(0) = 0$. These conditions are satisfied by the functions

$$\Psi_n'(r) \propto \frac{1}{r} \sin\left(n \frac{\pi r}{R}\right) \quad \text{at } n \geq 1.$$

The lowest level corresponds to $n = 1$, i.e.

$$E_{01} = \frac{\pi^2 \hbar^2}{2mR^2}.$$

Summing the electron energy and the free energy of the surface, $E_{\text{surf}} = 4\pi\sigma R^2$, we get the total energy of the system. Finding the minimum with respect to R yields the equilibrium radius of the cavity:

$$R = \left(\frac{\pi \hbar^2}{8\sigma m}\right)^{1/4} \approx 20 \text{ \AA}.$$

5.35. The potential energy of an atom in a non-uniform field $U(r)$ is $-\frac{1}{2}\alpha E^2(r)$ and therefore the force acting per unit volume is

$$F = \frac{1}{2} N_A \alpha \frac{\rho}{\mu} \nabla E^2(r),$$

where μ is the molar mass. In our problem this force is directed radially to centre. By putting down the conditions for the liquid equilibrium we can easily ensure that $dp/dr = F$. Integrating this equation with the boundary condition for infinity, $p(\infty) = 0$, we get

$$p(r) = \frac{1}{2} N_A \alpha \frac{\rho}{\mu} E^2(r).$$

It follows from the α value given in this relationship that the permittivity $\epsilon \approx 1$, therefore $E^2 = e^2/r^4$. The boundary condition on the ball surface $p(R) = p_s$ thus yields

$$R = \left(\frac{1}{2} N_A \alpha \frac{e^2 \rho}{\mu p_s}\right)^{1/4} \approx 7 \text{ \AA}.$$

VI. Radiation

6.1. $T^4 = \frac{1}{2} \frac{R^2 H^2}{(H^2 + r^2)^2} T_0^4$, where r is the distance from the disc centre projection. Lambert's law was used to solve this problem.

6.2. $\frac{W_{\text{corp}}}{W_{\text{rad}}} = \frac{2n(m_e + m_p)v^3}{\alpha^2 \sigma T^4} \approx 6 \times 10^{-8}.$

$$6.3. \bar{n} = \frac{4}{3} \frac{\sigma T^3}{ch} \approx 1.8 \times 10^7 \text{ cm}^{-3}.$$

$$6.4. p = \frac{32}{3} \frac{\sigma T^4}{c} \approx 1.3 \times 10^{-6} \text{ atm.}$$

6.5. Since the properties of equilibrium radiation are independent of system's volume, we must have $U(V, T) = Vu(T)$ and $S(V, T) = Vs(T)$. From the thermodynamic identities

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V, \quad T \left(\frac{\partial S}{\partial T}\right)_V = C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

combined with the equation of state we get

$$s = \frac{1}{3} \frac{du}{dT}, \quad \frac{ds}{dT} = \frac{1}{T} \frac{du}{dT}.$$

The solution of this system yields $u(T) = aT^4$ and $s(T) = 4/3 aT^3$. Since the integral radiation density $u(T)$ is related to the integral flux density σT^4 by the equation $\sigma T^4 = \frac{c}{4} u(T)$, which is analogous to the gas-kinetic formula for molecular flux, we get $\sigma = \frac{c}{4} a$.

$$6.6. C_p = \infty, VT^3 = \text{const.}$$

6.7. $\frac{\omega_{\text{ind}}}{\omega_{\text{sp}}} = \bar{n}(E) = \frac{1}{e^{E/kT} - 1}$, where $\bar{n}$ is the average Planck occupation number for an oscillator. For an atom $E \gg kT$ and hence

$$\frac{\omega_{\text{ind}}}{\omega_{\text{sp}}} \approx e^{-E/kT} \approx 1.6 \times 10^{-17},$$

while for spin $E \ll kT$ and

$$\frac{\omega_{\text{ind}}}{\omega_{\text{sp}}} \approx \frac{kT}{E} = \frac{kT}{2\mu B} \approx 2.2 \times 10^3.$$

6.8. Spontaneous radiation emerges due to the perturbing effect of zero oscillations of an electromagnetic field on a molecule. Therefore the probability that spontaneous radiation will occur is proportional to the area S under the absorption spectral curve of the molecule (this area being a measure of the coupling between the "intermolecular oscillator" and the electromagnetic field) multiplied by the average spectral density $\rho_0(\omega)$ of the energy of the electromagnetic zero oscillations at the frequency corresponding to a molecular transition and at the point in space at which the molecule is located. In a vacuum the number of normal

oscillators within a given frequency range is calculated in the same way that the Rayleigh-Jeans radiation formula is derived, hence

$$\omega_{sp}^0 \propto AS\rho_0(\omega) = AS \frac{\hbar\omega^3}{2\pi^2c^3},$$

where A is a constant.

In the cavity, the actual frequency range contains one normal oscillation with width $\sim \omega/Q$. Therefore, if only the width of molecular levels Γ is greater than ω/Q , the density ρ_0 is of the order of $\frac{\hbar\omega/2}{V\omega/Q}$. Because of the change in the spontaneous lifetime, the width Γ in the cavity may be significantly different from the width in a vacuum. However, it is essential that the amplitude in resonance and the width both vary simultaneously, as in a conventional damped oscillator, viz. in such a way that the area under the absorption curve remains unchanged. Therefore

$$\omega_{sp} \propto AS \frac{\hbar Q}{2V} \quad \text{and} \quad \frac{\omega_{sp}}{\omega_{sp}^0} \propto \pi^2 \frac{Qc^3}{V\omega^3}.$$

$$6.9. \quad \frac{n_{exc}}{n_{tot}} = \frac{1}{2} \left[1 - \frac{\ln(1-\tau)}{2\alpha L} \right] \approx 0.505.$$

VII. Solid State

7.1. According to the data in the problem, the wave vectors of the incident and scattered X-rays are $\mathbf{k} = k(1, 0, 0)$ and $\mathbf{k}' = 1/3 k(1, 2, 2)$, respectively, where $k = 2\pi\nu/c$. Therefore, the wave vector change due to the scattering is

$$\mathbf{k}' - \mathbf{k} = \frac{2}{3} k(-1, 1, 1).$$

On the other hand, in Bragg scattering

$$\mathbf{k}' - \mathbf{k} = 2\pi\mathbf{b},$$

where $\mathbf{b}$ is the integer vector of the reciprocal lattice. In a cubic face-centered lattice the basis vectors of the direct lattice are usually chosen as $\mathbf{a}_1 = \frac{1}{\sqrt{2}}(1, 1, a0)$ and analogously $\mathbf{a}_2$ and $\mathbf{a}_3$, where a is the interatomic distance. Thus, the basis vectors of the reciprocal lattice are constructed according to the type $\mathbf{b}_1 = \frac{1}{a\sqrt{2}}(1, 1, 1)$. Whence it can be seen that in first order interference the vector $\mathbf{b}$ should

simply be one of the minimum vectors $\mathbf{b}_l$ and hence

$$\frac{2}{3} k = \frac{2\pi}{a \sqrt{2}}.$$

Finally,

$$a = \frac{3c}{2 \sqrt{2} \nu} = 2.9 \times 10^{-8} \text{ cm.}$$

7.2. By solving the equations of the natural oscillations of the chain we get two frequency branches:

$$\omega_{1,2}^2(k) = \frac{\gamma}{\mu} \left[1 \pm \left(1 - \frac{4\mu \sin^2 ka}{M_1 + M_2} \right)^{1/2} \right], \quad |ka| \leq \frac{\pi}{2}.$$

On the lower branch, for $|ka| \ll 1$ the frequency $\omega_1 \propto k$ and hence this branch is acoustic. On the upper branch, $\omega_2^2(k=0) = 2\gamma/\mu$ and the branch is called optical. A gap exists between the branches, since $\omega_2(\pi/2a) > \omega_1(\pi/2a)$. At $M_1 = M_2$ the frequencies

$$\omega_{1,2}^2(k) = \frac{\gamma}{\mu} (1 \pm \cos ka)$$

and the gap vanishes. Thus the spatial period of the chain is halved giving rise to a single branch with

$$\omega^2 = \frac{\gamma}{\mu} (1 - \cos ka)$$

in twice the range of wave numbers, $|ka| \leq \pi$.

7.3. By solving the equations of forced oscillations of the chain and calculating dipole moment, we can find the polarizability

$$\alpha(\omega) = \frac{e^2}{\mu(2\gamma/\mu - \omega^2)}.$$

The polarizability diverges at the frequencies of long-wave optical oscillations, $\omega = (2\gamma/\mu)^{1/2}$.

7.4. The laws of conservation of energy and momentum in neutron scattering are

$$\frac{\hbar^2 \mathbf{k}^2}{2M} = \frac{\hbar^2 (\mathbf{k}')^2}{2M} \pm \hbar \omega(\mathbf{q}), \quad \mathbf{k} = \mathbf{k}' + \mathbf{q} + 2\pi \mathbf{b},$$

where the upper sign corresponds to the emission and the lower one to the absorption of a phonon. In the second expression, $\mathbf{b}$ is an arbitrary integer vector of the reciprocal lattice. By measuring $\mathbf{k}$ and $\mathbf{k}'$, it is possible to determine $\omega(\mathbf{q})$ directly from the first expression and $\mathbf{q}$ from the second one.

Hence the vector $\mathbf{b}$ can be determined if $\mathbf{q}$ belongs to the first Brillouin zone. This enables the function $\omega(\mathbf{q})$ to be reconstructed.

7.5. Using the general oscillation equation

$$M\ddot{x} = -\frac{dU(x)}{dx}$$

we can calculate $U(x)$ in the harmonic approximation. If we take into account that the change in U during the displacement of a nucleus is related to the interaction between the electron shells, then we have to estimate $U(x) \propto E_{\text{at}}(x/a)^2$. Whence $\omega^2 \propto E_{\text{at}}/Ma^2$ and since $E_{\text{at}} \propto \hbar^2/ma^2$, we finally get

$$\hbar\omega \propto \left(\frac{m}{M}\right)^{1/2} \cdot E_{\text{at}}.$$

The estimate for x_0 is obtained immediately if we allow that $U(x_0) \propto \hbar\omega$. Thus, the smallness of the amplitudes ($x_0 \ll a$), which is a necessary condition for the existence of molecules and crystals, is the result of the smallness of the "adiabatic parameter", $(m/M)^{1/2} \propto 10^{-3}$.

7.6. $C = \frac{2\pi^2}{15} \left(\frac{2}{s_1^3} + \frac{1}{s_2^3}\right) \left(\frac{kT}{\hbar}\right)^3 kV$, where V is the crystal volume.

7.7. At low temperatures only the phonons with momenta $q \leq q_0 = kT/\hbar s$ are excited. The volume of the region in q -space corresponding to these phonons is $\propto q_0^3$ in the two-dimensional case and $\propto q_0$ in the one-dimensional case. Therefore the thermal energies are $U \propto T^3$ and T^2 , respectively, while the heat capacities are proportional to T^2 and T , respectively.

7.8. By putting down the oscillation equation for an anharmonic oscillator,

$$M\ddot{x} = -\gamma x + \frac{1}{2}\delta x^2,$$

it is possible, as in problem 7.5, to estimate $\gamma \sim E_{\text{at}}/a^3$ and $\delta \sim E_{\text{at}}/a^3$. By averaging this equation over an oscillation period ($0 < t < T$) we have, by definition,

$$\ddot{\bar{x}} = \frac{1}{T} [\dot{x}(T) - \dot{x}(0)] = 0,$$

hence $\bar{x} = (\delta/2\gamma) \bar{x}^2$. However, given weak anharmonicity, we can take $\bar{x}^2$ in the harmonic approximation, i.e. $\gamma \bar{x}^2/2 =$

$kT/2$. Therefore

$$\bar{x} = \frac{\delta}{2\gamma^2} kT$$

and the expansion coefficient

$$\alpha = \frac{\delta k}{2\gamma^2 a} \sim \frac{k}{k_{at}} \sim 10^{-8} \text{ K}^{-1}$$

is independent of the temperature within the whole of the classical region, i.e. for which $kT \gg \hbar\omega$.

7.9. $S = R \ln(2I + 1)$ so that $S_{He} = R \ln 2$, $S_{Ar} = 2R \ln 2$. As usual, the entropy of a fully ordered state is chosen as the origin.

$$7.10. \kappa = \frac{d \ln n}{dp} = \frac{3}{5p} = \frac{3}{(3\pi^2)^{1/3}} \frac{m}{\hbar^3 n^{2/3}} \approx 1.5 \times 10^{-8} \text{ atm}^{-1}.$$

$$7.11. v_{Fz} = \frac{E_F a}{\hbar} = 2.3 \times 10^7 \text{ cm/s}, \quad v_{Fx} = v_{Fy} = 0.$$

$$7.12. v_F = \sqrt{\frac{2\pi a}{V}} \cdot \frac{\hbar}{m} = 1.05 \times 10^8 \text{ cm/s}, \quad \varphi = \frac{eH}{mc} \cos \theta. \text{ To solve the problem we use the equation of motion } \mathbf{v} = \partial E(\mathbf{p}) / \partial \mathbf{p}, \quad \dot{\mathbf{p}} = \mathbf{F}.$$

7.13. The potential distribution inside a conductor is governed by Poisson's equation:

$$\Delta \varphi = 4\pi e (n - n_0) / \epsilon,$$

where ϵ is the permittivity due to the electrons from the filled bands (viz. all electrons except the conduction ones).

In thermodynamic equilibrium, the chemical potential level ζ of electrons is constant throughout the crystal. However, because of the electric field the bottom of the conduction band, which simply corresponds to the potential energy of an electron, changes from point to point (Fig. 75) giving $E_F = E_F(x)$. If the band bending is small compared to E_F , then

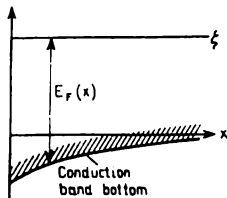


Fig. 75

$$n - n_0 \approx \frac{dn}{dE_F} \Delta E_F = \frac{dn}{dE_F} e\varphi.$$

By combining these equations we get

$$\Delta \varphi = \frac{4\pi e^2}{\epsilon} \frac{dn}{dE_F} \varphi.$$

The solution of the latter equation is

$$\varphi \propto \exp\left(\pm \frac{x}{l_{TF}}\right), \quad \text{where} \quad l_{TF} = \left(\frac{e}{4\pi e^2} \frac{dE_F}{dn}\right)^{1/2}.$$

Given the quadratic dispersion law: $E_F \propto p_F^2$, $n \propto p_F^3$ (p_F is the Fermi momentum of electrons), and therefore

$$\frac{dE_F}{dn} = \frac{2}{3} \cdot \frac{E_F}{n},$$

hence,

$$l_{TF} = \left(\frac{eE_F}{6\pi e^2 n}\right)^{1/2}.$$

Given the indicated parameter values, $l_{TF} \approx 4 \times 10^{-9}$ cm for metals and 1.5×10^{-6} cm for semiconductors. In the first case the value of l_{TF} demonstrates that an external field is practically fully screened within even one lattice constant and does not penetrate the metal.

7.14. Using the formula for an electron heat capacity $c_{el} = \frac{\pi^2 k^2 T}{2 E_F} n$ and the formula for a lattice heat capacity (see Problem 7.6) we get

$$kT = \sqrt{\frac{5}{4} \frac{(\hbar s)^2 n}{E_F}}.$$

Using the estimate for s from Problem 7.5 and assuming $E_F \sim E_{at}$ we get

$$kT \sim \left(\frac{n}{M}\right)^{3/4} E_{at},$$

whence $T \sim 10$ K.

7.15. $T \sim \Theta$, where $\Theta \sim \left(\frac{n}{M}\right)^{1/3} \frac{E_{at}}{k}$ is the Debye temperature of the crystal. The problem has to be solved in the same way as 7.14.

7.16. According to Maxwell's equations, the condition $\mathbf{H} = \mathbf{B} = 0$ results in

$$\text{curl } \mathbf{E} = 0, \quad \dot{\mathbf{D}} + 4\pi \mathbf{j} = 0.$$

It follows from the first relation that the electric field of such a wave is longitudinal, viz. the amplitude E_0 in the equation

$$\mathbf{E}(\mathbf{r}, t) = E_0 \exp[i(\mathbf{k}\mathbf{r} - \omega t)]$$

is parallel to $\mathbf{k}$. The vectors $\mathbf{D}$ and $\mathbf{j}$ are related to $\mathbf{E}$ by the material equations $\mathbf{D} = \epsilon \mathbf{E}$ and $\mathbf{j} = \sigma \mathbf{E}$, where ϵ is the

permittivity not related to the conduction electrons, while σ is the conductivity. For band electrons, $m^* \dot{\mathbf{v}} = e\mathbf{E}$, where $\mathbf{v}$ is the average velocity of the electron flux in an external field. Whence

$$\mathbf{j}_0 = en\mathbf{v}_0 = i \frac{e^2 n}{m^* \omega} \mathbf{E}_0;$$

substituting this into $\dot{\mathbf{D}} + 4\pi\mathbf{j} = 0$ yields the frequency of the natural oscillations, viz.

$$\omega_p^2 = \frac{4\pi e^2 n}{em^*}$$

For metals, $\omega_p \sim 2 \times 10^{16} \text{ s}^{-1}$, $\hbar\omega_p \sim 10 \text{ eV}$.

7.17. In order to introduce a complex permittivity $\epsilon(\omega)$ the Maxwell equations need to be rewritten as follows

$$\text{curl } \mathbf{H} = \frac{1}{c} \dot{\mathbf{E}} + \frac{4\pi}{c} \mathbf{j} \equiv \frac{\epsilon(\omega)}{c} \dot{\mathbf{E}}.$$

Then, in accordance with the material equations (see Problem 7.16),

$$\epsilon(\omega) = \epsilon + i \frac{4\pi\sigma}{\omega}.$$

By introducing a "friction" into the equation of electron motion to describe the finite mean free time, viz.

$$m^* \dot{\mathbf{v}} = -\frac{m^*}{\tau} \mathbf{v} + e\mathbf{E},$$

we get

$$\sigma(\omega) = \frac{e^2 n \tau}{m^* (1 - i\omega\tau)}, \quad \epsilon(\omega) = \epsilon \left[1 - \frac{\omega_p^2 / \omega^2}{1 + (i/\omega\tau)} \right].$$

Starting from the experimental values of the static electroconductivity of metals we can easily estimate that within the optical frequency range $\omega\tau \gg 1$, and therefore

$$\epsilon(\omega) = \epsilon (1 - \omega_p^2 / \omega^2).$$

When $\omega < \omega_p$ we have $\epsilon(\omega) < 0$, viz. total ("metallic") reflection takes place. In the ultraviolet range, for which $\omega > \omega_p$ (see Problem 7.16), $\epsilon(\omega) > 0$ and transmission is much higher.

7.18. At moderate frequencies ($\omega\tau \ll 1$) the conductivity σ is real, and so the conduction current significantly exceeds the displacement current (for $\omega_p^2 \tau / \omega \gg 1$ see Problem 7.17).

Then the complex wave number is

$$k = \frac{\omega}{c} \sqrt{\varepsilon(\omega)} = \frac{1+i}{\delta},$$

where $\delta = \frac{c}{\sqrt{2\pi\omega\sigma}}$ is the penetration depth.

7.19. The equation of an electron's motion is

$$m^* \dot{\mathbf{v}} = e\mathbf{E} + \frac{e}{c} \mathbf{v} \times \mathbf{H} - \frac{m^*}{\tau} \mathbf{v}, \quad \mathbf{H} \parallel z, \quad \mathbf{E} \perp \mathbf{H}.$$

We need only search for the mean drift velocity $\mathbf{v}$ and leave out the rotation around the $\mathbf{H}$ direction, which does not contribute to the average flux. Hence we assume $\dot{\mathbf{v}} = 0$. Then

$$j_x = \sigma \frac{E_x + \omega_c \tau E_y}{1 + (\omega_c \tau)^2}, \quad j_y = \sigma \frac{E_y - \omega_c \tau E_x}{1 + (\omega_c \tau)^2}, \quad j_z = 0,$$

where $\sigma = e^2 n \tau / m^*$ is the conductivity at $H = 0$, $\omega_c = eH/m^*c$ the cyclotron frequency.

At $E_y = 0$ the longitudinal current is

$$j_x = \frac{\mu}{1 + (\omega_c \tau)^2} E_x$$

and the way it falls with increasing H defines the magnetoresistance. The transverse current,

$$j_y = -\sigma \frac{\omega_c \tau}{1 + (\omega_c \tau)^2} E_x,$$

is the Hall current. These conditions are observed with a Corbino disc.

The current in the plate and along the normal is absent ($j_y = 0$), therefore the Hall field is $E_y = \omega_c \tau E_x$, and $j_x = \sigma E_x$. It is necessary to stress the total absence of magnetoresistance, the reason being that the current not only flows due to the external field E_x , but also due to the Hall field E_y . In strong fields ($\omega_c \tau \gg 1$) the second contribution predominates.

7.20. $\varphi_E = \varphi - e\sqrt{eE}$, $\mathcal{J}(E, T) = \mathcal{J}_0(T) e^{e\sqrt{eE}/kT}$.

7.21. The spread of charge is described by the continuity equation, $\dot{\rho} + \text{div } \mathbf{j} = 0$, and the material equation, $\mathbf{j} = \sigma \mathbf{E}$. In good conductors, the conductivity change associated with excess carriers forming a space charge is negligibly small compared to equilibrium conductivity, therefore $\sigma = \text{const}$. Since both $D = eE$, where e is the permit-

tivity associated with the electrons of filled bands, and $\operatorname{div} \mathbf{D} = 4\pi\rho$, we finally get

$$\dot{\rho} + \frac{4\pi\sigma}{\epsilon} \rho = 0.$$

The solution of this equation is

$$\rho(r, t) = \rho_0(r) e^{-t/\tau_M},$$

where $\tau_M = \epsilon/4\pi\sigma$ and $\rho_0(r)$ is the charge distribution at $t = 0$. It is clear from the solution that the density ρ , having not been deformed, falls exponentially in time so that those spatial regions in which the charge was absent at $t = 0$ remain forever neutral. Therefore the current $\mathbf{j}$ flows, naturally, throughout the entire space. For germanium, $\tau_M \approx 10^{-10}$ s.

7. 2. In contrast with Problem 7.13, the equilibrium electron distribution is given by $n(r) = n_0 e^{-U(r)/kT}$, where $U(r) = -e\varphi(r)$ and n_0 is the electron concentration outside the space charge region. The field weakness means that $|e\varphi| \ll kT$. Thus Poisson's equation reduces to

$$\Delta\varphi = \frac{4\pi e^2 n_0}{\epsilon kT} \varphi.$$

Its solutions are

$$\varphi \propto \exp\left\{\pm \frac{x}{l_D}\right\}, \quad \text{where } l_D = \left(\frac{\epsilon kT}{4\pi e^2 n_0}\right)^{1/2}.$$

At the given parameter values, $l_D \sim 5 \times 10^{-5}$ cm.

7.23. By solving the equation of motion

$$\frac{d\mathbf{p}}{dt} = \frac{e}{c} \mathbf{v} \times \mathbf{H}, \quad \mathbf{v} = \frac{\partial E(\mathbf{p})}{\partial \mathbf{p}}$$

we get

$$\omega_c^2 = \frac{e^2}{c^2} \left(\frac{H_x^2}{m_y m_z} + \frac{H_y^2}{m_x m_z} + \frac{H_z^2}{m_x m_y} \right).$$

$$7.24. \quad \sigma(\omega) = \frac{\sigma_0}{1 + (\omega - \omega_c)^2 \tau^2}, \quad \text{where } \omega_c = \frac{eH}{m^* c},$$

$$\sigma_0 = \frac{e^2 n \tau}{m^*}, \quad \text{and } n \text{ is the concentration.}$$

7.25. It follows from $\dot{p}_x = eE$ that

$$k_x = \frac{p_x}{\hbar} = \frac{eE}{\hbar} (t - t_0),$$

whence, by allowing for $\dot{x} = v_x$ and $v_x = \partial \mathcal{E} / \partial p_x$,

$$x(t) = x_0 + \frac{\mathcal{E}_0}{eE} \cos \left\{ \frac{eEa}{\hbar} (t - t_0) \right\}.$$

Thus, the electron oscillates around an equilibrium position x_0 with a frequency $\omega_E = eEa/\hbar$ and the average current vanishes. The reason is that the kinetic energy $\mathcal{E}_0 \cos(k_x a)$ is a bounded function of k_x . As a result, the motion of an electron with a potential energy $U(x) = -eEx$ is limited to a finite region (due to the law of conservation of energy). 7.26. In the equation of an electron's motion,

$$\dot{p}_x = \hbar \dot{k}_x = eE - \gamma v_x = eE + \gamma a \mathcal{E}_0 \sin(k_x a),$$

the variables are separated and this equation can immediately be integrated via elementary functions. The form of the solution differs depending on the value of the parameter $\Gamma = \gamma a \mathcal{E}_0 / eE$.

(a) $\Gamma > 1$. The solution has the following form:

$$\frac{\cos \{(k_x a - \varphi)/2\}}{\sin \{(k_x a + \varphi)/2\}} = \exp \left\{ -\sqrt{\Gamma^2 - 1} \frac{eEa}{\hbar} (t - t_0) \right\},$$

$$\varphi = \arcsin \frac{1}{\Gamma} \quad (0 < \varphi < \pi/2).$$

As $t \rightarrow \infty$ the solution becomes $\Gamma \sin k_x a = -1$, which corresponds to stationary values of k_x and v_x . This solution can be obtained directly by assuming that in the initial equation $\dot{k}_x = 0$, and so $v_x = eE/\gamma$ and $\sigma = e^2 n/\gamma$. For $\Gamma \gg 1$ we find that $k_x a \ll 1$, viz. the electron remains near the bottom of the band. It is therefore possible, as usual (e.g. Problem 7.19), to set $\gamma = m^*/\tau$ and then σ takes the standard form, $\sigma = e^2 n \tau / m^*$.

(b) $\Gamma < 1$. The solution has the form:

$$\tan \frac{k_x a}{2} = \sqrt{1 - \Gamma^2} \tan \left\{ \sqrt{1 - \Gamma^2} \frac{eEa}{2\hbar} (t - t_0) \right\} - \Gamma.$$

It follows from this expression that as t rises so does the wave number k_x ; the oscillations being superimposed thereon. The tangent on the right-hand side goes to infinity for the time intervals $\Delta t = \hbar / eEa \sqrt{1 - \Gamma^2}$. For each such a time interval the phase of the argument in the left-hand side tangent is augmented by π and hence k_x is increased by $\Delta k_x = 2\pi/a$. Integrating the initial equation of motion

over Δt we find that

$$\hbar \Delta k_x = eE \Delta t - \gamma \int v_x dt = eE \Delta t - \gamma \Delta x,$$

where Δx is the electron displacement in time Δt . The ratio $\Delta x/\Delta t$ is the average drift velocity while the average current density is

$$j = \frac{e^2 n E}{\gamma} [1 - \sqrt{1 - \Gamma^2}].$$

In strong fields, when $\Gamma \ll 1$, we get

$$j \approx \frac{e^2 n \Gamma^2 E}{2\gamma} \propto \frac{\gamma}{E},$$

viz. the final current only arises due to the "friction" force. This current increases with decreasing mean free time.

7.27. According to the problem, the electron is slow, viz. ka and $k'a \ll 1$, where k and k' are the electron wave vectors before and after a scattering and a is the lattice constant. Analogously, for long-wave phonons $qa \ll 1$ as well. Therefore the law of conservation of momentum can be written as $k = k' \pm q$ (see the solution of Problem 7.4, where b must be zero due to the smallness of k , k' , and q). For the angle θ between p and q we get

$$\cos \theta = \frac{s}{v} \pm \frac{q}{2k},$$

where the upper sign pertains to omission, and the lower one to the absorption of a phonon.

$$7.28. \sigma \sim \frac{e^2 n d^2 v}{6kT} \approx 1 \text{ S/m.}$$

7.29. By solving this problem as we did for Problem 7.21 we get for the relaxation time

$$\tau_M = \frac{4\pi}{e} \frac{d}{dE} \{E\sigma(E)\}.$$

If $\tau_M > 0$, the fluctuation is damped and this occurs for $E < E_0$. If $E > E_0$ the fluctuation gets worse over time.

7.30. The volume V in conventional r -space and volume V_k in a wave-vector space contain $VV_k/(2\pi)^3$ states. Since for nondegenerate electrons the occupation numbers $n(\xi) = \exp\left(\frac{\xi - \mathcal{E}}{kT}\right)$, then the electron concentration is

$$n = 2 \int \frac{d^3 k}{(2\pi)^3} \exp \frac{\xi - \mathcal{E}}{kT} = 2 \frac{(2\pi m k T)^{3/2}}{\hbar^3} \exp \frac{\xi}{kT} = Q \exp \frac{\xi}{kT},$$

where ζ is calculated from the bottom of the electron band ($\zeta < 0$). The quantity Q is called the band's statistical factor.

7.31. According to Problem 7.30, the electron concentration at an arbitrary ζ is $n = Q_n e^{\zeta/kT}$. In an analogous way, the hole concentration is

$$p = Q_p \exp\left(-\frac{2\Delta + \zeta}{kT}\right),$$

where Q_p is the band's statistical factor for holes and 2Δ the forbidden gap width. Hence

$$np = Q_n Q_p e^{-2\Delta/kT}$$

is independent of ζ and hence, of the impurity concentration too. Yet in an intrinsic semiconductor $n = p = n_i$ due to electric neutrality. Therefore, in the general case $np = n_i^2$.

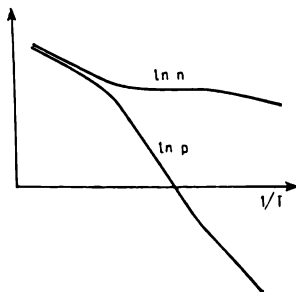


Fig. 76

7.32. At moderate temperatures the electrons are only exchanged between donors and the electron band. For electric neutrality we must have

$$Q_n e^{\zeta/kT} + \frac{N}{\exp\left(-\frac{2\Delta_1 + \zeta}{kT}\right) + 1} = N,$$

where N is the donor concentration. To realize this condition it is

sufficient to assess the number of electrons in the band and that in the donors. Thus a quadratic equation for $e^{\zeta/kT}$ is obtained, which is easily solved. At low temperatures, such as $n \ll N$, we have $n = \sqrt{NQ_n} e^{-\Delta_1/kT}$ while at higher temperatures $n \approx N$ all the impurities are ionized (impurity depletion region). The hole concentration p in terms of n is $np = n_i^2$ (see Problem 7.31). At higher temperatures the excitation of the electrons from a filled band predominates when $n \approx p \approx n_i \approx \sqrt{Q_n Q_p} e^{-\Delta/kT}$, this taking place at $n_i(T) \gg N$. The temperature dependence of n and p is depicted in Fig. 76.

7.33. $Ry^* = \frac{mc^4}{2\hbar^2 e^2}$, $a^* = \frac{\hbar^2 e}{me^2}$, $M = m_n^* + m_p^*$, where m is the reduced mass:

$$\frac{1}{m} = \frac{1}{m_n^*} + \frac{1}{m_p^*};$$

$$Ry^* \sim 0.7 \times 10^{-2} \text{ eV}, \quad a^* \sim 10^{-6} \text{ cm}.$$

7.34. The diffusion hole flux is $j = -D \frac{dp}{dx}$ and the continuity equation for the hole flux is

$$\frac{dj}{dx} + \frac{p}{\tau} = 0;$$

here p/τ is the number of recombinations per unit time per unit volume. These relations suggest the diffusion equation

$$D \frac{d^2 p}{dx^2} - \frac{p}{\tau} = 0$$

which has the solutions $p \propto \exp(\pm x/L)$, where $L = \sqrt{D\tau}$ is the diffusion displacement length. By applying the Einstein relation $eD = kT\mu$, the parameter values given yield $L \approx 0.2 \text{ cm}$.

7.35. In superconductors $B \approx H$ and the displacement current can be neglected as compared with the current of the superconducting electrons. Therefore the Maxwell equations take the form:

$$\text{curl } \mathbf{E} = -\frac{1}{c} \dot{\mathbf{H}}, \quad (1)$$

$$\text{curl } \mathbf{H} = \frac{4\pi}{c} \mathbf{j}. \quad (2)$$

The equation of electron motion $m\dot{\mathbf{v}} = e\mathbf{E}$ in conjunction with $\mathbf{j} = en\mathbf{v}$ yields $\mathbf{E} = \frac{m}{e^2 n} \frac{d\mathbf{j}}{dt}$. Substituting these into (1) we get $\text{curl } \mathbf{j} = -\frac{e^2 n}{mc} \mathbf{H}$. Eliminating $\mathbf{j}$ using (2) yields an equation for $\mathbf{H}$, i.e.

$$\text{curl curl } \mathbf{H} = -\frac{\mathbf{H}}{\lambda^2}, \quad \lambda = \sqrt{\frac{mc^2}{4\pi e^2 n}} = \frac{c}{\omega_p}.$$

By transforming the left-hand side and taking into account that $\text{div } \mathbf{H} = 0$ we finally get

$$\Delta \mathbf{H} = \frac{\mathbf{H}}{\lambda^2}.$$

For a flat metal surface this has the solution $H \propto \exp(\pm x/\lambda)$, viz. the magnetic field decays exponentially with depth into the superconductor. Since the external field's frequency does not enter λ , the result shows that the magnetic field is constant. The depth asked for is $\lambda \sim 10^{-5}$ – 10^{-6} cm.

7.36. The ${}^3\text{He}$ nucleus contains three nucleons and its spin is $s = 1/2$. Since the spin of the electron shell is zero, an ${}^3\text{He}$ atom is a fermion. Using the formulas for a degenerate Fermi-gas we get for the heat capacity per mole,

$$C = \frac{\pi^2}{2} R \frac{T}{E_F},$$

where the Fermi-energy is $E_F = \frac{\hbar^2}{2M} \left(\frac{3\pi^2 N_A}{V} \right)^{2/3}$; here M is the atomic mass and N_A is Avogadro's number. By defining Θ so that $C = RT/\Theta$ we find that $\Theta \sim 1$ K.

VIII. Nuclear Physics

8.1. The particle mass in this experiment is E_{cm} , which corresponds to the position of the maximum. The half-width of the resonance curve at half the height is $\Delta \approx 2$ MeV. The lifetime can be found from the relationship $\tau \approx \frac{\hbar}{\Delta E} \approx 3 \times 10^{-22}$ s. To find the mass in another experiment, let us use the relativistic invariance of the squared four-momentum:

$$m^2 c^4 = (E_1 + E_2)^2 - (\mathbf{p}_1 + \mathbf{p}_2)^2 c^2 = 4E^2 - 3p^2 c^2,$$

since by definition $E_1 = E_2 = E$ and consequently $p_1 = p_2 = p$, with $\cos(\mathbf{p}_1, \mathbf{p}_2) = \cos \varphi = 1/2$. Since e^+ and e^- are ultrarelativistic particles ($v \approx c$), we have $E \approx pc$ and $mc^2 \approx 3.1$ GeV.

8.2. A proton consists of three quarks, $q_p q_p q_n$. The spins of two are parallel while the spin of the third is antiparallel to the other two. Ω -hyperon has strangeness $S = -3$ and can only be composed of the three identical quarks, $q_\Lambda q_\Lambda q_\Lambda$, whose spins are parallel (the Ω -hyperon's spin is $3/2 \hbar$). The π^+ -meson and K^+ -meson consist of the quarks $q_p q_n$ and $q_p \bar{q}_\Lambda$, and J -meson of the quarks $q_c \bar{q}_c$. The π - and K -mesons both have a quark and an antiquark with antiparallel spins, and the J -meson has a quark and an antiquark with parallel spins. The composition of the Ω -hyperon exemplifies the fact that in the simplest quark model the quarks do not obey Pauli's principle.

8.3. $T = Mc^2 - (m_{\Lambda} + 3m_{\pi})c^2 = 714 \text{ MeV}$, this accounting for the fraction $\delta = T/Mc^2 = 0.30$. Here m_{Λ} and m_{π} are the Λ -particle and pion masses, respectively.

8.4. The total energy released in an annihilation is $E = 2M_{\pi}c^2$. The π^0 -meson energy is at maximum if the momenta of all three pions are collinear with $\mathbf{p}_{\pi^+} = \mathbf{p}_{\pi^-}$. Thus

$$E_{\pi^0} = E_{\pi^+} \quad \text{and} \quad \mathbf{p}_{\pi^0} = 2\mathbf{p}_{\pi^+}. \quad (1)$$

(If $\mathbf{p}_{\pi^+} \neq \mathbf{p}_{\pi^-}$, then the $\pi^+\pi^-$ centre of mass moves slower than in the case of equal momenta, since a fraction of energy is expended for the pions to move relative to each other.) The law of conservation of energy is

$$2M_{\pi}c^2 = E_{\pi^0} + 2E_{\pi^+} = E_{\pi^0} + 2\sqrt{p_{\pi^+}^2c^2 + m_{\pi^+}^2c^4},$$

and using (1) we get

$$E_{\pi^0} = \frac{4M_{\pi}^2c^4 - 3m_{\pi^+}^2c^4}{4M_{\pi}c^2} \approx 0.92 \text{ GeV}.$$

8.5. The number of interactions in a volume dV in time dt is

$$d\eta = \sigma u_{12} \rho_1 \rho_2 dV dt,$$

where $u_{12} = |\mathbf{u}_1 - \mathbf{u}_2|$, and the densities of the particles are $\rho_1 = \rho_2 = n/c$ (n is the flux, c the velocity of light). In our case the difference in the velocities of particles, $u_{12} = 2c$, since $\mathbf{u}_1 = -\mathbf{u}_2$ and $|\mathbf{u}_1| = |\mathbf{u}_2| \approx c$. The number of counts (allowing for the detection efficiency, $\epsilon = 0.1$) is:

$$K = \epsilon \frac{d\eta}{dt} = 2\epsilon\sigma \frac{n^2}{c} dV.$$

Taking into account that $dV = Sl$ we find that

$$n = \left(\frac{Kc}{2\epsilon l \sigma S} \right)^{1/2}$$

and, finally, the current in the storage ring is

$$I = enS = \left(\frac{KSce^2}{2\epsilon l \sigma} \right)^{1/2} \approx 2 \text{ A}.$$

8.6. Emission starts at a particle velocity $v = c/n$. The appropriate kinetic energy is

$$T = m_0c^2 \left[\frac{1}{\sqrt{1-(v^2/c^2)}} - 1 \right] = m_0c^2 \left[\frac{n}{\sqrt{n^2-1}} - 1 \right] \approx 4.8 \text{ GeV}.$$

The refractive index n will be found from $n^2 = 1 + 4\pi\alpha N$, where N is the number of atoms per unit volume, and α the polarizability of the nitrogen molecules. Since at a constant temperature $N \propto p$, viz. N is proportional to pressure, and $n^2 - 1 \sim 2(n - 1)$, we must have

$$\frac{n-1}{n_0-1} \approx \frac{p}{p_0},$$

where $p_0 = 1$ atm.

8.7. Radiation from kaons in the counter only appears when $\frac{c}{v_K} > n > \frac{c}{v_\pi}$, where v_K and v_π are the velocities of kaons and pions that have momentum p and total energy E . Since $\beta = v/c = pc/E$, we find that

$$\sqrt{1 + \frac{m_K^2 c^4}{p^2 c^2}} \approx \sqrt{2} > n > \sqrt{1 + \frac{m_\pi^2 c^4}{p^2 c^2}} = \sqrt{1.1},$$

where $m_K c^2$ and $m_\pi c^2$ are the kaon and pion masses, respectively.

8.8. The path-length of about 70% pions before they decay is ($v \approx c$):

$$L \approx c\tau_0 \frac{E_\pi}{m_\pi c^2} \approx 600 \text{ m.}$$

8.9. $T_p = 300$ MeV.

8.10. The number of pions reaching the detector is $n = N\Omega e^{-t/\tau}$, where N is the number of all the pions generated, Ω the solid angle of the detector from the point of pion generation, $\Omega = S/L^2$, t a pion's flight time along path L , and τ the pion lifetime in the laboratory frame,

$$\tau = \tau_0 \frac{E}{m_\pi c^2} \approx \tau_0 \left(1 + \frac{T}{m_\pi c^2} \right),$$

where T is the kinetic energy. Finally, we get

$$N = n \frac{L^2}{S} \exp \left(\frac{L}{c\tau_0} \frac{m_\pi c^2}{\sqrt{T^2 + 2m_\pi c^2 T}} \right) \approx 1.4 \times 10^6 \text{ particle/s.}$$

8.11. The decrease in the number of protons due to the interaction in the nitrogen layer dx (g/cm^2) is

$$-dN = N \frac{dx}{\Lambda},$$

where Λ is the mean free path prior to the interaction. If x_0 is the thickness of Earth's atmosphere ($x_0 \approx 1000 \text{ g/cm}^2$),

then the solution of the above equation yields

$$N(x=x_0) = N(x=0) \exp\left(-\frac{x_0}{\Lambda}\right).$$

Here $\Lambda = A/\sigma N_A$, where N_A is Avogadro's number, A the relative atomic mass of nitrogen, $\sigma = \pi R^2$, and R the radius of a nitrogen nucleus. Whence,

$$N(x=x_0)/N(x=0) = 7 \times 10^{-6}.$$

8.12. The number of $^{37}_{18}\text{Ar}$ atoms is

$$n = \Phi \sigma t N_{\text{Cl}},$$

where $t = 1 \text{ year} = 3.2 \times 10^7 \text{ s}$, N_{Cl} the number of chlorine atoms, and $\Phi = \frac{N}{4\pi R^2}$ the neutrino flux on the Earth.

Whence

$$N_{\text{Cl}} = \frac{4\pi R^2 n}{N \sigma t}$$

and finally we find that the experiment needs carbon tetrachloride in amount of

$$M = \frac{\mu N_{\text{Cl}}}{4} \frac{1}{\eta} = \mu N_{\text{Cl}} \approx 560 \text{ t},$$

where μ is the molar mass of carbon tetrachloride.

8.13. To begin with, let us determine how much energy could be released in the centre of mass system (c.m.s.). By definition, the system's velocity is

$$\beta_0 = \frac{\sum_{i=1}^2 p_i c}{\sum_{i=1}^2 E_i},$$

where p_i and E_i are the momenta and energies of the protons in the laboratory system prior to the reaction. In our case

$$\beta_0 = \frac{pc}{\sqrt{p^2 c^2 + m_p^2 c^4} + m_p c^2} \approx 0.82.$$

Whence the energy of each of the colliding protons in the c.m.s. (the energies are equal since the momenta and masses of colliding particles are equal) is

$$E = \frac{m_p c^2}{\sqrt{1-\beta_0^2}} = \gamma m_p c^2 = 1.8 \text{ GeV}.$$

The total energy in the c.m.s. that can be expended to produce new particles is

$$2E - 2m_p c^2 \approx 1.6 \text{ GeV},$$

since both protons (or other baryons) remain after the reaction because of the conservation of the baryon number. Most pions are produced when all these particles are at rest in the c.m.s. while nucleons remain to be baryons. Therefore we finally have

$$n = \frac{2E - 2m_p c^2}{m_\pi c^2} \approx 11.$$

8.14. We find the muon's kinetic energy from the laws of conservation of energy and momentum:

$$T_\mu = \frac{m_K^2 c^4 + m_\mu^2 c^4}{2m_K c^2}.$$

The lifetime is

$$\tau = \frac{E}{m_\mu c^2} \tau_0 = \frac{T_\mu + m_\mu c^2}{m_\mu c^2} \tau_0 = 5.4 \times 10^{-6} \text{ s}.$$

8.15. The angular momentum received by the disc is equal to the total moment of the absorbed particles, i.e.

$$L = \frac{1}{4} \frac{N\hbar}{2}.$$

The coefficient $1/4$ arises because of the allowance for the angles of incidence of the electrons onto the disc:

$$\frac{1}{4} = \frac{1}{4\pi} \int_0^{\pi/2} \cos \theta d\Omega = -\frac{1}{2} \int_0^{\pi/2} \cos \theta \cdot d \cos \theta.$$

The disc is rotated by an angle

$$\varphi = \frac{L}{I} = \frac{N\hbar}{8I} \approx 5 \times 10^{-11} \text{ rad}.$$

8.16. As the magnetization proceeds, the number of spins of fluorine nuclei directed along the field N_1 and against the field N_2 is different and their difference is

$$\begin{aligned} \Delta N = N_1 \left[1 - \exp \left(-\frac{2\mu B}{kT} \right) \right] &= N_1 \frac{2\mu B}{kT} \\ &= \frac{N_0}{2} \frac{2\mu B}{kT} = N_0 \frac{\mu B}{kT} \ll 1, \end{aligned}$$

(where μ is the magnetic moment of fluorine nuclei, k is Boltzmann's constant, and N_0 the number of fluorine nuclei). When the field is switched off half the excess nuclei $\Delta N/2$ will change orientation and the sample will receive an angular momentum of

$$L = \frac{1}{2} \Delta N \frac{\hbar}{2} = \frac{1}{4} \frac{\mu B \hbar}{kT} N_0, \quad N_0 = 2N_A,$$

where N_A is Avogadro's number. Whence

$$\mu = L \frac{2kT}{N_A \hbar B} = 1.31 \times 10^{-24} \text{ J/T} = 2.62 \mu_N.$$

8.17. The dependence of half-life on energy (in MeV) is given by Geiger-Nuttall rule

$$\log T_{1/2} \approx C + \frac{D}{E^{1/4}}$$

The constants C and D will be found from the data in the problem: $D = 149$ and $C = -57$. The answer is $T_{1/2} \sim 10^{24}$ years.

8.18. To begin with, we find the α -particle range in aluminium, R_{Al} . Since the energy ionization losses are proportional to NZ , where N is the number of atoms per cubic centimetre and Z the nucleus charge, we have

$$\frac{R_{Al}}{R_{air}} = \frac{(NZ)_{air}}{(NZ)_{Al}}.$$

The number of nuclear reactions occurring in a thin layer dx is $d\nu = n_\alpha N_{Al} \sigma dx = n_p$, and $dx \approx R_{Al}$. Finally, we get

$$\sigma = \frac{n_p}{n_\alpha N_{Al} R_{Al}} = \frac{n_p Z_{Al}}{n_\alpha (NZ)_{air} R_{air}} = 4 \times 10^{-2} \text{ barn}.$$

8.19. The number of uranium fission events is $n = N/E \approx 3.1 \times 10^{16} \text{ s}^{-1}$, where E is the average total energy released per uranium fission event. The antineutrino flux beyond the shielding is

$$\Phi_{\bar{\nu}} \approx \frac{\langle n_{\bar{\nu}} \rangle n}{4\pi L^2} = 6 \times 10^{10} \text{ cm}^{-2} \cdot \text{s}^{-1}.$$

The fraction of the energy carried away by the antineutrino flux is

$$\frac{\langle E_{\bar{\nu}} \rangle \langle n_{\bar{\nu}} \rangle}{E} \approx 0.05.$$

8.20. Since γ -quanta are scattered at 180° , the π^0 -meson is at rest in the laboratory system. Therefore

$$p_n = p_d = p \text{ and } E_n + m_p c^2 = E_d + m_n c^2,$$

or

$$\sqrt{p^2 c^2 + m^2 c^4} + m_p c^2 = \sqrt{p^2 c^2 + m_d^2 c^4} + m_n c^2.$$

Neglecting the deuteron's binding energy and the difference between the proton's and the neutron's masses, viz. assuming $m_n = m_p = m$, $m_d = 2m$, $m_n = \mu$, we get

$$\sqrt{p^2 c^2 + m^2 c^4} + m c^2 = \sqrt{p^2 c^2 + 4m^2 c^4} + \mu c^2,$$

$$T_n = E_n - m c^2 = \frac{3m^2 c^4 - (m - \mu)^2 c^4}{2(m - \mu) c^2} - m c^2 \approx 300 \text{ MeV}.$$

The nonrelativistic approach yields

$$\frac{p^2}{2m} = \frac{p^2}{4m} + \mu c^2,$$

whence $T_n \approx 2\mu c^2 = 270 \text{ MeV}$, this being less accurate.
 8.21. $N(t)$ is the number of neutrons in the vessel at time t . Then the number of neutron collisions per unit area of the vessel's internal surface is $\nu N/4V$. The number of neutrons in the vessel falls for two reasons, viz. decay and leakage through the foil. This decrease is

$$-dN = N \frac{dt}{\tau} + \frac{N}{V} \frac{\nu}{4} S dt.$$

The solution of this equation yields

$$N = N_0 \exp \left[-\frac{t}{(1/\tau) + \nu S/4V} \right].$$

Using the data in the problem,

$$(1/\tau + \nu S/4V)^{-1} = \tau/2,$$

whence

$$S = \frac{4V}{\nu \tau} \approx 0.8 \text{ mm}^2.$$

8.22. The binding energy of nucleus, viz. that from Weizsäcker's formula is

$$U = 14.0A - 13.0A^{2/3} - 0.584 \frac{Z^2}{A^{1/3}} + \text{smaller terms}.$$

The nuclear radius is known to be $R_N = r_0 A^{1/3}$, where $r_0 \approx 1.25 \times 10^{-13}$ cm. Therefore

$$U \text{ (MeV)} \approx 14.0A - \frac{13.0}{4\pi r_0^2} S - 0.584 \frac{Z^2}{A^{1/3}} + \dots$$

where S is the nucleus's surface area. The coefficient of surface tension is

$$\sigma = \left| \frac{\partial U}{\partial S} \right| \approx \frac{13.0}{4\pi r_0^2} \text{ MeV/cm}^2 \approx 1.1 \times 10^{17} \text{ J/m}^2.$$

8.23. The ultimate velocity v_0 is determined from the condition

$$\frac{v_0}{v_{th}} = \sin i \approx 3 \times 10^{-3},$$

where $v_{th} = \sqrt{\frac{2kT}{m}}$. Neutrons for which $v < v_0$ will be trapped within the cavity. Their relative number is

$$\eta = \frac{\int_0^{v_0} \exp\left(-\frac{mv^2}{2kT}\right) v^3 dv}{\int_0^\infty \exp\left(-\frac{mv^2}{2kT}\right) v^3 dv} \approx \frac{\frac{v_0^4}{4}}{\frac{\sqrt{\pi}}{4} (v_{th}^2)^{3/2}} = \frac{4}{3\sqrt{\pi}} \left(\frac{v_0}{v_{th}}\right)^3 \approx 2 \times 10^{-8}.$$

8.24. The lifetime is

$$\tau = \frac{1}{n\sigma v_{th}} \approx 1.7 \times 10^{-4} \text{ s},$$

where n is the hydrogen concentration and v_{th} the neutron's thermal velocity (see the solution of Problem 8.23).

8.25. The total number of spin states for a system of two particles with spins I and $i = 1/2$ (in $\hbar$ units) is

$$(2I + 1)(2i + 1) \text{ or } 2(2I + 1).$$

The number of states with the total spin $I_s = I + 1/2$ is $2I_s + 1 = 2(I + 1)$, and the number of states with $I_s = I - 1/2$ is $2I$. Thus the scattering probability with parallel orientations of I and i is $\frac{I+1}{2I+1} = \frac{5}{8}$, and that with the antiparallel one is $\frac{I}{2I+1} = \frac{3}{8}$.

8.26. From the laws of conservation of energy and momen-

tum,

$$E_{\gamma} = T_N + T_n - Q, \quad \frac{E_{\gamma}}{c} = p_N + p_n, \quad \text{where}$$

$$Q = (mc^2)_{\gamma_{L1}} - [(mc^2)_{e_{L1}} + (mc^2)_n],$$

we find the required quantities

$$T_N = p_N^2/2m_N \approx 1 \text{ MeV} \text{ and } T_n = p_n^2/2m_n \approx 6 \text{ MeV}.$$

8.27. Since the binding energy of nucleons is low, it is possible when estimating to assume that the longitudinal momentum of the nucleons is $p_l \approx \frac{E_{\gamma}}{c}$. The transverse momentum of the nucleons inside the nucleus is due to the Fermi-motion of the nucleons. The Fermi-momentum will be estimated from the uncertainty relation

$$p_F \sim \Delta p \sim \frac{\hbar}{r_d} \sim 100 \text{ MeV}.$$

The estimate for the angle of scattering of the nucleons will be obtained from the relation

$$\varphi \approx \frac{p_F}{p_l} \approx 0.2 \text{ rad}.$$

8.28. At the entry to the neutronguide the neutrons for which $v < v_b$ will have a Maxwellian distribution of velocities, while those for which $v > v_b$ will be absent. Owing to the elasticity of the scattering on the walls and neglecting the flux onto the detector, the neutrons will have an equilibrium distribution of

$$dN(v, z) \propto \exp\left(-\frac{mv^2}{2kT} + \frac{mgz}{kT}\right) v^2/dV$$

given that

$$E = \frac{mv^2}{2} + mgz < E_b^2, \quad E_b^2 = mv_b^2/2.$$

Thus there will be no neutrons with $E > E_b^2$. Here z is the height up the neutronguide. For ultracold neutrons, the exponent can be changed for unity and the neutron flux is

$$d\Phi \propto v dN \propto v^3 dv \propto E dE.$$

The total flux at a height z is

$$\Phi(z) = \int d\Phi \propto \int_{mgz}^{E_b} E dE \propto E_b^2 - (mgz)^2.$$

This flux vanishes at $h = z_{\max}$

$$h = E_b^2/mg = E_b^2/2g \approx 5 \text{ m}.$$

APPENDICES

I. Constants and Tables

1. Fundamental physical constants

SI	CGS
Velocity of light in vacuum, $c \approx 3 \times 10^8$ m/s	3×10^{10} cm/s
Elementary charge, $e = 1.6 \times 10^{-19}$ C	4.8×10^{-10} CGSE
Mass of a hydrogen atom, $m_H = 1.67 \times 10^{-27}$ kg	1.67×10^{-24} g
Boltzmann's constant, $k = 1.38 \times 10^{-23}$ J/K	1.38×10^{-16} erg/K
Planck's constant, $h = h/2\pi = 1.05 \times 10^{-34}$ J·s	1.05×10^{-27} erg·s
Universal gas constant, $R = 8.31$ J/(K·mol)	8.31×10^7 erg/(K·m)
Gravitational constant, $\gamma = 6.67 \times 10^{-11}$ N·m ² /kg ²	6.67×10^{-8} dyn·cm ² /g ²
Avogadro's number,	$N_A = 6.02 \times 10^{23}$ mol ⁻¹
Loschmidt's number,	$n_0 = 2.69 \times 10^{19}$ cm ⁻³

2. Electric, atomic, and nuclear constants

SI	CGS
Electric constant, $\epsilon_0 = 8.85 \times 10^{-12}$ F/m	
Magnetic constant, $\mu_0 = 1.26 \times 10^{-6}$ H/m	
Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8}$ W/(m ² ·K ⁴)	5.67×10^{-5} erg/(s·cm ² ·K ⁴)

SI	CGS
Fine structure constant,	$\alpha = e^2/\hbar c \approx 1/137$
Electron rest mass,	$m_e c^2 = 0.511 \text{ MeV}$
Proton rest mass,	$m_p c^2 = 938 \text{ MeV}$
Deuteron rest mass,	$m_d c^2 = 1876 \text{ MeV}$
α -particle at rest mass,	$m_\alpha c^2 = 3726 \text{ MeV}$
Rydberg's constant for an infinitely heavy mass, $R_\infty = 10\,973\,731 \text{ m}^{-1}$	$109737.31 \text{ cm}^{-1}$
Ionization potential of hydrogen atom,	$I = 13.6 \text{ eV}$
Bohr radius	
$r_B = 0.53 \times 10^{-10} \text{ m}$	$0.53 \times 10^{-8} \text{ cm}$
Bohr magneton,	
$\mu_B = \frac{e\hbar}{2m_e c} = 9.27 \times 10^{-24} \text{ J/T}$	$0.927 \times 10^{-20} \text{ erg/G}$
Nuclear magneton,	
$\mu_N = \frac{e\hbar}{2m_p c} = 5.05 \times 10^{-27} \text{ J/T}$	$5.05 \times 10^{-24} \text{ erg/G}$
Classical radius of the electron,	
$r_0 = \frac{e^2}{m_e c^2} = 2.82 \times 10^{-13} \text{ m}$	$2.82 \times 10^{-13} \text{ cm}$
Atomic mass unit,	
10^{-27} kg	10^{-24} g
Compton wavelength of the electron,	
$\lambda_e = \frac{h}{m_e c} = 3.86 \times 10^{-13} \text{ m}$	$3.86 \times 10^{-11} \text{ cm}$
Compton wavelength of the pion	
$\lambda_\pi = \frac{h}{m_\pi c} = 1.4 \times 10^{-13} \text{ m}$	$1.4 \times 10^{-13} \text{ cm}$
Compton wavelength of the nucleon,	
$\lambda_n = \frac{h}{m_n c} = 0.2 \times 10^{-13} \text{ m}$	$0.2 \times 10^{-13} \text{ cm}$
Nuclear time $\tau_N = R_N/c \approx 0.5 \times 10^{-23} \text{ s}$, where R_N is the range of the nuclear force	
The temperature $293.4 \text{ K} = 20.4^\circ \text{C}$ corresponds to $kT = 0.0253 \text{ eV}$	
At an energy of $\frac{m_n v^2}{2} = 0.0253 \text{ eV}$ a neutron has a velocity	
$v_0 = 2200 \text{ m/s}$	
Wavelength of a neutron, $\lambda_n = 0.287/\mathcal{E}^{1/2}$ ($\mathcal{E}$ in eV, λ_n in 10^{-8} cm).	

3. Some astronomical constants

Sun's mass

$$M_S = 1.99 \times 10^{33} \text{ g}$$

Energy radiated by the Sun per second (luminous emittance)

$$L_S = 3.86 \times 10^{36} \text{ W} = 3.86 \times 10^{33} \text{ erg/s}$$

Sun's radius

$$R_S = 6.96 \times 10^{10} \text{ cm}$$

Angular radius of the Sun at the average distance from the Earth

$$\alpha_R = 4.65 \times 10^{-3} \text{ rad}$$

Sun's temperature near its surface

$$T_S = 5.5 \times 10^3 \text{ K}$$

Earth's mass

$$M_E = 5.98 \times 10^{24} \text{ kg}$$

Earth's radius (at the equator)

$$R_E = 6.38 \times 10^8 \text{ cm}$$

Earth's temperature (average)

$$T_E = 300 \text{ K}$$

Average orbital velocity of the Earth

$$v_E = 3 \times 10^6 \text{ cm/s}$$

II. Some Non-System Units

SI		CGS
Units of length:		
light year	$9.5 \times 10^{15} \text{ m}$	$9.5 \times 10^{17} \text{ cm}$
parsec (ps)	$3.1 \times 10^{16} \text{ m}$	$3.1 \times 10^{18} \text{ cm}$
ångström (Å)	10^{-10} m	10^{-8} cm
fermi	10^{-15} m	10^{-13} cm
Unit of area:		
barn	10^{-28} m^2	10^{-28} cm^2
Unit of time:		
year	$3.16 \times 10^7 \text{ s}$	
Unit of energy:		
electron-volt, 1 eV = $1.6 \times 10^{-19} \text{ J}$ = $1.6 \times 10^{-12} \text{ erg}$		

III. Units of Some Physical Quantities in the SI and CGS Systems and Their Relationships

Physical quantity	SI			CGS			Conversion factors
	Unit		Dimension	Unit		Dimension	
	Name	Notation		Name	Notation		
Frequency	hertz	Hz	T^{-1}	hertz	Hz	T^{-1}	1
Density	kilogram per cubic metre	kg/m^3	$L^{-3}M$	gram per cubic centimetre	g/cm^3	$L^{-3}M$	$1 \text{ kg/m}^3 = 10^3 \text{ g/cm}^3$
Velocity	metre per second	m/s	LT^{-1}	centimetre per second	cm/s	LT^{-1}	$1 \text{ m/s} = 10^3 \text{ cm/s}$
Acceleration	metre per square second	m/s^2	LT^{-2}	centimetre per square second	cm/s^2	LT^{-2}	$1 \text{ m/s}^2 = 10^3 \text{ cm/s}^2$
Angular velocity	radian per second	rad/s	T^{-1}	radian per second	rad/s	T^{-1}	1
Angular acceleration	radian per square second	rad/s^2	T^{-2}	radian per square second	rad/s^2	T^{-2}	1
Force	newton	N	LMT^{-2}	dyne	dyn	LMT^{-2}	$1 \text{ N} = 10^5 \text{ dyn}$
Work, energy, quantity of heat	joule	J	L^2MT^{-2}	erg	erg	L^2MT^{-2}	$1 \text{ J} = 10^7 \text{ erg}$
Power	watt	W	L^2MT^{-3}	erg per second	erg/s	L^2MT^{-3}	$1 \text{ W} = 10^7 \text{ erg/s}$

Quantity of electricity, electric charge	coulomb	C	TI	CGS unit	CGS	$L^2/M^1/T^{-1}$	$1 \text{ C} = 3 \cdot 10^9 \text{ CGS}$
Electric voltage, potential difference, e.m.f.	volt	V	$L^2MT^{-3}I^{-1}$	CGS-unit of voltage	CGS	$L^2/M^1/T^{-1}$	$1 \text{ V} = \frac{1}{300} \text{ CGS}$
Electric field strength	volt per metre	V/m	$LMT^{-3}I^{-1}$	CGS-unit of strength	CGS	$L^{-1}/M^1/T^{-1}$	$1 \text{ V/m} = \frac{1}{3 \times 10^4} \text{ CGS}$
Electric resistance	ohm	Ω	$L^2MT^{-3}I^{-2}$	CGS-unit of resistance	CGS	$L^{-1}T$	$1 \Omega = \frac{1}{9 \times 10^{11}} \text{ CGS}$
Inductance	henry	H	$L^2MT^{-3}I^{-2}$	CGS-unit of inductance	CGS	L	$1 \text{ H} = \frac{1}{9 \times 10^{11}} \text{ CGS}$
Capacitance	farad	F	$L^{-2}M^{-1}T^4I^2$	centimetre	cm	L	$1 \text{ F} = 9 \times 10^{11} \text{ cm}$
Magnetic flux	weber	Wb	$L^2MT^{-3}I^{-1}$	maxwell	Mx	$L^2/M^1/T^{-1}$	$1 \text{ Wb} = 10^8 \text{ Mx}$
Magnetic induction	tesla	T	$MT^{-3}I^{-1}$	gauss	G	$L^{-1}/M^1/T^{-1}$	$1 \text{ T} = 10^4 \text{ G}$
Magnetic field strength	ampere per metre	A/m	$L^{-1}I$	oersted	Oe	$L^{-1}/M^1/T^{-1}$	$1 \text{ A/m} = 4\pi \cdot 10^{-3} \text{ Oe}$
Light flux	lumen	lm	J	lumen	lm	J	1
Luminance	candela per square metre	cd/m^2	$L^{-3}J$	candela per square centimetre	cd/cm^2	$L^{-3}J$	$1 \text{ cd/m}^2 = 10^{-4} \text{ cd/cm}^2$
Illuminance	lux	lx	$L^{-3}J$	phot	ph	$L^{-3}J$	$1 \text{ lx} = 10^{-8} \text{ ph}$

IV. Elementary Particle Table *

Particle	Symbol**	Mass		Lifetime, s	Spin, $\hbar$	Charges					Isospin		Strangeness	Dominant decay path
		MeV	m_e			$\frac{1}{3}$	$\frac{2}{3}$	$\frac{4}{3}$	$\frac{5}{3}$	$\frac{8}{3}$	T	T_z		
Photon	γ	0	0	∞	1	0	0	0	0	0				
Neutrino	$\bar{\nu}$	0	0	∞	$1/2$	0	+1	0						
Electron	e^-	0.511	1	∞	$1/2$	-1	+1	0						
Muon	μ^-	105.86	206.77	$2.1 \cdot 10^{-6}$	$1/2$	-1	+1	0						$\mu^- \rightarrow e^- + \nu_\mu + \bar{\nu}_e$
Pi-mesons	$\pi^+ \pi^-$	139.6	273.2	$2.55 \cdot 10^{-8}$	0	+1	0	0	0	0	1	+1	0	$\pi^+ \rightarrow \mu^+ + \nu_\mu$
	π^0	135.0	264.2	$2 \cdot 10^{-16}$	0	0	0	0	0	0	1	0	0	$\pi^0 \rightarrow 2\gamma$
K-mesons	$K^+ K^-$	493.8	986.3	$1.23 \cdot 10^{-8}$	0	+1	0	0	0	0	$1/2$	$+1/2$	+1	$K^+ \rightarrow \mu^+ + \nu_\mu$
	$\bar{K}^0$	498.0	974.5	$K_1^0 1 \cdot 10^{-10}$ $K_2^0 6 \cdot 10^{-8}$	0	0	0	0	0	0	$1/2$	$-1/2$	+1	$K_1^0 \rightarrow 2\pi$ $K_2^0 \rightarrow 3\pi$

IV. (Continued)

Particle	Symbol**	Mass		Lifetime, s	Spin, s	Charges			Isospin		Strangeness	Dominant decay path
		MeV	m_e			Electric, e	Lepton, l	Baryon, b	T	T_z		
Proton	p $\tilde{p}$	938.26	1836.1	∞	1/2	+1	0	+1	1/2	+1/2	0	
Neutron	n $\tilde{n}$	939.55	1838.6	$1 \cdot 10^3$	1/2	0	0	+1	1/2	-1/2	0	$n \rightarrow p + e^- + \bar{\nu}_e$
Lambda-hyperons	Λ^0 $\tilde{\Lambda}^0$	1115.4	2182.8	$2.6 \cdot 10^{-10}$	1/2	0	0	+1	0	0	-1	$\Lambda^0 \rightarrow p + \pi^-$
	Σ^+ $\tilde{\Sigma}^-$	1189.4	2328	$0.8 \cdot 10^{-10}$	1/2	+1	0	+1	1	+1	-1	$\Sigma^+ \rightarrow N + \pi^+$
Sigma-hyperons	Σ^- $\tilde{\Sigma}^+$	1197	2342	$1.6 \cdot 10^{-10}$	1/2	-1	0	+1	1	-1	-1	$\Sigma^- \rightarrow n + \pi^-$
	Σ^0 $\tilde{\Sigma}^0$	1192	2333	$< 10^{-16}$	1/2	0	0	+1	1	0	-1	$\Sigma^0 \rightarrow \Lambda^0 + \gamma$
Xi-hyperons	Ξ^- $\tilde{\Xi}^+$	1321	2585	$1.7 \cdot 10^{-10}$	1/2	-1	0	+1	1/2	-1/2	-2	$\Xi^- \rightarrow \Lambda^0 + \pi^-$
	Ξ^0 $\tilde{\Xi}^0$	1314	2572	$3 \cdot 10^{-10}$	1/2	0	0	+1	1/2	+1/2	-2	$\Xi^0 \rightarrow \Lambda^0 + \pi^0$
Omega-hyperon	Ω^- $\tilde{\Omega}^+$	1675	3278	$\sim 10^{-10}$	3/2	-1	0	+1	0	0	-3	$\Omega^- \rightarrow \Xi + \pi, \Lambda^0 + K^-$

* An antiparticle has the same values of mass, lifetime, spin, and isospin T as the basic particle but its values of electric, lepton, and baryon charges, its isospin projection T_z , and strangeness S are all opposite in sign.

** The symbols of corresponding antiparticles are indicated on the right.

V. Prefixes and Powers for Decimal Multiple and Fractional Units

Name	Notation	Multiplier	Name	Notation	Multiplier
tera	T	10^{12}	centi	c	10^{-2}
giga	G	10^9	milli	m	10^{-3}
mega	M	10^6	micro	μ	10^{-6}
kilo	k	10^3	nano	n	10^{-9}
hecto	h	10^2	pico	p	10^{-12}
deca	da	10^1	femto	f	10^{-15}
deci	d	10^{-1}	atto	a	10^{-18}

Periodic Table

Period	Series	Groups of				
		I	II	III	IV	V
1	1	H 1.00797 hydrogen				
2	2	Li 6.939 lithium	Be 9.0122 beryllium	B 10.811 boron	C 12.01115 carbon	N 14.0067 nitrogen
3	3	Na 22.9898 sodium	Mg 24.31 magnesium	Al 26.9815 aluminium	Si 28.086 silicon	P 30.9738 phosphorus
4	4	K 39.102 potassium	Ca 40.08 calcium	Sc 44.956 scandium	Ti 47.90 titanium	V 50.942 vanadium
	5	Cu 63.546 copper	Zn 65.37 zinc	Ga 69.72 gallium	Ge 72.59 germanium	As 74.92 arsenic
5	6	Rb 85.47 rubidium	Sr 87.62 strontium	Y 88.905 yttrium	Zr 91.22 zirconium	Nb 92.908 niobium
	7	Ag 107.868 silver	Cd 112.40 cadmium	In 114.82 indium	Sn 118.69 tin	Sb 121.75 antimony
6	8	Cs 132.905 caesium	Ba 137.34 barium	[*] La 138.91 lanthanum	Hf 178.49 hafnium	Ta 180.95 tantalum
	9	Au 196.967 gold	Hg 200.59 mercury	Tl 204.37 thallium	Pb 207.19 lead	Bi 208.98 bismuth
7	10	Fr (223) francium	Ra (226) radium	[**] Ac (227) actinium	Ku (260) kurchatovium	

* Lanthanides

58 Ce 140.12 cerium	59 Pr 140.91 praseodymium	60 Nd 144.24 neodymium	61 Pm (145) promethium	62 Sm 150.35 samarium	63 Eu 151.96 europium	64 Gd 157.25 gadolinium
65 Tb 158.92 terbium	66 Dy 162.50 dysprosium	67 Ho 164.93 holmium	68 Er 167.26 erbium	69 Tm 168.93 thulium	70 Yb 173.04 ytterbium	71 Lu 174.97 lutetium

The bracketed integer indicates the mass number of the most stable radio

of Chemical Elements

elements						
VI		VII		VIII		0
						He ² 4.0026 helium
O ⁸ 15.9994 oxygen		F ⁹ 18.9984 fluorine				Ne ¹⁰ 20.183 neon
S ¹⁶ 32.064 sulphur		Cl ¹⁷ 35.453 chlorine				Ar ¹⁸ 39.948 argon
24 Cr 51.996 chromium		25 Mn 54.938 manganese	26 Fe 55.847 iron	27 Co 58.933 cobalt	28 Ni 58.71 nickel	
Se ³⁴ 78.96 selenium		Br ³⁵ 79.90 bromine				Kr ³⁶ 83.80 krypton
42 Mo 95.94 molybdenum		43 Tc (99) technetium	44 Ru 101.07 ruthenium	45 Rh 102.905 rhodium	46 Pd 106.4 palladium	
Te ⁵² 127.80 tellurium		I ⁵³ 126.904 iodine				Xe ⁵⁴ 131.30 xenon
74 W 183.85 tungsten		75 Re 186.2 rhenium	76 Os 190.2 osmium	77 Ir 192.2 iridium	78 Pt 195.09 platinum	
Po ⁸⁴ (210) polonium		At ⁸⁵ (210) astatine				Rn ⁸⁶ (222) radon

** Actinides

90 Th 232.038 thorium	91 Pa (231) protactinium	92 U 238.03 uranium	93 Np (237) neptunium	94 Pu (242) plutonium	95 Am (243) americium	96 Cm (247) curium
97 Bk (247) berkelium	98 Cf (249) californium	99 Es (254) einsteinium	100 Fm (253) fermium	101 Md (256) mendelevium	102 (No) (256) (nobelium)	103 Lr (257) lawrencium

isotope.

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